MATH2023: Multivariable Calculus

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Contents

1 Vectors and the Geometry of Space **4**
1.1 Three-dimensional coordinate systems . . . 4
1.1.1 Cartesian coordinates . . . 4
1.1.2 The distance formula . . . 4
1.1.3 Surfaces described by equations . . . 5
1.2 Cylinders and quadric surfaces . . . 5
1.2.1 Cylinders . . . 5
1.2.2 Quadric surfaces . . . 6
1.3 Vectors . . . 8
1.3.1 Geometric definition . . . 8
1.3.2 Component form . . . 8
1.4 The dot product . . . 9
1.4.1 Definition and algebraic properties . . . 9
1.4.2 The dot product and the cosine . . . 9
1.4.3 Projections . . . 10
1.5 The cross product . . . 11
1.5.1 Geometric definition . . . 11
1.5.2 Further properties . . . 12
1.6 Equations of lines and planes . . . 13
1.6.1 Lines . . . 13
1.6.2 Planes . . . 14
1.6.3 Distance from a point to a plane . . . 15
1.7 Parametric curves and their derivatives . . . 15
1.8 Parameterization table . . . 17

2 Differential Calculus for Multivariable Functions **17**
2.1 Functions of several variables . . . 17
2.1.1 The graph . . . 18
2.1.2 Level curves and level surfaces . . . 18
2.2 Limits and continuity . . . 18
2.2.1 The definition . . . 18
2.2.2 Properties of limits . . . 19
2.2.3 Continuity . . . 20
2.3 Extra: analysis in brief . . . 20
2.3.1 Basic $\mathbb{R}^n$ topology . . . 20
2.3.2 Bolzano–Weierstrass theorem . . . 22
2.3.3 Topology and the extreme value theorem . . . 22
2.3.4 Uniform continuity . . . 23

2.4 Partial derivatives . . . 23
2.4.1 Definition . . . 23
2.4.2 Implicit differentiation . . . 24
2.4.3 Higher derivatives . . . 25
2.5 Tangent planes and linear approximation . . . 26
2.5.1 Tangent plane to a graph . . . 26
2.5.2 Tangent plane to a level surface . . . 26
2.5.3 Linear approximation and differentiability . . . 27
2.5.4 Algebraic properties of derivatives . . . 28
2.6 The chain rule . . . 29
2.7 Directional derivatives and the gradient . . . 30
2.7.1 Bounds and steepest ascent . . . 31
2.8 Maximum and minimum values . . . 32
2.8.1 Local extrema . . . 32
2.8.2 The second derivative test . . . 33
2.8.3 Absolute extrema . . . 34
2.9 Lagrange multipliers . . . 35
2.9.1 The method . . . 35
2.9.2 Two constraints . . . 37
2.10 Extra: the Hessian and the multivariable mean value theorem . . . 37
2.10.1 The Hessian matrix . . . 37
2.10.2 The multivariable mean value theorem . . . 39
2.10.3 More on the gradient . . . 39
2.10.4 Extra: divergence, curl and the Laplacian . . . 40

3 Multiple Integrals **42**
3.1 Double integrals over rectangles . . . 42
3.1.1 The definition . . . 42
3.1.2 Properties . . . 43
3.2 Iterated integrals and Fubini's theorem . . . 44
3.3 Double integrals over general regions . . . 45
3.3.1 Type I and type II regions . . . 45
3.4 Double integrals in polar coordinates . . . 47
3.4.1 Polar rectangles . . . 47
3.5 Applications of double integrals . . . 49
3.6 Surface area . . . 50
3.6.1 The formula . . . 50
3.7 Triple integrals . . . 51
3.7.1 Definition and Fubini's theorem . . . 51
3.8 Triple integrals in cylindrical and spherical coordinates . . . 52
3.8.1 Cylindrical coordinates . . . 52
3.8.2 Spherical coordinates . . . 52
3.9 Change of variables . . . 53
3.9.1 Transformations and the Jacobian . . . 53
3.9.2 The change of variables formula . . . 54

4 Vector Calculus **55**
4.1 Vector fields . . . 55
4.2 Line integrals . . . 56
4.2.1 Line integrals of scalar functions . . . 56
4.2.2 Line integrals of vector fields . . . 57

4.3 The fundamental theorem for line integrals . . . 58
4.3.1 Path independence and conservative fields . . . 59
4.4 Green's theorem . . . 61
4.4.1 Statement . . . 61
4.4.2 Proof for regions of both types . . . 62
4.4.3 Extended versions: regions with holes . . . 63
4.4.4 Limitations of Green's theorem and winding numbers . . . 64
4.4.5 Self-intersecting curves . . . 64
4.5 Curl and divergence . . . 65
4.6 Parametric surfaces and surface integrals . . . 66
4.6.1 Parametric surfaces . . . 66
4.6.2 Surface integrals of scalar functions . . . 66
4.6.3 Oriented surfaces and flux . . . 67
4.7 Stokes' theorem . . . 68
4.8 The divergence theorem . . . 69
4.8.1 Limitations of the divergence theorem . . . 70
4.9 Useful table of the integral theorems . . . 72

1 Vectors and the Geometry of Space

Single-variable calculus operates on subsets of $\mathbb{R}$; multivariable calculus operates on subsets of $\mathbb{R}^n$, and in this course mostly on $\mathbb{R}^2$ and $\mathbb{R}^3$. This chapter sets up the geometry of $\mathbb{R}^3$: coordinates, vectors, dot and cross products, lines and planes, and parametrized curves.

1.1 Three-dimensional coordinate systems

1.1.1 Cartesian coordinates

Points in space

To locate a point in space we need three numbers: we choose an origin O, three mutually perpendicular coordinate axes (the x-, y- and z-axes) meeting at O, and a common unit of length. The triple (a, b, c) records the signed distances from the three coordinate planes

$$(a,b,c) \in \mathbb{R}^3 \quad \Longleftrightarrow \quad \begin{cases} \text{distance } a \text{ from the } yz\text{-plane,} \\ \text{distance } b \text{ from the } xz\text{-plane,} \\ \text{distance } c \text{ from the } xy\text{-plane.} \end{cases}$$

The three coordinate planes divide space into eight **octants**; the **first octant** is the set of points with $x, y, z \geq 0$.

Orientation: the right-hand rule

The orientation of the axes is fixed by convention: if the fingers of the right hand curl from the positive x-axis towards the positive y-axis, then the thumb points along the positive z-axis. Equivalently, the ordered triple $(\vec{i}, \vec{j}, \vec{k})$ is *positively oriented*.

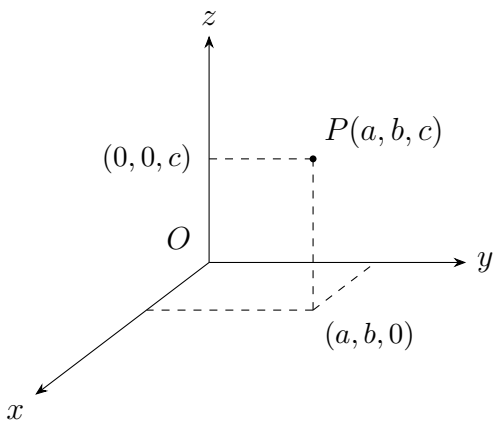


Figure 1: Locating a point by three coordinates.

1.1.2 The distance formula

Distance in $\mathbb{R}^3$

The distance between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$\boxed{|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

In particular the distance from $P(x, y, z)$ to the origin is $\sqrt{x^2 + y^2 + z^2}$.

Proof. Let $A(x_2, y_1, z_1)$ and $B(x_2, y_2, z_1)$ be the auxiliary points obtained by moving one coordinate at a time. The segment P_1A is parallel to the x-axis, AB to the y-axis and BP_2 to the z-axis, and the three segments are mutually perpendicular. Applying Pythagoras twice in the right triangles $\triangle P_1AB$ and $\triangle P_1BP_2$:

$$|P_1B|^2 = |P_1A|^2 + |AB|^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2,$$

$$|P_1P_2|^2 = |P_1B|^2 + |BP_2|^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2.$$

Taking square roots gives the formula. ■

Sphere

The sphere with centre $C(h, k, l)$ and radius r is the set

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2.$$

The **standard equation** is obtained by expanding:

$$x^2 + y^2 + z^2 + Dx + Ey + Fz + G = 0.$$

Completing the square recovers the centre and the radius; if the right-hand side is negative the set is empty, and if it is 0 the set is a single point.

Worked example: a sphere

Describe the surface $x^2 + y^2 + z^2 - 4x + 6y - 2z + 5 = 0$.
Grouping and completing squares,

$$(x^2 - 4x + 4) + (y^2 + 6y + 9) + (z^2 - 2z + 1) = 4 + 9 + 1 - 5 = 9,$$

so $(x - 2)^2 + (y + 3)^2 + (z - 1)^2 = 9$: a sphere of radius 3 centred at $(2, -3, 1)$.

1.1.3 Surfaces described by equations

A single equation in x, y, z generally describes a surface (one equation removes one degree of freedom), two equations describe a curve, and three equations generally describe finitely many points.

- $y = 5$: the plane parallel to the xz-plane through $(0, 5, 0)$.
- $x^2 + y^2 = 1$ in $\mathbb{R}^3$: the (infinite) circular cylinder obtained by translating the unit circle along the z-axis.
- $x^2 + y^2 = 1$ together with $z = 2$: the circle of radius 1 at height $z = 2$, obtained as an intersection of a cylinder and a plane.
- $z = x^2 + y^2$: an upward paraboloid; $z^2 = x^2 + y^2$: a double cone; $x^2 + y^2 - z^2 = 1$: a hyperboloid of one sheet.

1.2 Cylinders and quadric surfaces

1.2.1 Cylinders

Cylinder

A **cylinder** is the surface formed by all lines parallel to a fixed line (the **generatrix**) and passing through a given curve (the **directrix**).

Example: parabolic cylinder $z = x^2$

For each fixed value of y the cross-section in the xz-plane is the parabola $z = x^2$. The surface is therefore obtained by translating this parabola along the y-axis, and it extends infinitely in that direction.

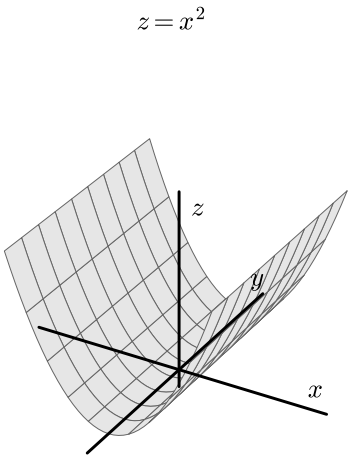


Figure 2: The parabolic cylinder $z = x^2$, generated by translating the parabola along the y-axis.

Example: circular cylinder $x^2 + y^2 = 1$

The surface consists of all points whose projection onto the xy-plane lies on the circle $x^2 + y^2 = 1$. Each horizontal slice (fixed z) is a circle of radius 1, so the surface extends infinitely along the z-axis.

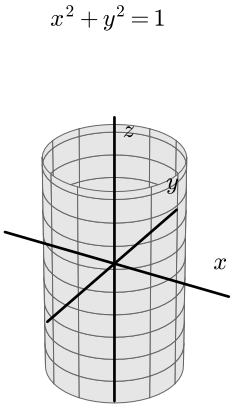


Figure 3: The circular cylinder $x^2 + y^2 = 1$, generated by translating the unit circle along the z-axis.

1.2.2 Quadric surfaces

Quadric surface

A **quadric surface** is a surface in $\mathbb{R}^3$ defined by a second-degree polynomial equation in x, y, z:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0.$$

By a translation or a rotation of the coordinates it can be brought into one of the forms

$$Ax^2 + By^2 + Cz^2 + J = 0, \qquad Ax^2 + By^2 + Iz = 0.$$

For $a, b, c \in \mathbb{R} \setminus \{0\}$ there are six kinds of quadric surfaces:

- (a) **Ellipsoid**: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$;
- (b) **elliptic paraboloid**: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = z$;
- (c) **hyperbolic paraboloid**: $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = z$;
- (d) **elliptic cone**: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z^2}{c^2}$;
- (e) **hyperboloid of one sheet**: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$;
- (f) **hyperboloid of two sheets**: $\dfrac{z^2}{c^2} - \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$.

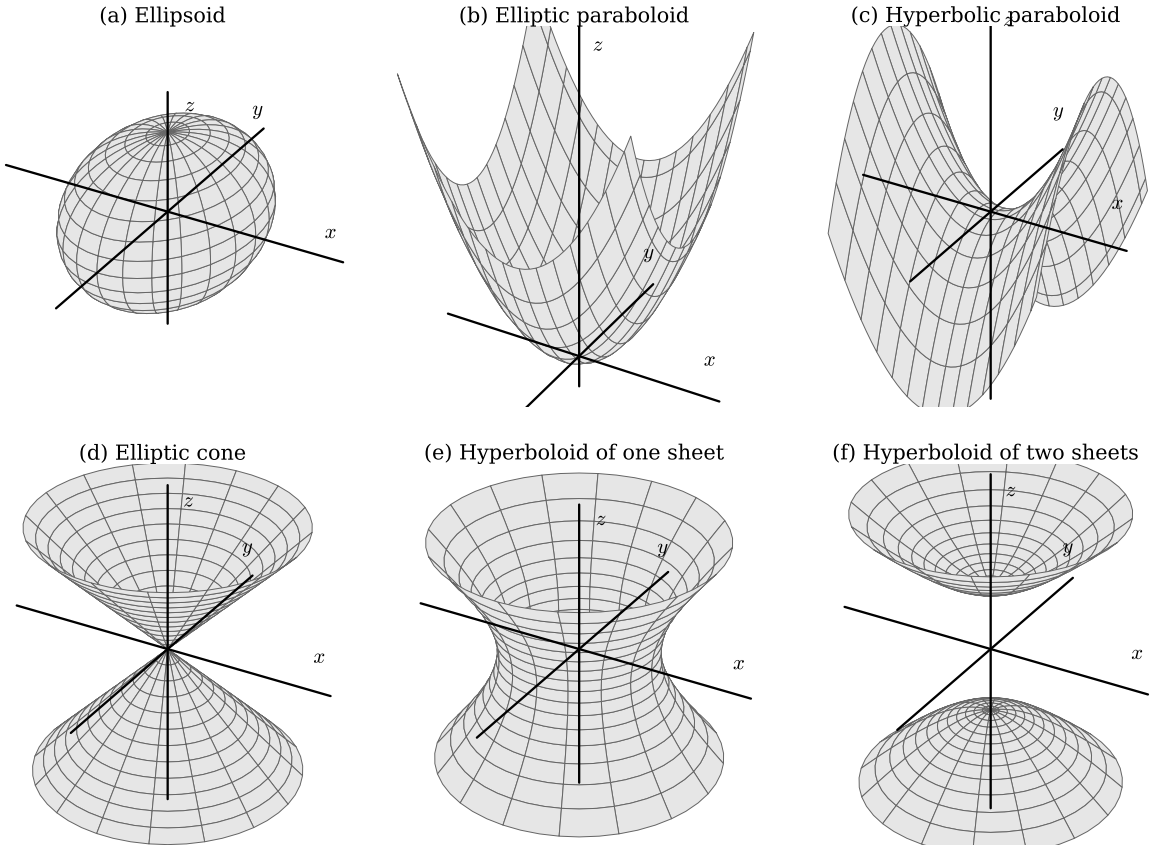


Figure 4: The six quadric surfaces.

Cross sections of quadric surfaces

The type of a quadric is recognised from its traces in the three coordinate planes:

Surface	xy-plane	yz-plane	xz-plane
Ellipsoid	ellipse	ellipse	ellipse
Elliptic paraboloid	ellipse ($z > 0$)	parabola	parabola
Hyperbolic paraboloid	hyperbola ($z \neq 0$)	parabola	parabola
Elliptic cone	ellipse ($z \neq 0$)	hyperbola ($x \neq 0$) two lines ($x = 0$)	hyperbola ($y \neq 0$) two lines ($y = 0$)
Hyperboloid of one sheet	ellipse	hyperbola	hyperbola
Hyperboloid of two sheets	ellipse ($\lvert z\rvert > \lvert c\rvert$)	hyperbola	hyperbola

1.3 Vectors

1.3.1 Geometric definition

Vectors

A **vector** is a directed line segment considered *up to translation*: two vectors are equal precisely when they have the same length and the same direction. The vector with initial point A and terminal point B is written $\overrightarrow{AB}$. The **zero vector** $\vec{0}$ has length 0 and no direction.

Vector operations are defined geometrically:

- **Addition (triangle law):** $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$. Equivalently (parallelogram law), $\overrightarrow{AB} + \overrightarrow{AD}$ is the diagonal of the parallelogram spanned by the two vectors.
- **Subtraction:** $\overrightarrow{OA} - \overrightarrow{OB} = \overrightarrow{BA}$.
- **Scalar multiplication:** $c\,\overrightarrow{AB}$ has length $|c|\,|\overrightarrow{AB}|$; its direction is that of $\overrightarrow{AB}$ if $c > 0$, opposite if $c < 0$, and it is $\vec{0}$ if $c = 0$.

1.3.2 Component form

Place the initial point of a vector $\vec{v}$ at the origin; its terminal point $P(x, y, z)$ gives the **position vector** $\vec{v} = \overrightarrow{OP} = \langle x, y, z\rangle$. We use angle brackets $\langle\,\cdot\,\rangle$ for vectors and parentheses $(\,\cdot\,)$ for points.

Algebra of components

For $\vec{u} = \langle u_1, u_2, u_3\rangle$, $\vec{v} = \langle v_1, v_2, v_3\rangle$ and $c \in \mathbb{R}$:

$$\vec{u} + \vec{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3\rangle, \qquad c\vec{u} = \langle cu_1, cu_2, cu_3\rangle,$$

$$|\vec{u}| = \sqrt{u_1^2 + u_2^2 + u_3^2}, \qquad \overrightarrow{AB} = \langle x_B - x_A,\ y_B - y_A,\ z_B - z_A\rangle.$$

All operations are performed *componentwise*.

Remark. The last formula is a consequence of $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$.

Standard basis vectors

Write

$$\vec{i} = \langle 1, 0, 0\rangle, \qquad \vec{j} = \langle 0, 1, 0\rangle, \qquad \vec{k} = \langle 0, 0, 1\rangle.$$

Every vector has a *unique* expression

$$\vec{v} = \langle x, y, z\rangle = x\vec{i} + y\vec{j} + z\vec{k}.$$

A **unit vector** has length 1; the unit vector in the direction of $\vec{v} \neq \vec{0}$ is

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|}.$$

Proof of uniqueness. If $x\vec{i}+y\vec{j}+z\vec{k} = x'\vec{i}+y'\vec{j}+z'\vec{k}$, then componentwise $(x-x')\vec{i}+(y-y')\vec{j}+(z-z')\vec{k}=\vec{0}$, i.e. $\langle x-x', y-y', z-z'\rangle = \langle 0,0,0\rangle$, hence $x=x'$, $y=y'$, $z=z'$. ∎

1.4 The dot product

1.4.1 Definition and algebraic properties

Dot product

For $\vec{u}=\langle u_1,u_2,u_3\rangle$ and $\vec{v}=\langle v_1,v_2,v_3\rangle$, the **dot product** (or scalar product) is the *number*

$$\boxed{\vec{u}\cdot\vec{v} = u_1v_1+u_2v_2+u_3v_3}$$

The same definition applies in $\mathbb{R}^2$ with two components.

Algebraic properties

For all vectors $\vec{u},\vec{v},\vec{w}$ and scalars c:

(i) $\vec{u}\cdot\vec{u} = |\vec{u}|^2$;
(ii) $\vec{u}\cdot\vec{v} = \vec{v}\cdot\vec{u}$;
(iii) $\vec{u}\cdot(\vec{v}+\vec{w}) = \vec{u}\cdot\vec{v}+\vec{u}\cdot\vec{w}$;
(iv) $(c\vec{u})\cdot\vec{v} = c(\vec{u}\cdot\vec{v})$;
(v) $\vec{u}\cdot\vec{v} = \dfrac{1}{4}\big(|\vec{u}+\vec{v}|^2-|\vec{u}-\vec{v}|^2\big)$ (polarization identity);
(vi) $\vec{0}\cdot\vec{v}=0$.

Proof. (i) $\vec{u}\cdot\vec{u}=u_1^2+u_2^2+u_3^2=|\vec{u}|^2$. (ii) Multiplication of numbers is commutative. (iii) Expand:

$$\vec{u}\cdot(\vec{v}+\vec{w}) = \sum_{i=1}^{3} u_i(v_i+w_i) = \sum_{i=1}^{3} u_iv_i + \sum_{i=1}^{3} u_iw_i.$$

(iv) is immediate from $\sum u_i(cv_i) = c\sum u_iv_i$. (v) Expand both squares using (iii) and (ii):

$$|\vec{u}+\vec{v}|^2 = |\vec{u}|^2+2\vec{u}\cdot\vec{v}+|\vec{v}|^2, \qquad |\vec{u}-\vec{v}|^2 = |\vec{u}|^2-2\vec{u}\cdot\vec{v}+|\vec{v}|^2,$$

and subtract. (vi) is clear. ∎

1.4.2 The dot product and the cosine

Geometric meaning

If $\theta\in[0,\pi]$ is the angle between $\vec{u}$ and $\vec{v}$, then

$$\boxed{\vec{u}\cdot\vec{v} = |\vec{u}|\,|\vec{v}|\cos\theta}$$

Consequently

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{|\vec{u}|\,|\vec{v}|}, \qquad \vec{u}\perp\vec{v} \iff \vec{u}\cdot\vec{v}=0.$$

Proof (Stewart). Consider the triangle with sides $\vec{u}$, $\vec{v}$ and $\vec{u}-\vec{v}$, and let θ be the angle between $\vec{u}$ and $\vec{v}$. The law of cosines gives

$$|\vec{u}-\vec{v}|^2 = |\vec{u}|^2+|\vec{v}|^2-2|\vec{u}|\,|\vec{v}|\cos\theta.$$

On the other hand, expanding with the algebraic properties,

$$|\vec{u}-\vec{v}|^2 = (\vec{u}-\vec{v})\cdot(\vec{u}-\vec{v}) = |\vec{u}|^2 - 2\vec{u}\cdot\vec{v} + |\vec{v}|^2.$$

Comparing the two expressions yields $-2\vec{u}\cdot\vec{v} = -2|\vec{u}||\vec{v}|\cos\theta$, which is the claim. ∎

Cauchy–Schwarz inequality

For all $\vec{u},\vec{v}\in\mathbb{R}^3$,

$$|\vec{u}\cdot\vec{v}| \le |\vec{u}|\,|\vec{v}|,$$

with equality if and only if one vector is a scalar multiple of the other.

Proof. By the theorem, $|\vec{u}\cdot\vec{v}| = |\vec{u}||\vec{v}||\cos\theta| \le |\vec{u}||\vec{v}|$ since $|\cos\theta|\le 1$. Equality holds iff $\cos\theta = \pm 1$, i.e. $\theta = 0$ or π, i.e. the vectors are parallel. ∎

Worked examples

(a) $\vec{u}=\langle 1,1,1\rangle$, $\vec{v}=\langle 1,-1,0\rangle$: $\vec{u}\cdot\vec{v}=0$, so $\theta=\pi/2$. (b) $\vec{u}=\langle 1,0\rangle$, $\vec{v}=\langle 2,2\rangle$: $\cos\theta = \dfrac{2}{1\cdot 2\sqrt{2}} = \dfrac{\sqrt{2}}{2}$, so $\theta=\pi/4$. (c) The **direction angles** α,β,γ of $\vec{v}$ are the angles with the positive coordinate axes; then

$$\cos\alpha = \frac{v_1}{|\vec{v}|},\quad \cos\beta = \frac{v_2}{|\vec{v}|},\quad \cos\gamma = \frac{v_3}{|\vec{v}|},\qquad \cos^2\alpha+\cos^2\beta+\cos^2\gamma = 1.$$

Application: the law of cosines and Thales' theorem

Since $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$, taking the dot product with itself gives

$$|BC|^2 = |AC|^2 + |AB|^2 - 2\,\overrightarrow{AC}\cdot\overrightarrow{AB} = |AC|^2 + |AB|^2 - 2|AC||AB|\cos\theta.$$

For $\theta=\pi/2$ this reduces to Pythagoras' theorem.
Thales. Let ABC be a triangle inscribed in a circle of radius r and centre O, with AC a diameter. Then $\overrightarrow{OC} = -\overrightarrow{OA}$ and

$$\begin{aligned}\overrightarrow{BA}\cdot\overrightarrow{BC} &= (\overrightarrow{OA}-\overrightarrow{OB})\cdot(\overrightarrow{OC}-\overrightarrow{OB})\\ &= (-\overrightarrow{OA}-\overrightarrow{OB})\cdot(\overrightarrow{OA}-\overrightarrow{OB}) = |\overrightarrow{OB}|^2 - |\overrightarrow{OA}|^2 = r^2 - r^2 = 0.\end{aligned}$$

Hence the angle at B is a right angle.

1.4.3 Projections

Scalar and vector projection

Let $\vec{a}\neq\vec{0}$. The **vector projection** of $\vec{b}$ onto $\vec{a}$ is the vector

$$\boxed{\operatorname{proj}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|^2}\,\vec{a} = (\vec{a}\cdot\hat{a})\,\hat{a}}$$

and the **scalar projection** (component) is the signed length

$$\boxed{\operatorname{comp}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|} = \vec{b}\cdot\hat{a}}$$

Then $\operatorname{proj}_{\vec{a}}\vec{b} = \big(\operatorname{comp}_{\vec{a}}\vec{b}\big)\hat{a}$, and $\operatorname{comp}_{\vec{a}}\vec{b} > 0$ when the projection points along $\vec{a}$, < 0 when it points opposite to $\vec{a}$.

Proof. Let S be the foot of the perpendicular from the terminal point of $\vec{b}$ to the line through $\vec{a}$. Then $\operatorname{proj}_{\vec{a}}\vec{b} = \overrightarrow{PS}$ is parallel to $\vec{a}$, so $\overrightarrow{PS} = t\vec{a}$ for some $t\in\mathbb{R}$. The vector

$\vec{b} - t\vec{a} = \overrightarrow{SQ}$ is perpendicular to $\vec{a}$, hence

$$0 = (\vec{b} - t\vec{a}) \cdot \vec{a} = \vec{a} \cdot \vec{b} - t|\vec{a}|^2 \quad \Longrightarrow \quad t = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2}.$$

The signed length is $|\overrightarrow{PS}| = |t|\,|\vec{a}|$ with the sign of t, i.e. $\operatorname{comp}_{\vec{a}} \vec{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$. ■

Worked example: work

A constant force $\vec{F} = \langle 3, 3, 2 \rangle$ N moves a particle from $P(1, 1, 0)$ to $Q(6, 6, 0)$ (metres). The displacement is $\vec{D} = \overrightarrow{PQ} = \langle 5, 5, 0 \rangle$ and the work is

$$W = \vec{F} \cdot \vec{D} = 15 + 15 + 0 = 30 \text{ J}.$$

In general $W = \vec{F} \cdot \vec{D} = |\vec{F}||\vec{D}| \cos\theta$, which may be negative.

1.5 The cross product

1.5.1 Geometric definition

Cross product

For $\vec{u}, \vec{v} \in \mathbb{R}^3$, the **cross product** $\vec{u} \times \vec{v}$ is the vector which is
- perpendicular to both $\vec{u}$ and $\vec{v}$;
- oriented by the right-hand rule (curl the fingers from $\vec{u}$ to $\vec{v}$; the thumb points along $\vec{u} \times \vec{v}$);
- of length $|\vec{u} \times \vec{v}| = |\vec{u}|\,|\vec{v}| \sin\theta$, where θ is the angle between $\vec{u}$ and $\vec{v}$.

Thus $|\vec{u} \times \vec{v}|$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$.

Algebraic definition

For $\vec{u} = \langle u_1, u_2, u_3 \rangle$, $\vec{v} = \langle v_1, v_2, v_3 \rangle$,

$$\boxed{\vec{u} \times \vec{v} = \langle u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1 \rangle}$$

Proof that the two definitions agree. Write $\vec{u} = u_1\vec{i} + u_2\vec{j} + u_3\vec{k}$ and $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$. The cross product is distributive and satisfies $\vec{i} \times \vec{j} = \vec{k}$, $\vec{j} \times \vec{k} = \vec{i}$, $\vec{k} \times \vec{i} = \vec{j}$, $\vec{i} \times \vec{i} = \vec{j} \times \vec{j} = \vec{k} \times \vec{k} = \vec{0}$ and $\vec{v} \times \vec{u} = -\vec{u} \times \vec{v}$ (each of these is read off from the geometric definition). Expanding bilinearly,

$$\vec{u} \times \vec{v} = u_1v_1(\vec{i} \times \vec{i}) + u_1v_2(\vec{i} \times \vec{j}) + u_1v_3(\vec{i} \times \vec{k}) + u_2v_1(\vec{j} \times \vec{i}) + \cdots$$

$$= u_1v_2\vec{k} - u_1v_3\vec{j} - u_2v_1\vec{k} + u_2v_3\vec{i} + u_3v_1\vec{j} - u_3v_2\vec{i} = \langle u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1 \rangle.$$

That this vector is perpendicular to $\vec{u}$ and $\vec{v}$ and has the stated length is exactly the content of the next two results. ■

Magnitude and the sine

$$\boxed{|\vec{u} \times \vec{v}|^2 = |\vec{u}|^2|\vec{v}|^2 - (\vec{u} \cdot \vec{v})^2} \qquad \Longrightarrow \qquad |\vec{u} \times \vec{v}| = |\vec{u}|\,|\vec{v}| \sin\theta.$$

Proof. Expanding both sides in components, the identity

$$(u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2$$
$$= (u_1^2 + u_2^2 + u_3^2)(v_1^2 + v_2^2 + v_3^2) - (u_1v_1 + u_2v_2 + u_3v_3)^2$$

holds (Lagrange's identity; verify by expanding). By the previous section,

$\vec{u}\cdot\vec{v} = |\vec{u}||\vec{v}|\cos\theta$, so

$$|\vec{u}\times\vec{v}|^2 = |\vec{u}|^2|\vec{v}|^2\big(1-\cos^2\theta\big) = |\vec{u}|^2|\vec{v}|^2\sin^2\theta.$$

Since $\theta\in[0,\pi]$, $\sin\theta\geq 0$ and the claim follows. ■

Perpendicularity

$(\vec{u}\times\vec{v})\cdot\vec{u} = 0$ and $(\vec{u}\times\vec{v})\cdot\vec{v} = 0$.

Proof. Using the algebraic formula,

$$(\vec{u}\times\vec{v})\cdot\vec{u} = u_1(u_2v_3-u_3v_2)+u_2(u_3v_1-u_1v_3)+u_3(u_1v_2-u_2v_1) = 0,$$

since every monomial appears twice with opposite signs. The other identity is similar (or follows from $\vec{v}\times\vec{u} = -\vec{u}\times\vec{v}$). ■

Determinant mnemonic

$$\vec{u}\times\vec{v} = \begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{vmatrix} = \vec{i}\begin{vmatrix}u_2 & u_3\\ v_2 & v_3\end{vmatrix} - \vec{j}\begin{vmatrix}u_1 & u_3\\ v_1 & v_3\end{vmatrix} + \vec{k}\begin{vmatrix}u_1 & u_2\\ v_1 & v_2\end{vmatrix}.$$

Expanding along the first row reproduces the algebraic formula.

Worked example

$\langle 1,2,3\rangle\times\langle 4,5,6\rangle = \langle 2\cdot 6-3\cdot 5,\ 3\cdot 4-1\cdot 6,\ 1\cdot 5-2\cdot 4\rangle = \langle -3,6,-3\rangle$.

1.5.2 Further properties

Properties

(i) $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$ (anticommutativity);
(ii) $(\vec{a}+\vec{b})\times\vec{c} = \vec{a}\times\vec{c}+\vec{b}\times\vec{c}$ (distributivity);
(iii) $\vec{a}\times\vec{0} = \vec{0}$, $\vec{a}\times\vec{a} = \vec{0}$;
(iv) $\vec{i}\times\vec{j} = \vec{k}$, $\vec{j}\times\vec{k} = \vec{i}$, $\vec{k}\times\vec{i} = \vec{j}$;
(v) $\vec{a}\times\vec{b} = \vec{0}$ iff $\vec{a}$ and $\vec{b}$ are parallel (or one of them is $\vec{0}$);
(vi) $|\vec{a}\times\vec{b}|$ is the area of the parallelogram spanned by $\vec{a},\vec{b}$; $\dfrac{1}{2}|\vec{a}\times\vec{b}|$ is the area of the triangle they span.

Proof. (i)–(iv) follow by direct computation from the algebraic definition. For (v): if $\vec{a},\vec{b}\neq\vec{0}$, then $\vec{a}\times\vec{b} = \vec{0} \iff |\vec{a}||\vec{b}|\sin\theta = 0 \iff \sin\theta = 0 \iff \theta\in\{0,\pi\}$. (vi) is the geometric definition together with the formula for the area of a parallelogram (base times height $= |\vec{a}|\,|\vec{b}|\sin\theta$). ■

Scalar triple product and volume

For $\vec{u},\vec{v},\vec{w}\in\mathbb{R}^3$, the number

$$\vec{u}\cdot(\vec{v}\times\vec{w}) = \begin{vmatrix}u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\end{vmatrix}$$

is the **scalar triple product**, and its absolute value is the volume of the parallelepiped spanned by $\vec{u},\vec{v},\vec{w}$.

Proof. The base of the parallelepiped is the parallelogram spanned by $\vec{v}$ and $\vec{w}$, whose area is $|\vec{v}\times\vec{w}|$. The height is the length of the projection of $\vec{u}$ onto the normal direction

$\vec{v} \times \vec{w}$, namely

$$h = |\operatorname{comp}_{\vec{v}\times\vec{w}} \vec{u}| = \frac{|\vec{u} \cdot (\vec{v} \times \vec{w})|}{|\vec{v} \times \vec{w}|}.$$

Hence $V = (\text{base}) \cdot h = |\vec{u} \cdot (\vec{v} \times \vec{w})|$. The determinant formula is the expansion of $\vec{u} \cdot (\vec{v} \times \vec{w})$. ■

Worked example: volume

For $A(1,2,3)$, $B(4,5,6)$, $C(2,3,5)$, $D(4,7,8)$, the parallelepiped spanned by $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ has volume

$$\left| \det \begin{pmatrix} 3 & 3 & 3 \\ 1 & 1 & 2 \\ 3 & 5 & 5 \end{pmatrix} \right| = |-6| = 6.$$

Torque

If a force $\vec{F}$ acts at a point with position vector $\vec{r}$ relative to a pivot, the **torque** is $\vec{\tau} = \vec{r} \times \vec{F}$; its magnitude $|\vec{r}||\vec{F}| \sin\theta$ measures the turning effect.

1.6 Equations of lines and planes

1.6.1 Lines

Vector, parametric and symmetric equations

The line L through $P_0(x_0, y_0, z_0)$ parallel to $\vec{v} = \langle a, b, c \rangle \neq \vec{0}$ consists of the points

$$\vec{r}(t) = \vec{r}_0 + t\vec{v}, \qquad t \in \mathbb{R},$$

i.e.

$$\boxed{x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct}$$

(**parametric equations**). Eliminating t gives the **symmetric equations**

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \qquad (abc \neq 0),$$

with the obvious modification when some component of $\vec{v}$ vanishes, e.g. $a = 0$ gives $x = x_0$ and $\dfrac{y - y_0}{b} = \dfrac{z - z_0}{c}$.

Justification. $P \in L$ iff $\overrightarrow{P_0P}$ is parallel to $\vec{v}$, i.e. $\overrightarrow{P_0P} = t\vec{v}$ for a unique t. Reading this componentwise gives the parametric equations; solving each for t (when the corresponding component of $\vec{v}$ is nonzero) and equating gives the symmetric equations. ■

Worked example

The line through $P_1(1,2,3)$ and $P_2(4,5,6)$ has direction $\overrightarrow{P_1P_2} = \langle 3,3,3 \rangle$, so

$$\vec{r}(t) = \langle 1 + 3t,\ 2 + 3t,\ 3 + 3t \rangle, \qquad \text{symmetric form } x - 1 = y - 2 = z - 3.$$

Relative position of two lines in $\mathbb{R}^3$

There are four possibilities: *identical*, *parallel*, *intersecting*, *skew* (non-parallel and non-intersecting).

- If the direction vectors are parallel, the lines are parallel or identical; test whether a point of one lies on the other.
- If the direction vectors are not parallel, solve $\vec{r}_1(t) = \vec{r}_2(s)$; a solution means they intersect, no solution means they are skew.

Worked example: skew lines

$\alpha(t) = \langle 1+t, -2+3t, 4-t\rangle$ and $\beta(s) = \langle 2s, 3+s, -3+4s\rangle$ have directions $\langle 1,3,-1\rangle$ and $\langle 2,1,4\rangle$, which are not parallel. Equating the first two coordinates gives $t = 11/5$, $s = 8/5$; substituting into the third coordinate gives $4-11/5 = 9/5$ but $-3+32/5 = 17/5$. No common point exists, so the lines are skew.

1.6.2 Planes

Point–normal equation

The plane through $P_0(x_0, y_0, z_0)$ with normal vector $\vec{n} = \langle a, b, c\rangle \neq \vec{0}$ is

$$\boxed{a(x-x_0) + b(y-y_0) + c(z-z_0) = 0}$$

equivalently $ax + by + cz + d = 0$ with $d = -(ax_0 + by_0 + cz_0)$.

Proof. A point $P(x,y,z)$ lies on the plane iff $\overrightarrow{P_0P}$ is perpendicular to $\vec{n}$, i.e. $\vec{n} \cdot \overrightarrow{P_0P} = 0$. Expanding,

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0.$$

Conversely, any equation $ax+by+cz+d = 0$ with $(a,b,c) \neq \vec{0}$ describes a plane: pick any solution P_0 (solve for one variable in terms of the other two), then $d = -(ax_0+by_0+cz_0)$ and the equation becomes $a(x-x_0)+b(y-y_0)+c(z-z_0) = 0$, i.e. the plane through P_0 with normal $\langle a,b,c\rangle$. ∎

Worked examples

(a) Through $P(1,2,3)$ with normal $\langle 4,5,6\rangle$:

$$4(x-1) + 5(y-2) + 6(z-3) = 0 \iff 4x + 5y + 6z - 32 = 0.$$

(b) Through $P_0(1,2,3)$ spanned by $\langle 1,3,4\rangle$ and $\langle 1,4,3\rangle$: take $\vec{n} = \langle 1,3,4\rangle \times \langle 1,4,3\rangle = \langle -7,1,1\rangle$, so

$$-7(x-1) + (y-2) + (z-3) = 0 \iff -7x + y + z + 2 = 0.$$

(c) Through three points P, Q, R: take $\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$.

Relative position of two planes

Two planes are *identical*, *parallel* (and distinct), or *intersect in a line*. Compare normal vectors $\vec{n}_1, \vec{n}_2$: if they are parallel the planes are parallel or identical (test one point); otherwise they intersect in a line. The **angle** between two planes is the acute angle between their normals,

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}.$$

Worked example

For $x+y+z = 1$ and $x-2y+3z = 1$, $\vec{n}_1 = \langle 1,1,1\rangle$, $\vec{n}_2 = \langle 1,-2,3\rangle$,

$$\cos\theta = \frac{|1-2+3|}{\sqrt{3}\sqrt{14}} = \frac{2}{\sqrt{42}}, \qquad \theta \approx 72^\circ.$$

1.6.3 Distance from a point to a plane

Distance formula

The distance from $P_0(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$ is

$$\boxed{D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}}$$

Proof. Let Q_0 be the foot of the perpendicular from P_0 to the plane, so that the required distance is $|P_0Q_0|$. Choose any point $P_1(x_1, y_1, z_1)$ *on* the plane, so $ax_1 + by_1 + cz_1 + d = 0$. The vector $\overrightarrow{P_1P_0}$ projects onto the normal direction with signed length

$$|P_0Q_0| = \left|\operatorname{comp}_{\vec{n}} \overrightarrow{P_1P_0}\right| = \left|\overrightarrow{P_1P_0} \cdot \frac{\vec{n}}{|\vec{n}|}\right|.$$

Now

$$\begin{aligned}\overrightarrow{P_1P_0} \cdot \vec{n} &= a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1) \\ &= (ax_0 + by_0 + cz_0) - (ax_1 + by_1 + cz_1) = ax_0 + by_0 + cz_0 + d,\end{aligned}$$

where the last step uses $ax_1 + by_1 + cz_1 = -d$. Dividing by $|\vec{n}| = \sqrt{a^2 + b^2 + c^2}$ gives the formula; it does not depend on the auxiliary point P_1. ∎

Worked example: distance between parallel planes

For $x - y + 2z = 4$ and $2x - 2y + 4z = 20$ (i.e. $x - y + 2z = 10$), pick $P(4, 0, 0)$ on the first plane. Then

$$D = \frac{|4 - 10|}{\sqrt{1 + 1 + 4}} = \frac{6}{\sqrt{6}} = \sqrt{6}.$$

1.7 Parametric curves and their derivatives

Parametric curves

A **parametric curve** in $\mathbb{R}^3$ is a map $\vec{r} : \mathbb{R} \to \mathbb{R}^3$,

$$\vec{r}(t) = \langle f(t), g(t), h(t)\rangle = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k},$$

written also as the system $x = f(t)$, $y = g(t)$, $z = h(t)$. The variable t is the **parameter**; the curve itself is the *image* of $\vec{r}$. The domain may be any interval, and the target may be $\mathbb{R}^2$.

Examples

(a) Circle of radius r centred at (x_0, y_0): $\vec{r}(t) = \langle x_0 + r\cos t,\, y_0 + r\sin t\rangle$. (b) $\vec{r}(t) = \langle r\cos at, r\sin at\rangle$: circle of radius $|r|$, traversed counterclockwise if $a > 0$, clockwise if $a < 0$, with period $2\pi/|a|$. (c) The graph of $f : \mathbb{R} \to \mathbb{R}$ is the parametric curve $\vec{r}(t) = \langle t, f(t)\rangle$; thus graphs are special parametric curves. (d) The line of the previous section is the parametric curve $\vec{r}(t) = \vec{r}_0 + t\vec{v}$.

The cycloid

A circle of radius r rolls without slipping along the x-axis. Let P start at the origin and let θ be the angle through which the circle has rolled. The centre is at $C(r\theta, r)$, and

$$\overrightarrow{CP} = \langle -r\sin\theta,\, -r\cos\theta\rangle,$$

hence

$$\vec{r}(\theta) = \langle r\theta - r\sin\theta,\; r - r\cos\theta\rangle = \langle r(\theta - \sin\theta),\; r(1 - \cos\theta)\rangle.$$

Derivative and tangent vector

If $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$ with x, y, z differentiable, then

$$\vec{r}\,'(t) = \langle x'(t), y'(t), z'(t)\rangle.$$

$\vec{r}\,'(t_0) \neq \vec{0}$ is a **tangent vector** to the curve at $\vec{r}(t_0)$, pointing in the direction of increasing t; the **tangent line** is

$$\vec{R}(s) = \vec{r}(t_0) + s\,\vec{r}\,'(t_0).$$

For a plane curve with $x'(t_0) \neq 0$, the slope of the tangent is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y'(t_0)}{x'(t_0)}.$$

Justification. Differentiate the components; the difference quotient $\dfrac{1}{h}(\vec{r}(t_0+h) - \vec{r}(t_0))$ is a secant vector, and its limit is the tangent vector. By the chain rule, $y = y(t(x))$ gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$. ■

Worked example: the cycloid

For $\vec{r}(\theta) = \langle r(\theta - \sin\theta), r(1-\cos\theta)\rangle$,

$$\vec{r}\,'(\theta) = \langle r(1-\cos\theta),\ r\sin\theta\rangle.$$

The tangent is horizontal when $y' = 0$ and $x' \neq 0$: $\sin\theta = 0$ and $\cos\theta \neq 1$, i.e. $\theta = (2k+1)\pi$. At $\theta = 2k\pi$ both components vanish; the slope is found by a limit,

$$\lim_{\theta\to 2k\pi} \frac{\sin\theta}{1-\cos\theta} = \lim_{\theta\to 2k\pi} \frac{\cos\theta}{\sin\theta} = +\infty,$$

so the tangent is vertical there (a cusp).

Arc length

If $\vec{r}(t)$, $t \in [a,b]$, is continuously differentiable and is traversed exactly once, the **arc length** is

$$\boxed{s = \int_a^b |\vec{r}\,'(t)|\,\mathrm{d}t = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,\mathrm{d}t}$$

The **arc length function** is $s(t) = \displaystyle\int_a^t |\vec{r}\,'(u)|\,\mathrm{d}u$, and $\dfrac{\mathrm{d}s}{\mathrm{d}t} = |\vec{r}\,'(t)|$ is the **speed**.

Justification. Divide $[a,b]$ by $a = t_0 < t_1 < \cdots < t_n = b$ and replace the arc over $[t_{i-1}, t_i]$ by the chord

$$|\vec{r}(t_i) - \vec{r}(t_{i-1})| \approx |\vec{r}\,'(t_i^*)|\,\Delta t,$$

the approximation coming from the linear approximation $\vec{r}(t_{i-1}+\Delta t) \approx \vec{r}(t_{i-1}) + \vec{r}\,'(t_{i-1})\Delta t$. Summing gives a Riemann sum for $\displaystyle\int_a^b |\vec{r}\,'(t)|\,\mathrm{d}t$, and passing to the limit gives the formula. ■

Worked example

For $\vec{r}(t) = \langle \cos t, \sin t, t\rangle$, $0 \le t \le 2\pi$: $|\vec{r}\,'(t)| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$, so $s = 2\sqrt{2}\,\pi$.

1.8 Parameterization table

Parametrizations of standard curves and surfaces

Plane curves.

Shape	Parametrization	Domain
Ellipse	$\langle a\cos t,\ b\sin t\rangle$	$t \in [0, 2\pi]$
Line	$\langle x_0 + t(x_1 - x_0),\ y_0 + t(y_1 - y_0)\rangle$	$t \in \mathbb{R}$
Spiral	$\langle t\cos t,\ t\sin t\rangle$	$t \geq 0$
Lemniscate	$\left\langle \dfrac{\cos t}{1+\sin^2 t},\ \dfrac{\cos t \sin t}{1+\sin^2 t} \right\rangle$	$t \in [0, 2\pi]$
Cycloid	$\langle r(t - \sin t),\ r(1 - \cos t)\rangle$	$t \in \mathbb{R}$

Surfaces in $\mathbb{R}^3$.

Surface	Parametrization	Domain
Ellipsoid	$\langle a\sin\theta\cos\phi,\ b\sin\theta\sin\phi,\ c\cos\theta\rangle$	$\theta \in [0, \pi],\ \phi \in [0, 2\pi]$
Elliptic paraboloid	$\left\langle x,\ y,\ \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} \right\rangle$	$x, y \in \mathbb{R}$
Hyperbolic paraboloid	$\left\langle x,\ y,\ \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} \right\rangle$	$x, y \in \mathbb{R}$
Elliptic cone	$\langle r\cos\theta,\ r\sin\theta,\ kr\rangle$	$r \in \mathbb{R},\ \theta \in [0, 2\pi]$
Hyperboloid of one sheet	$\langle a\cosh r\cos\theta,\ a\cosh r\sin\theta,\ b\sinh r\rangle$	$r \in \mathbb{R},\ \theta \in [0, 2\pi]$
Hyperboloid of two sheets	$\langle a\sinh r\cos\theta,\ a\sinh r\sin\theta,\ \pm b\cosh r\rangle$	$r \in \mathbb{R},\ \theta \in [0, 2\pi]$

2 Differential Calculus for Multivariable Functions

This chapter is dual to the theory of parametric curves: instead of mapping a one-dimensional space into $\mathbb{R}^2$ or $\mathbb{R}^3$, we map a two- or three-dimensional space into $\mathbb{R}$. We study limits, continuity, partial derivatives, tangent planes, the chain rule, directional derivatives, extrema and constrained extrema.

2.1 Functions of several variables

Definition

A **function of two variables** is a map $f : D \to \mathbb{R}$ where $D \subseteq \mathbb{R}^2$. When D is not specified it is understood to be the largest set on which the formula makes sense. The **domain** is D and the **range** (image) is $\{f(x, y) : (x, y) \in D\}$.
Similarly, a function of three variables is a map $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^3$, and a function of n variables is a map $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^n$.

Worked examples: domain and range

(a) $f(x, y) = 1 - 3x - 2y$: domain $\mathbb{R}^2$, range $\mathbb{R}$.
(b) $f(x, y) = \sqrt{9 - x^2 - y^2}$: the formula is defined iff $x^2 + y^2 \leq 9$, so D is the closed disk of radius 3 centred at the origin; since $0 \leq \sqrt{9 - x^2 - y^2} \leq 3$, the range is $[0, 3]$.
(c) $f(x, y) = \ln(x^2 + y^2)$: defined on $\mathbb{R}^2 \setminus \{(0, 0)\}$.

2.1.1 The graph

Graph

The **graph** of $f : D \to \mathbb{R}$ is

$$\{(x, y, z) \in \mathbb{R}^3 : (x, y) \in D,\ z = f(x, y)\}.$$

It satisfies the vertical line test for lines perpendicular to the xy-plane. The graph of a function of three variables lives in $\mathbb{R}^4$ and cannot be drawn.

Examples

(a) $z = 1-3x-2y \iff 3x+2y+z = 1$: a plane through $(1/3, 0, 0)$, $(0, 1/2, 0)$, $(0, 0, 1)$.
(b) $z = \sqrt{9 - x^2 - y^2} \iff x^2 + y^2 + z^2 = 9$ with $z \geq 0$: the northern hemisphere of the sphere of radius 3.

2.1.2 Level curves and level surfaces

Level sets

Let $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^2$ and $k \in \mathbb{R}$. The **level curve** of f at level k is

$$\{(x, y) \in D : f(x, y) = k\} = f^{-1}(k).$$

Geometrically it is the projection onto the xy-plane of the intersection of the graph with the horizontal plane $z = k$. A **contour map** is a drawing of several level curves.
For $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^3$, the **level surface** at level k is $f^{-1}(k) \subseteq \mathbb{R}^3$.

Examples

(a) $f(x, y) = 9 - x^2 - y^2$: the level curve at k is $x^2 + y^2 = 9 - k$, a circle of radius $\sqrt{9-k}$ when $k < 9$, a point when $k = 9$, empty when $k > 9$.
(b) $f(x, y) = c$ (constant): the level curve at $k \neq c$ is empty, and at $k = c$ is all of $\mathbb{R}^2$.
(c) $f(x, y, z) = x^2 + y^2 + z^2$: the level surface at $k > 0$ is the sphere of radius $\sqrt{k}$; at $k = 0$ it is the single point O.

2.2 Limits and continuity

2.2.1 The definition

Limit

Let $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^n$ and let P_0 be a limit point of D. We say

$$\lim_{P \to P_0} f(P) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$|f(P) - L| < \epsilon \qquad \text{whenever } P \in D \text{ and } 0 < |PP_0| < \delta.$$

In coordinates for $D \subseteq \mathbb{R}^2$, with $P_0 = (x_0, y_0)$: for every $\epsilon > 0$ there is $\delta > 0$ such that

$$|f(x, y) - L| < \epsilon \qquad \text{whenever } 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

Two-path test (non-existence)

If $f(P) \to L_1$ as $P \to P_0$ along a curve C_1 and $f(P) \to L_2$ along another curve C_2, with $L_1 \neq L_2$, then $\lim_{P \to P_0} f(P)$ does not exist. To *prove* that a limit exists one must use the definition (or the formal properties below).

Worked examples

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2-y^2}{x^2+y^2}$ does not exist: along $y=0$ the value is 1, along $x=0$ it is -1.

(b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2+y^4}$ does not exist. Along every line $y=kx$,

$$\frac{k^2x^3}{x^2+k^4x^4} = \frac{k^2x}{1+k^4x^2} \longrightarrow 0,$$

but along the parabola $x=y^2$,

$$\frac{y^4}{y^4+y^4} = \frac{1}{2}.$$

This shows that checking all lines is *not* sufficient.

2.2.2 Properties of limits

Limit laws

If $\lim\limits_{P\to P_0} f(P) = L$ and $\lim\limits_{P\to P_0} g(P) = M$, then

$$\lim(f \pm g) = L \pm M, \qquad \lim(fg) = LM, \qquad \lim \frac{f}{g} = \frac{L}{M} \ \ (M \neq 0).$$

Proof of the sum rule. Let $\epsilon > 0$. Choose $\delta_1, \delta_2 > 0$ such that

$$|f(P) - L| < \frac{\epsilon}{2} \ \ (0 < |PP_0| < \delta_1), \qquad |g(P) - M| < \frac{\epsilon}{2} \ \ (0 < |PP_0| < \delta_2).$$

Let $\delta = \min\{\delta_1, \delta_2\}$. For $0 < |PP_0| < \delta$, both inequalities hold, hence by the triangle inequality

$$|(f+g)(P) - (L+M)| \leq |f(P) - L| + |g(P) - M| < \epsilon.$$

The difference rule is identical; the product and quotient rules are proved by the standard ϵ-δ arguments. ∎

Bootstrapping

$\lim\limits_{(x,y,z)\to(a,b,c)} x = a$, $\lim y = b$, $\lim z = c$, so by the limit laws, for every polynomial P,

$$\lim_{(x,y,z)\to(a,b,c)} P(x,y,z) = P(a,b,c).$$

If P, Q are polynomials with $Q(a,b,c) \neq 0$, then

$$\lim_{(x,y,z)\to(a,b,c)} \frac{P(x,y,z)}{Q(x,y,z)} = \frac{P(a,b,c)}{Q(a,b,c)}.$$

Squeeze theorem

If $f(P) \leq g(P) \leq h(P)$ for all P in a ball around P_0, and

$$\lim_{P\to P_0} f(P) = \lim_{P\to P_0} h(P) = L,$$

then $\lim\limits_{P\to P_0} g(P) = L$.

Proof. Given $\epsilon > 0$, choose δ so that $|f(P)-L| < \epsilon$ and $|h(P)-L| < \epsilon$ for $0 < |PP_0| < \delta$. Then $L - \epsilon < f(P) \leq g(P) \leq h(P) < L + \epsilon$, hence $|g(P) - L| < \epsilon$. ∎

Worked examples

(a) $\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2) = 4 - 4 = 0.$

(b) $\lim_{(x,y)\to(0,0)} \dfrac{x^2y}{x^2+y^2}$. We claim the limit is 0. Indeed

$$0 \le \left|\frac{x^2y}{x^2+y^2}\right| \le \frac{x^2|y|}{x^2} = |y| \le \sqrt{x^2+y^2} \longrightarrow 0,$$

so the squeeze theorem applies. Note that $|y| \le \sqrt{x^2+y^2}$ because $y^2 \le x^2+y^2$.

2.2.3 Continuity

Continuity

f is **continuous at** P_0 if $\lim_{P\to P_0} f(P) = f(P_0)$. f is continuous on D if it is continuous at every point of D.

Properties

Sums, differences, products and quotients of functions continuous at P_0 are continuous at P_0 (in the quotient case provided the denominator does not vanish at P_0). Compositions of continuous functions are continuous. Consequently all polynomials are continuous everywhere and all rational functions are continuous on their domains.

Examples

(a) $f(x,y) = \dfrac{x^2-y^2}{x^2+y^2}$ is continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$.
(b) $f(x,y) = e^{x^2-y^2}$ is continuous on $\mathbb{R}^2$ as a composition of a polynomial and the exponential.

2.3 Extra: analysis in brief

This subsection puts the topology behind the extreme value theorem on a firm footing. It is not needed for computation but explains *why* continuous functions on closed bounded sets behave so well.

2.3.1 Basic $\mathbb{R}^n$ topology

Open ball, interior point, open set

Let $D \subseteq \mathbb{R}^n$.
- The **open ball** of radius $r > 0$ centred at $\vec{x}_0$ is $B(\vec{x}_0, r) = \{\vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{x}_0\| < r\}$.
- A point $\vec{x}_0 \in D$ is an **interior point** of D if some open ball around it lies entirely in D: $\exists r > 0$ with $B(\vec{x}_0, r) \subseteq D$.
- D is **open** if all its points are interior points.

Example

For $S = \{(x,y) \in \mathbb{R}^2 : 0 \le x, y \le 1\}$ the boundary points are not interior. If $0 < x, y < 1$, then $r = \min\{x, y, 1-x, 1-y\} > 0$ gives $B((x,y), r) \subseteq S$.

Properties of open sets

Let $D_i \subseteq \mathbb{R}^n$ be open. Then (i) every open ball $B(\vec{x}_0, r)$ is open; (ii) the union $\bigcup_i D_i$ is open; (iii) the intersection $\bigcap_{i=1}^{k} D_i$ of *finitely many* open sets is open.

Proof. (i) If $\vec{x}_1 \in B(\vec{x}_0, r)$ put $r_1 = r - \|\vec{x}_1 - \vec{x}_0\| > 0$; then $B(\vec{x}_1, r_1) \subseteq B(\vec{x}_0, r)$.

(ii) If $\vec{x}_0 \in \bigcup_i D_i$ then $\vec{x}_0 \in D_{i_0}$ for some i_0; by openness $B(\vec{x}_0, r) \subseteq D_{i_0} \subseteq \bigcup_i D_i$.

(iii) If $\vec{x}_0 \in \bigcap_{i=1}^{k} D_i$, pick $r_i > 0$ with $B(\vec{x}_0, r_i) \subseteq D_i$ and put $r = \min\{r_1, \dots, r_k\} > 0$; then $B(\vec{x}_0, r) \subseteq \bigcap_{i=1}^{k} D_i$. ■

Boundary, closed set, closure

Let $D \subseteq \mathbb{R}^n$.

- $\vec{x}_0$ is a **boundary point** of D if every ball around it meets both D and its complement: for all $r > 0$, $B(\vec{x}_0, r) \cap D \neq \varnothing$ and $B(\vec{x}_0, r) \cap (\mathbb{R}^n \setminus D) \neq \varnothing$. The set of all such points is the **boundary** ∂D.
- D is **closed** if it contains all its boundary points, i.e. $\partial D \subseteq D$.
- The **closure** of D is $\overline{D} = D \cup \partial D$.

Sequential characterization

D is open $\iff$ whenever $\vec{x}_i \to \vec{x} \in D$, all but finitely many $\vec{x}_i$ lie in D. D is closed $\iff$ whenever $\vec{x}_i \in D$ and $\vec{x}_i \to \vec{x}$, then $\vec{x} \in D$.

Proof (sketch)

For openness: if D were not open there would be $\vec{x}_0 \in D \cap \partial D$, and choosing $\vec{x}_i \in B(\vec{x}_0, 1/i) \cap (\mathbb{R}^n \setminus D)$ gives a sequence in the complement converging to a point of D, contradicting the condition; the converse is immediate from the definition of interior point. For closedness: if $\vec{x}_i \in D$, $\vec{x}_i \to \vec{x} \notin D$ and D is closed, then some ball around $\vec{x}$ misses D, contradicting convergence; conversely if D is not closed pick $\vec{x}_0 \in \overline{D} \setminus D$ and a sequence $\vec{x}_i \in B(\vec{x}_0, 1/i) \cap D$.

Properties of closed sets

Let $D_i \subseteq \mathbb{R}^n$ be closed. Then (i) the closure $\overline{D}$ of any set is closed; (ii) the intersection $\bigcap_i D_i$ of any collection of closed sets is closed; (iii) the union of *finitely many* closed sets is closed.

Proof. (i) $\overline{D} = D \cup \partial D$ contains all its boundary points. (ii) If $\vec{x}_i \in \bigcap_i D_i$ and $\vec{x}_i \to \vec{x}$, then $\vec{x} \in D_i$ for every i, so $\vec{x} \in \bigcap_i D_i$. (iii) If $\vec{x}_i \in \bigcup_{j=1}^{k} D_j$ and $\vec{x}_i \to \vec{x}$, one of the D_j (say D_{j_0}) contains infinitely many terms; a subsequence lies in D_{j_0} and converges to $\vec{x}$, so $\vec{x} \in D_{j_0} \subseteq \bigcup_{j=1}^{k} D_j$. ■

Bounded and compact sets

$D \subseteq \mathbb{R}^n$ is **bounded** if there is $M > 0$ with $\|\vec{x}\| < M$ for all $\vec{x} \in D$ (equivalently $D \subseteq B(\vec{0}, k)$ for some $k \in \mathbb{N}$). D is **compact** if it is closed *and* bounded.

Axiom of completeness

Every non-empty subset $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound (supremum) in $\mathbb{R}$.

Remark. Closed and bounded are independent notions: a set can be closed but unbounded, bounded but not closed, both, or neither.

2.3.2 Bolzano–Weierstrass theorem

Bolzano–Weierstrass theorem

Every bounded sequence in $\mathbb{R}$ has a convergent subsequence. The same holds in $\mathbb{R}^n$.

Proof (bisection, $n = 1$). Let (x_i) satisfy $a_1 \le x_i \le b_1$. Divide $I_1 = [a_1, b_1]$ into two equal halves; at least one contains infinitely many terms, call it I_2, with $\ell(I_2) = \frac{1}{2}\ell(I_1)$. Inductively obtain $I_1 \supset I_2 \supset \cdots$ with $\ell(I_k) \to 0$. By completeness there is $x \in \bigcap_k I_k$. Choose $n_1 < n_2 < \cdots$ with $x_{n_k} \in I_k$; then $|x_{n_k} - x| \le \ell(I_k) \to 0$, so $x_{n_k} \to x$. ∎

Two proofs in $\mathbb{R}^n$

(Diagonal subsequence.) Write $\vec{x}_i = \langle x_i^{(1)}, \ldots, x_i^{(n)} \rangle$. Each coordinate sequence is bounded, so successively extract subsequences making the first, then the second, …, then the n-th coordinate converge; the final subsequence converges coordinatewise. *(Nested cubes.)* Enclose $(\vec{x}_i)$ in $[-R, R]^n = Q_1$, repeatedly subdivide into 2^n equal closed subcubes keeping one with infinitely many terms, and use completeness to find $\vec{x} \in \bigcap_k Q_k$; then $\|\vec{x}_{n_k} - \vec{x}\| \to 0$.

Examples

(a) $x_n = (-1)^n$ is bounded but not convergent; its even and odd subsequences converge to 1 and -1.
(b) $x_n = \sin n$ is bounded; since $\{\sin n\}$ is dense in $[-1, 1]$, every $L \in [-1, 1]$ is a subsequential limit.
(c) $x_n = \langle \cos n, \sin n \rangle$ lies on the unit circle, which is compact, so some subsequence converges to a point of S^1.

2.3.3 Topology and the extreme value theorem

Topological continuity

If $\vec{F} : D \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ is continuous and surjective and D is compact, then V is compact.

Proof. *V bounded:* if not, pick $\vec{y}_k \in V$ with $\|\vec{y}_k\| > k$ and $\vec{x}_k \in D$ with $\vec{F}(\vec{x}_k) = \vec{y}_k$. Since $(\vec{x}_k)$ is bounded, BWT gives $\vec{x}_{k_j} \to \vec{x} \in D$; continuity gives $\vec{F}(\vec{x}_{k_j}) \to \vec{F}(\vec{x})$, but $\|\vec{F}(\vec{x}_{k_j})\| = \|\vec{y}_{k_j}\| \to \infty$, a contradiction.

V closed: if $\vec{y}_i \in V$, $\vec{y}_i \to \vec{y}$, choose $\vec{x}_i \in D$ with $\vec{F}(\vec{x}_i) = \vec{y}_i$; a convergent subsequence $\vec{x}_{i_j} \to \vec{x} \in D$ gives $\vec{F}(\vec{x}_{i_j}) \to \vec{F}(\vec{x})$, while $\vec{F}(\vec{x}_{i_j}) = \vec{y}_{i_j} \to \vec{y}$, so $\vec{y} = \vec{F}(\vec{x}) \in V$. ∎

Extreme value theorem

If $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is continuous and D is compact, then there exist $\vec{x}_m, \vec{x}_M \in D$ with

$$f(\vec{x}_m) \le f(\vec{x}) \le f(\vec{x}_M) \qquad \text{for all } \vec{x} \in D.$$

Proof. By the previous theorem $f(D) \subseteq \mathbb{R}$ is compact, hence closed and bounded, so $m = \inf f(D)$ and $M = \sup f(D)$ exist and belong to $f(D)$ (closedness). Thus $m = f(\vec{x}_m)$ and $M = f(\vec{x}_M)$ for some $\vec{x}_m, \vec{x}_M \in D$. ■

2.3.4 Uniform continuity

Uniform continuity

$\vec{F} : D \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is **uniformly continuous** on D if

$$\forall \epsilon > 0 \ \exists \delta > 0 : \quad \vec{x}, \vec{y} \in D,\ \|\vec{x} - \vec{y}\| < \delta \implies \|\vec{F}(\vec{x}) - \vec{F}(\vec{y})\| < \epsilon.$$

Remark. The difference from ordinary continuity is that δ depends only on ϵ, not on the point $\vec{x}$.

Continuous on compact ⇒ uniformly continuous

Proof. For each $\epsilon > 0$ and each $\vec{x} \in D$ let $\delta_{\vec{x}} > 0$ be such that $\|\vec{x} - \vec{y}\| < \delta_{\vec{x}} \Rightarrow \|\vec{F}(\vec{x}) - \vec{F}(\vec{y})\| < \epsilon$. Since D is compact, $\delta = \inf_{\vec{x} \in D} \delta_{\vec{x}}$ exists and is positive, so $\|\vec{x} - \vec{y}\| < \delta$ gives the conclusion for all $\vec{x}, \vec{y} \in D$. ■

Matrix norm

For $A \in \mathbb{R}^{m \times n}$ define $\|A\| = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2}$. Then $\|A\vec{x}\| \le \|A\|\,\|\vec{x}\|$ for every $\vec{x} \in \mathbb{R}^n$.

Proof. Writing $A = [a_{ij}]$ and applying the Cauchy–Schwarz inequality to each row,

$$\|A\vec{x}\|^2 = \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} a_{ij} x_j \Big)^2 \le \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} a_{ij}^2 \Big) \Big(\sum_{j=1}^{n} x_j^2 \Big) = \|A\|^2 \|\vec{x}\|^2.$$

■

Every linear map is uniformly continuous

Proof. Let $\vec{F}(\vec{x}) = A\vec{x}$. Then $\|\vec{F}(\vec{x}) - \vec{F}(\vec{y})\| = \|A(\vec{x} - \vec{y})\| \le \|A\|\,\|\vec{x} - \vec{y}\|$. Given $\epsilon > 0$ take $\delta = \epsilon / \|A\|$ (if $A = \vec{0}$ any δ works). ■

2.4 Partial derivatives

2.4.1 Definition

Partial derivatives

Let f be defined near (a, b). The **partial derivative with respect to** x at (a, b) is

$$\boxed{f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}}$$

and the **partial derivative with respect to** y is

$$f_y(a,b) = \lim_{h\to 0} \frac{f(a,b+h) - f(a,b)}{h}.$$

In practice: to compute f_x, treat y as a constant and differentiate in x; to compute f_y, treat x as a constant.

Notation: $f_x = \dfrac{\partial f}{\partial x} = \partial_x f$, and similarly f_y, f_z.

Geometric interpretation

Let S be the graph $z = f(x,y)$ and $P(a,b,c)$ a point on it. The plane $y = b$ cuts S in the curve C_1 (the trace in the x-direction); the plane $x = a$ cuts S in the curve C_2. Then

$$f_x(a,b) = \text{slope of the tangent to } C_1 \text{ at } P,$$
$$f_y(a,b) = \text{slope of the tangent to } C_2 \text{ at } P.$$

Worked example

For $f(x,y) = x^3 + x^2y^3 - 2y^2$,

$$f_x = 3x^2 + 2xy^3, \qquad f_y = 3x^2y^2 - 4y,$$

so $f_x(2,1) = 12 + 4 = 16$ and $f_y(2,1) = 12 - 4 = 8$.

2.4.2 Implicit differentiation

Implicit functions

If an equation $F(x,y) = 0$ defines y implicitly as a differentiable function of x, then differentiating the identity $F(x, y(x)) = 0$ in x and using the chain rule gives

$$F_x + F_y \frac{\mathrm{d}y}{\mathrm{d}x} = 0 \qquad \Longrightarrow \qquad \boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y}}$$

provided $F_y \neq 0$. Similarly, if $F(x,y,z) = 0$ defines $z = z(x,y)$, then

$$\boxed{\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \qquad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}}$$

Worked examples

(a) $x^2 + y^2 = 1$. Differentiating in x with $y = y(x)$:

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \Longrightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}.$$

At $(0,1)$ the slope is 0; at $(1,0)$ the formula requires division by zero, and indeed near $(1,0)$, y is not a function of x.

(b) $x^3 + y^3 + z^3 + 6xyz + 4 = 0$. Differentiating in x (with y constant, $z = z(x,y)$):

$$3x^2 + 3z^2 z_x + 6yz + 6xyz_x = 0 \Longrightarrow z_x = -\frac{x^2 + 2yz}{z^2 + 2xy}.$$

Similarly

$$3y^2 + 3z^2 z_y + 6xz + 6xyz_y = 0 \Longrightarrow z_y = -\frac{y^2 + 2xz}{z^2 + 2xy}.$$

At $(-1,1,2)$ (which satisfies the equation): $z_x = -\dfrac{1+4}{4-2} = -\dfrac{5}{2}$ and $z_y = -\dfrac{1-4}{4-2} = \dfrac{3}{2}$.

2.4.3 Higher derivatives

Second-order partial derivatives

$$f_{xx} = (f_x)_x = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = (f_y)_y = \frac{\partial^2 f}{\partial y^2},$$
$$f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y\,\partial x}, \qquad f_{yx} = (f_y)_x = \frac{\partial^2 f}{\partial x\,\partial y}.$$

The mixed derivatives f_{xy} and f_{yx} are generally equal under a mild hypothesis.

Clairaut's theorem

Let f be defined on an open set $D \subseteq \mathbb{R}^2$. If f_{xy} and f_{yx} exist and are continuous on D, then

$$f_{xy}(x,y) = f_{yx}(x,y) \qquad \text{for all } (x,y) \in D.$$

The same statement holds for functions of three or more variables and for higher derivatives.

Proof. Let $(a,b) \in D$ and h,k be small nonzero numbers so that the rectangle with corners $(a,b),(a+h,b),(a,b+k),(a+h,b+k)$ lies inside D. Consider the **second difference quotient**

$$\Delta(h,k) = \frac{f(a+h,b+k) - f(a+h,b) - f(a,b+k) + f(a,b)}{hk}.$$

We evaluate $\Delta(h,k)$ in two ways.

First way: group the terms as $\big[f(a+h,b+k) - f(a+h,b)\big] - \big[f(a,b+k) - f(a,b)\big]$. Set

$$g(x) = f(x,b+k) - f(x,b).$$

Then

$$\Delta(h,k) = \frac{g(a+h) - g(a)}{hk}.$$

Since f_{xy} exists, g is differentiable in x, so the mean value theorem gives a point ξ between a and $a+h$ with

$$\frac{g(a+h) - g(a)}{h} = g'(\xi) = f_x(\xi, b+k) - f_x(\xi, b).$$

Now apply the mean value theorem in y to $f_x(\xi, y)$: there is a point η between b and $b+k$ such that

$$f_x(\xi,b+k) - f_x(\xi,b) = f_{xy}(\xi,\eta)\,k.$$

Thus $\Delta(h,k) = f_{xy}(\xi,\eta)$.

Second way: group the terms as $\big[f(a+h,b+k) - f(a,b+k)\big] - \big[f(a+h,b) - f(a,b)\big]$. Set

$$p(y) = f(a+h,y) - f(a,y).$$

Then $\Delta(h,k) = \big[p(b+k) - p(b)\big]/(hk)$. Applying the mean value theorem in y (to p) and then in x (to $f_y(x,\eta')$) gives points ξ' between a and $a+h$ and η' between b and $b+k$ such that

$$\Delta(h,k) = f_{yx}(\xi',\eta').$$

Comparing the two expressions, $f_{xy}(\xi,\eta) = f_{yx}(\xi',\eta')$ for all small h,k. As $(h,k) \to (0,0)$, both (ξ,η) and (ξ',η') tend to (a,b). By continuity of f_{xy} and f_{yx} at

(a,b), the limits exist and are equal:

$$f_{xy}(a,b) = \lim_{(h,k)\to(0,0)} f_{xy}(\xi,\eta) = \lim_{(h,k)\to(0,0)} f_{yx}(\xi',\eta') = f_{yx}(a,b).$$

■

Worked example

For $f(x,y) = x^3 + 2xy$: $f_x = 3x^2 + 2y$, $f_y = 2x$, hence $f_{xy} = 2 = f_{yx}$.
For $f(x,y,z) = \sin(2x + yz)$:

$$f_{xyz} = \frac{\partial^2}{\partial y\,\partial z}\big(2\cos(2x+yz)\big) = \frac{\partial}{\partial y}\big(2y\cos(2x+yz)\big) = 2\cos(2x+yz) - 2yz\sin(2x+yz).$$

2.5 Tangent planes and linear approximation

Throughout this section the first-order partial derivatives are assumed continuous.

2.5.1 Tangent plane to a graph

Tangent plane of $z = f(x,y)$

The tangent plane to $z = f(x,y)$ at (x_0, y_0, z_0), where $z_0 = f(x_0,y_0)$, is

$$\boxed{z - z_0 = f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)}$$

Derivation. The plane must contain the two tangent lines to the traces. The trace in $y = y_0$ has tangent vector $\langle 1, 0, f_x\rangle$ and the trace in $x = x_0$ has tangent vector $\langle 0, 1, f_y\rangle$ (evaluated at (x_0,y_0)). Hence a normal vector is

$$\vec{n} = \langle 1, 0, f_x\rangle \times \langle 0, 1, f_y\rangle = \langle -f_x, -f_y, 1\rangle,$$

and the plane through (x_0,y_0,z_0) with this normal is $f_x(x-x_0) + f_y(y-y_0) - (z-z_0) = 0$, which is the stated equation. ■

Worked example

For $z = 2x^2 + y^2$ at $(1,1,3)$: $f_x = 4x$, $f_y = 2y$, so $f_x(1,1) = 4$, $f_y(1,1) = 2$ and

$$z - 3 = 4(x-1) + 2(y-1) \iff z = 4x + 2y - 3.$$

2.5.2 Tangent plane to a level surface

Gradient is normal

Let $f : D \to \mathbb{R}$ with $D \subseteq \mathbb{R}^3$ open and let S be the level surface $f(x,y,z) = k$. If $\nabla f(x_0,y_0,z_0) \neq \vec{0}$, then $\nabla f(x_0,y_0,z_0)$ is perpendicular to the tangent plane of S at P_0, and the tangent plane is

$$\boxed{f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0}$$

where $\nabla f = \langle f_x, f_y, f_z\rangle$.

Proof. Let $\vec{r}(t)$ be any differentiable curve lying in S with $\vec{r}(0) = P_0$. Then $f(\vec{r}(t)) = k$ for all t, so differentiating and using the chain rule,

$$0 = \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t))\Big|_{t=0} = \nabla f(P_0)\cdot \vec{r}\,'(0).$$

Thus $\nabla f(P_0)$ is perpendicular to the tangent vector of every curve in S through P_0, i.e. perpendicular to the tangent plane. ∎

Worked example: tangent plane to the unit sphere

For $x^2+y^2+z^2=1$ and $P_0=(x_0,y_0,z_0)$ on it,

$$2x_0(x-x_0)+2y_0(y-y_0)+2z_0(z-z_0)=0 \iff x_0x+y_0y+z_0z=x_0^2+y_0^2+z_0^2=1.$$

2.5.3 Linear approximation and differentiability

Linearization

The **linearization** of f at (x_0,y_0) is

$$L(x,y)=f(x_0,y_0)+f_x(x_0,y_0)(x-x_0)+f_y(x_0,y_0)(y-y_0),$$

and $f(x,y)\approx L(x,y)$ is the **linear (tangent plane) approximation**. In vector form, with $\Delta P=(x-x_0,y-y_0)$ and the **total derivative**

$$Df(x_0,y_0)=\big(f_x(x_0,y_0),\ f_y(x_0,y_0)\big),$$

we have $L(P)=f(P_0)+Df(P_0)\cdot\Delta P$.

Differentiability

Let $D\subseteq\mathbb{R}^2$ be open and $P_0\in D$. Then f is **differentiable at** P_0 if there is a vector $Df(P_0)$ such that

$$\lim_{P\to P_0}\frac{|f(P)-f(P_0)-Df(P_0)\cdot(P-P_0)|}{|P-P_0|}=0.$$

When this holds, the partial derivatives exist at P_0 and $Df(P_0)=(f_x(P_0),f_y(P_0))$.

Remark. The definition says precisely that the linear approximation is correct up to an error which is $o(|P-P_0|)$.

Proof that $Df=(f_x,f_y)$. Take $P=(x_0+h,y_0)$, so $P-P_0=(h,0)$ and $|P-P_0|=|h|$. The quotient becomes

$$\frac{\big|f(x_0+h,y_0)-f(x_0,y_0)-Df_1\,h\big|}{|h|}\longrightarrow 0,$$

which says exactly that $Df_1=\lim_{h\to0}\dfrac{f(x_0+h,y_0)-f(x_0,y_0)}{h}=f_x(x_0,y_0)$. The same argument with $P=(x_0,y_0+k)$ gives $Df_2=f_y(x_0,y_0)$. ∎

Differentiability implies continuity

If f is differentiable at P_0, then f is continuous at P_0.

Proof. Write

$$f(P)-f(P_0)=Df(P_0)\cdot(P-P_0)+\varepsilon(P)\,|P-P_0|,\qquad \varepsilon(P)\to0\ \ (P\to P_0).$$

By the Cauchy–Schwarz inequality,

$$|f(P)-f(P_0)|\le|Df(P_0)|\,|P-P_0|+|\varepsilon(P)|\,|P-P_0|\longrightarrow0,$$

so $f(P)\to f(P_0)$. ∎

A sufficient condition for differentiability

If f_x and f_y exist in a disk around P_0 and are continuous at P_0, then f is differentiable at P_0.

Proof. Let $P_0 = (a, b)$ and $\Delta x, \Delta y$ be small. Write

$$f(a+\Delta x, b+\Delta y) - f(a,b) = \underbrace{\big[f(a+\Delta x, b+\Delta y) - f(a+\Delta x, b)\big]}_{\text{(I)}} + \underbrace{\big[f(a+\Delta x, b) - f(a,b)\big]}_{\text{(II)}}.$$

Applying the one-variable mean value theorem to (II) in the x-variable and to (I) in the y-variable,

$$\text{(II)} = f_x(a+\theta_1\Delta x, b)\,\Delta x, \qquad \text{(I)} = f_y(a+\Delta x, b+\theta_2\Delta y)\,\Delta y,$$

with $\theta_1, \theta_2 \in (0,1)$. By continuity of the partials at (a,b),

$$f_x(a+\theta_1\Delta x, b) = f_x(a,b) + \varepsilon_1, \qquad f_y(a+\Delta x, b+\theta_2\Delta y) = f_y(a,b) + \varepsilon_2,$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$. Hence

$$f(a+\Delta x, b+\Delta y) - f(a,b) = f_x(a,b)\Delta x + f_y(a,b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y.$$

Since $|\varepsilon_1\Delta x + \varepsilon_2\Delta y| \le (|\varepsilon_1| + |\varepsilon_2|)\sqrt{\Delta x^2 + \Delta y^2}$, the error is $o(|P - P_0|)$, which is exactly differentiability. ∎

Worked example

Show that $f(x,y) = xe^{xy}$ is differentiable at $(1,0)$ and estimate $f(1.1, -0.1)$.

$f_x = e^{xy} + xye^{xy} \Rightarrow f_x(1,0) = 1$ and $f_y = x^2e^{xy} \Rightarrow f_y(1,0) = 1$. Both partials are continuous everywhere, so f is differentiable everywhere. The linearization is

$$L(x,y) = f(1,0) + 1\cdot(x-1) + 1\cdot y = 1 + (x-1) + y = x + y,$$

hence $f(1.1, -0.1) \approx 1.0$. The true value is $1.1e^{-0.11} = 0.98542$.

2.5.4 Algebraic properties of derivatives

Rules for computing derivatives

Let f, g be differentiable and $c \in \mathbb{R}$. Then, differentiating in one variable while the others are held fixed,

$$(cf)_x = cf_x, \qquad (f \pm g)_x = f_x \pm g_x,$$
$$(fg)_x = f_x g + f g_x, \qquad \Big(\frac{f}{g}\Big)_x = \frac{f_x g - f g_x}{g^2} \quad (g \neq 0).$$

The same rules hold for ∂_y and ∂_z. Compositions are differentiated by the chain rule (Section 2.5).

Proof of the product rule. By definition of the partial derivative,

$$(fg)_x(x_0,y_0) = \lim_{h\to 0} \frac{f(x_0+h,y_0)g(x_0+h,y_0) - f(x_0,y_0)g(x_0,y_0)}{h}.$$

Add and subtract $f(x_0+h,y_0)g(x_0,y_0)$ in the numerator:

$$\begin{aligned}(fg)_x &= \lim_{h\to 0}\Big[f(x_0+h,y_0)\,\frac{g(x_0+h,y_0)-g(x_0,y_0)}{h} + g(x_0,y_0)\,\frac{f(x_0+h,y_0)-f(x_0,y_0)}{h}\Big] \\ &= f(x_0,y_0)g_x(x_0,y_0) + g(x_0,y_0)f_x(x_0,y_0) = f_x g + f g_x,\end{aligned}$$

where in the limit we used that f is continuous (being differentiable) so $f(x_0+h,y_0) \to f(x_0,y_0)$.

The scalar multiple and sum rules are immediate from the definition:

$$(cf)_x = \lim_{h\to 0} \frac{cf(x_0+h,y_0)-cf(x_0,y_0)}{h} = c\lim_{h\to 0}\frac{f(x_0+h,y_0)-f(x_0,y_0)}{h} = cf_x,$$

and similarly $(f \pm g)_x = f_x \pm g_x$.

For the quotient rule, write $f = f/g \cdot g$ and apply the product rule:

$$f_x = \left(\frac{f}{g}\right)_x g + \frac{f}{g}\, g_x \quad\Longrightarrow\quad \left(\frac{f}{g}\right)_x = \frac{f_x - f g_x/g}{g} = \frac{f_x g - f g_x}{g^2},$$

which is valid whenever $g \neq 0$. ■

2.6 The chain rule

Case 1: one independent variable

If $z = f(x,y)$ is differentiable and $x = g(t)$, $y = h(t)$ are differentiable, then

$$\boxed{\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial z}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial z}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}}$$

Proof. By differentiability of f,

$$\Delta z = f_x\,\Delta x + f_y\,\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y, \qquad \varepsilon_1,\varepsilon_2 \to 0.$$

Divide by $\Delta t \neq 0$:

$$\frac{\Delta z}{\Delta t} = f_x\frac{\Delta x}{\Delta t} + f_y\frac{\Delta y}{\Delta t} + \varepsilon_1\frac{\Delta x}{\Delta t} + \varepsilon_2\frac{\Delta y}{\Delta t}.$$

As $\Delta t \to 0$, $\Delta x \to 0$ and $\Delta y \to 0$ (continuity of g, h), so $\varepsilon_1, \varepsilon_2 \to 0$, while $\Delta x/\Delta t \to \mathrm{d}x/\mathrm{d}t$ and $\Delta y/\Delta t \to \mathrm{d}y/\mathrm{d}t$. Passing to the limit gives the formula. ■

Worked example

$z = x^2y + 3xy^4$, $x = \sin 2t$, $y = \cos t$. Then

$$\frac{\mathrm{d}z}{\mathrm{d}t} = (2xy + 3y^4)\cdot 2\cos 2t + (x^2 + 12xy^3)\cdot(-\sin t).$$

At $t = 0$: $x = 0$, $y = 1$, so $\dfrac{\mathrm{d}z}{\mathrm{d}t} = (0+3)\cdot 2 + 0 = 6$.

Case 2: two independent variables

If $z = f(x,y)$ and $x = x(s,t)$, $y = y(s,t)$, then

$$\boxed{\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}, \qquad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}}$$

Worked example

$z = \sin 2x \cos 3y$, $x = s + t$, $y = s - t$:

$$z_s = 2\cos 2x\cos 3y - 3\sin 2x \sin 3y, \qquad z_t = 2\cos 2x\cos 3y + 3\sin 2x\sin 3y.$$

General version

If $u = u(x_1,\dots,x_n)$ and each $x_k = x_k(t_1,\dots,t_m)$, then

$$\boxed{\frac{\partial u}{\partial t_i} = \sum_{k=1}^{n} \frac{\partial u}{\partial x_k}\frac{\partial x_k}{\partial t_i}}$$

This is the matrix product of the row vector $(\partial u/\partial x_1, \dots, \partial u/\partial x_n)$ with the matrix $(\partial x_k/\partial t_i)$: "rates of change multiply".

Worked example

$u = x^4y + y^2z^3$ with $x = rse^t$, $y = rs^2e^{-t}$, $z = r^2 s \sin t$. At $(r, s, t) = (2, 1, 0)$ we have $(x, y, z) = (2, 2, 0)$. Then

$$u_s = (4x^3y)\frac{\partial x}{\partial s} + (x^4 + 2yz^3)\frac{\partial y}{\partial s} + (3y^2z^2)\frac{\partial z}{\partial s},$$

and $\partial_s x = re^t = 2$, $\partial_s y = 2rse^{-t} = 4$, $\partial_s z = r^2 \sin t = 0$, so

$$u_s = \big(4 \cdot 8 \cdot 2\big) \cdot 2 + (16 + 0) \cdot 4 + 0 = 128 + 64 = 192.$$

2.7 Directional derivatives and the gradient

Directional derivative

Let f be defined on an open set containing (x_0, y_0) and let $\vec{u} = \langle a, b\rangle$ be a *unit* vector. The **directional derivative** of f at (x_0, y_0) in the direction $\vec{u}$ is

$$\boxed{D_{\vec{u}} f(x_0, y_0) = \lim_{t \to 0} \frac{f(x_0 + at, y_0 + bt) - f(x_0, y_0)}{t}}$$

The normalisation $|\vec{u}| = 1$ is needed so that the answer depends only on the direction, not on the magnitude of $\vec{u}$.

Computation via the gradient

If f is differentiable at (x_0, y_0), then for every unit vector $\vec{u}$,

$$\boxed{D_{\vec{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \vec{u}}$$

where $\nabla f = \langle f_x, f_y\rangle$ is the **gradient**. In three variables, $\nabla f = \langle f_x, f_y, f_z\rangle$ and

$$D_{\vec{u}} f = f_x a + f_y b + f_z c.$$

Proof. Let $\vec{r}(t) = \langle x_0 + at, y_0 + bt\rangle$, so $\vec{r}(0) = P_0$ and $\vec{r}\,'(0) = \langle a, b\rangle = \vec{u}$. We claim

$$D_{\vec{u}} f(P_0) = \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t))\Big|_{t=0}.$$

Indeed, writing out the definition of both sides,

$$D_{\vec{u}} f(P_0) = \lim_{t \to 0} \frac{f(P_0 + t\vec{u}) - f(P_0)}{t},$$

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t))\Big|_{t=0} = \lim_{t \to 0} \frac{f(x_0 + at, y_0 + bt) - f(x_0, y_0)}{t},$$

which are identical. Now, since f is *differentiable* at P_0 (and in particular has continuous partial derivatives in an open neighbourhood), the chain rule applies to the composite function $t \mapsto f(\vec{r}(t))$:

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t))\Big|_{t=0} = f_x(P_0)\, x'(0) + f_y(P_0)\, y'(0) = f_x(P_0)a + f_y(P_0)b = \nabla f(P_0) \cdot \vec{u}.$$

■

Warning.

The formula $D_{\vec{u}}f = \nabla f \cdot \vec{u}$ requires f to be differentiable at P_0, not merely to have partial derivatives there. There exist functions whose partial derivatives exist at a point but which are not differentiable (and not even continuous), and for which the directional derivative along some directions exists but is not given by the dot product with the gradient. The textbook proof of the mean value theorem for functions of two variables also depends on differentiability.

Worked examples

(a) With $\vec{u} = \langle 1, 0\rangle$: $D_{\vec{u}}f = f_x$. With $\vec{u} = \langle 0, 1\rangle$: $D_{\vec{u}}f = f_y$. Thus partial derivatives are the directional derivatives along the coordinate axes.
(b) $f(x, y) = x^3 - 3xy + 4y^2$ and $\vec{u} = \langle\sqrt{3}/2, 1/2\rangle$ (angle $\pi/6$ with the horizontal). Then

$$D_{\vec{u}}f = (3x^2 - 3y)\frac{\sqrt{3}}{2} + (-3x + 8y)\frac{1}{2},$$

and at $(1, 2)$: $D_{\vec{u}}f = (3 - 6)\dfrac{\sqrt{3}}{2} + (-3 + 16)\dfrac{1}{2} = \dfrac{13 - 3\sqrt{3}}{2}$.

2.7.1 Bounds and steepest ascent

The gradient points uphill

For f differentiable and $|\vec{u}| = 1$,

$$\boxed{-|\nabla f(P_0)| \le D_{\vec{u}}f(P_0) \le |\nabla f(P_0)|}$$

The maximum $|\nabla f(P_0)|$ is attained when $\vec{u}$ has the direction of $\nabla f(P_0)$, and the minimum $-|\nabla f(P_0)|$ when $\vec{u}$ has the opposite direction. These two directions are the directions of **steepest ascent** and **steepest descent**.

Proof. For any two vectors, the Cauchy–Schwarz inequality gives

$$-|\vec{u}||\vec{v}| \le \vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}|\cos\theta \le |\vec{u}||\vec{v}|,$$

with equality iff $\theta = \pi$ (resp. $\theta = 0$), i.e. iff the vectors are anti-parallel (resp. parallel). Apply this with $\vec{u}$ a unit vector and $\vec{v} = \nabla f(P_0)$:

$$-|\nabla f(P_0)| \le \nabla f(P_0) \cdot \vec{u} \le |\nabla f(P_0)|.$$

Equality on the right (the **maximum** directional derivative) occurs exactly when $\vec{u}$ points in the direction of $\nabla f(P_0)$, namely $\vec{u} = \nabla f(P_0)/|\nabla f(P_0)|$. Since the directional derivative measures the *instantaneous rate of change* of f per unit distance in the direction $\vec{u}$, this is the direction of **steepest ascent**. Similarly, the minimum value $-|\nabla f(P_0)|$ occurs when $\vec{u} = -\nabla f(P_0)/|\nabla f(P_0)|$, the direction of **steepest descent**. ■

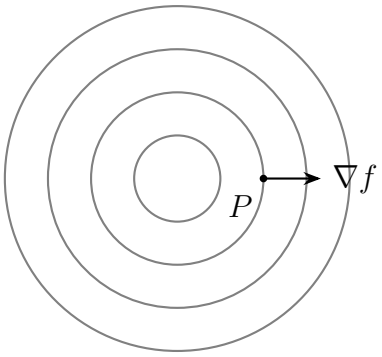
Gradient is normal to level sets

$\nabla f(P_0)$ is perpendicular to the level curve (in $\mathbb{R}^2$) or level surface (in $\mathbb{R}^3$) of f through P_0, provided $\nabla f(P_0) \neq \vec{0}$.

Proof. If $\vec{r}(t)$ is a curve in the level set with $\vec{r}(0) = P_0$, then $f(\vec{r}(t))$ is constant, so $0 = \dfrac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = \nabla f(P_0) \cdot \vec{r}\,'(0)$: the gradient is orthogonal to every tangent direction of the level set. ■

Worked example

For $f(x, y) = x^2 + y^2$ at (x_0, y_0), $\nabla f = \langle 2x_0, 2y_0\rangle$ is perpendicular to the circle $x^2 + y^2 = r^2$,



level curves $f = k$

Figure 5: The gradient is perpendicular to the level curves and points towards increasing f.

as expected.

2.8 Maximum and minimum values

2.8.1 Local extrema

Local extrema and critical points

f has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all (x, y) in some disk centred at (a, b); **local minimum** is defined with $\geq$. A point (a, b) is a **critical point** of f if $f_x(a, b) = f_y(a, b) = 0$, or if one of the partial derivatives does not exist. A critical point which is neither a local maximum nor a local minimum is a **saddle point**.

First derivative test

If f has a local maximum or minimum at (a, b) and $f_x(a, b), f_y(a, b)$ exist, then

$$\nabla f(a, b) = \vec{0}.$$

Proof. Let $\vec{r}(t)$ be any parametric curve with $\vec{r}(0) = (a, b)$, defined on an open interval around 0. Then $t \mapsto f(\vec{r}(t))$ has a local extremum at $t = 0$, so

$$0 = \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t))\Big|_{t=0} = \nabla f(a, b) \cdot \vec{r}\,'(0).$$

Since $\vec{r}\,'(0)$ can be any vector (take $\vec{r}(t) = (a, b) + t\vec{w}$), $\nabla f(a, b)$ is orthogonal to every vector, hence $\nabla f(a, b) = \vec{0}$. Taking $\vec{w} = \langle 1, 0\rangle$ and $\langle 0, 1\rangle$ gives $f_x(a, b) = f_y(a, b) = 0$. ■

Worked examples

(a) $f = x^2 - y^2$: $\nabla f = (2x, -2y) = \vec{0}$ only at $(0, 0)$. Along $y = 0$, $f = x^2$ has a minimum at 0; along $x = 0$, $f = -y^2$ has a maximum at 0. Hence $(0, 0)$ is a saddle point.
(b) $f = x^2 + y^2 - 2x - 6y + 14$: $f_x = 2x - 2$, $f_y = 2y - 6$, so $(1, 3)$ is the only critical point, with $f(1, 3) = 4$. Since

$$f = (x - 1)^2 + (y - 3)^2 + 4 \geq 4,$$

$(1, 3)$ is a global (hence local) minimum.

2.8.2 The second derivative test

Second derivative test

Let (a, b) be a critical point of f and suppose the second partial derivatives are continuous on a disk around (a, b). Put

$$D = D(a, b) = \det \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{yx}(a, b) & f_{yy}(a, b) \end{pmatrix} = f_{xx} f_{yy} - (f_{xy})^2.$$

Then
(a) if $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum;
(b) if $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum;
(c) if $D < 0$, then (a, b) is a saddle point;
(d) if $D = 0$, the test is inconclusive.

Proof. Let $\vec{u} = \langle h, k \rangle$ be a unit vector. We first compute the **second directional derivative** along $\vec{u}$, the derivative of $D_{\vec{u}} f$ in the direction $\vec{u}$.

By definition of directional derivative, $D_{\vec{u}} f = h f_x + k f_y$. Applying $D_{\vec{u}}$ a second time:

$$(D_{\vec{u}})^2 f = h\, D_{\vec{u}}(f_x) + k\, D_{\vec{u}}(f_y) = h(h f_{xx} + k f_{xy}) + k(h f_{xy} + k f_{yy}) = h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}.$$

Since (a, b) is a critical point, the Taylor expansion of f in any direction simplifies: the first-order terms vanish and the sign of $f(a+th, b+tk) - f(a, b)$ for small t is governed by $(D_{\vec{u}})^2 f$ evaluated at (a, b). Therefore the sign of this quadratic form determines whether (a, b) is a local extremum or a saddle.

Case 1: $f_{xx}(a, b) \neq 0$. Complete the square in the quadratic form:

$$\begin{aligned} (D_{\vec{u}})^2 f &= f_{xx} \left(h^2 + 2\frac{f_{xy}}{f_{xx}} hk + \frac{f_{xy}^2}{f_{xx}^2} k^2 \right) + \left(k^2 f_{yy} - \frac{f_{xy}^2}{f_{xx}} k^2 \right) \\ &= f_{xx} \left(h + \frac{f_{xy}}{f_{xx}} k \right)^2 + \frac{k^2}{f_{xx}} \left(f_{xx} f_{yy} - f_{xy}^2 \right) \\ &= f_{xx} \left(h + \frac{f_{xy}}{f_{xx}} k \right)^2 + \frac{k^2}{f_{xx}} D. \end{aligned}$$

Case 2: $f_{xx}(a, b) = 0$ *but* $f_{yy}(a, b) \neq 0$. The determinant condition $D \neq 0$ guarantees that at least one of f_{xx}, f_{yy} is nonzero (for if both are zero then $D = -f_{xy}^2 \leq 0$ and the case $D > 0$ cannot arise). So when $D > 0$ the quadratic form can always be completed in one variable or the other. For instance, if $f_{xx} = 0$ we complete in k:

$$(D_{\vec{u}})^2 f = f_{yy} \left(k + \frac{f_{xy}}{f_{yy}} h \right)^2 + \frac{h^2}{f_{yy}} D.$$

Now let's analyse the discriminant conditions.

(a) $D > 0$ *and* $f_{xx}(a, b) > 0$. Since the second partial derivatives are continuous on a disk around (a, b), the value (f_{xx}, f_{xy}, f_{yy}) at nearby points is arbitrarily close to its value at (a, b). The expression $D = f_{xx} f_{yy} - f_{xy}^2$ is a continuous function of the three entries, and if $D(a, b) > 0$ there exists a small disk where $D > 0$ everywhere. Similarly $f_{xx} > 0$ persists. In this disk, both terms in the completed square sum are non-negative (the square is always ≥ 0 and $k^2 D / f_{xx} \geq 0$ since $f_{xx} > 0$ and $D > 0$). The first term is zero only when $h = -\dfrac{f_{xy}}{f_{xx}} k$, but the second term is zero only when $k = 0$, which forces $h = 0$ (since $\vec{u}$ is unit). Hence the only way $(D_{\vec{u}})^2 f$ can be zero is if $\vec{u} = \vec{0}$, which is not a unit vector. Therefore $(D_{\vec{u}})^2 f > 0$ for *all* unit $\vec{u}$ near (a, b). This means that the restriction of f to every line through (a, b) is convex, so f has a local minimum.

(b) $D > 0$ and $f_{xx}(a,b) < 0$. Exactly the same argument applies, but with reversed sign: $(D_{\vec{u}})^2 f < 0$ for all unit $\vec{u}$, every line restriction is concave, so f has a local maximum. (If $f_{xx} = 0$ here, then f_{yy} is negative because $D > 0 = f_{xx}f_{yy} - f_{xy}^2 = -f_{xy}^2$ is impossible unless $f_{xy} = 0$, in which case $D = f_{xx}f_{yy} > 0$ with one factor nonzero; whichever is nonzero has the sign we need.)

(c) $D < 0$. The quadratic form is *indefinite*. Suppose for instance $f_{xx} > 0$. Then the first term in the completed square is a square with positive coefficient, and the second term $\dfrac{k^2}{f_{xx}}D$ is negative whenever $k \neq 0$ (since $D < 0$). By choosing $k = 0$ we get $(D_{\vec{u}})^2 f = f_{xx}h^2 > 0$ along the x-axis direction. By choosing $h = -\dfrac{f_{xy}}{f_{xx}}k$ (the direction of the zero of the square), the first term vanishes and we have $(D_{\vec{u}})^2 f = \dfrac{k^2}{f_{xx}}D < 0$. So $(D_{\vec{u}})^2 f$ takes both positive and negative values as $\vec{u}$ varies over unit vectors, which means that f increases along one line through (a,b) and decreases along another: (a,b) is a saddle point. The argument is symmetric when $f_{xx} < 0$ or when we complete with respect to f_{yy}.

(d) $D = 0$. The completed square has a vanishing second term and the test gives no information. Indeed, consider three functions all with critical point $(0,0)$ and all with $D(0,0) = 0$:

$$\begin{aligned} f(x,y) &= x^4 + y^4 && \text{has a local minimum,} \\ f(x,y) &= -x^4 - y^4 && \text{has a local maximum,} \\ f(x,y) &= x^4 - y^4 && \text{has a saddle point.} \end{aligned}$$

The second derivative test cannot distinguish these. ■

Worked example

$f = x^4 + y^4 - 4xy + 1$. Then $f_x = 4x^3 - 4y$, $f_y = 4y^3 - 4x$; setting both to zero gives $x = y$ and $x^4 = x$, i.e. $x \in \{0, 1, -1\}$. Moreover

$$f_{xx} = 12x^2, \quad f_{yy} = 12y^2, \quad f_{xy} = -4, \qquad D = 144x^2y^2 - 16.$$

$D(0,0) = -16 < 0$: saddle point. $D(1,1) = 128 > 0$ and $f_{xx}(1,1) = 12 > 0$: local minimum. $D(-1,-1) = 128 > 0$ and $f_{xx}(-1,-1) = 12 > 0$: local minimum.

2.8.3 Absolute extrema

Definitions

$P_0 \in D$ is an **absolute (global) minimum** of $f : D \to \mathbb{R}$ if $f(P) \geq f(P_0)$ for all $P \in D$; absolute maximum is defined with $\leq$.
A set $S \subseteq \mathbb{R}^n$ is **bounded** if there is M with $|OP| < M$ for all $P \in S$. A point P is a **limit point** of S if every ball around P contains a point of S different from P. S is **closed** if it contains all its limit points, equivalently if its complement is open.

Extreme value theorem

A continuous function on a closed and bounded subset of $\mathbb{R}^n$ attains both an absolute maximum and an absolute minimum.

Procedure for absolute extrema on a closed bounded region D

1. Find the values of f at all critical points in the *interior* of D.
2. Find the extreme values of f on the *boundary* of D (this may require parametrizing each smooth piece and using one-variable calculus; the boundary may itself have a boundary, so the step is recursive).

3. Compare all the values obtained.

Worked example 1: a rectangle

Find the absolute extrema of $f(x,y) = x^2 - 2xy + 2y$ on the rectangle $D = [0,3] \times [0,2]$.
Interior. $f_x = 2x - 2y = 0$, $f_y = -2x + 2 = 0$ give $(1,1) \in (0,3) \times (0,2)$, with $f(1,1) = 1$.
Boundary. On the four sides:

$$f(x,0) = x^2 \ (0 \le x \le 3): \ \min 0, \ \max 9; \qquad f(0,y) = 2y: \ \min 0, \ \max 4;$$

$$f(x,2) = (x-2)^2 \ (0 \le x \le 3): \ \min 0, \ \max 4; \qquad f(3,y) = 9 - 4y: \ \min 1, \ \max 9.$$

Compare. The absolute minimum is 0 (at $(0,0)$ and $(2,2)$), the absolute maximum is 9 (at $(3,0)$).

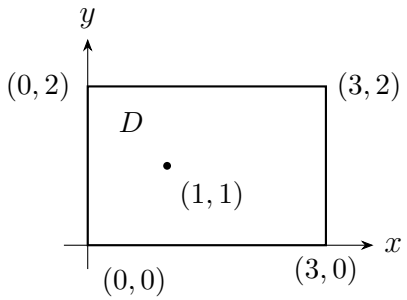


Figure 6: The rectangle $[0,3] \times [0,2]$ with its interior critical point.

Worked example 2: a disk

Find the absolute extrema of $f(x,y) = x^2 + y^2 - 2x + 2y + 5$ on the closed disk $x^2 + y^2 \le 4$.
Interior. $f_x = 2x - 2 = 0$, $f_y = 2y + 2 = 0$ give $(1,-1)$, which lies in the disk, and $f(1,-1) = 3$.
Boundary. Parametrize $\vec{r}(t) = \langle 2\cos t, 2\sin t \rangle$, $0 \le t \le 2\pi$. Then

$$g(t) = f(\vec{r}(t)) = 4\cos^2 t + 4\sin^2 t - 4\cos t + 4\sin t + 5 = 9 - 4\cos t + 4\sin t,$$

so $g'(t) = 4\sin t + 4\cos t = 0$, i.e. $\tan t = -1$, giving $t = -\pi/4$ or $t = 3\pi/4$. Hence $(x,y) = (\sqrt{2}, -\sqrt{2})$ or $(-\sqrt{2}, \sqrt{2})$, with

$$f(\sqrt{2}, -\sqrt{2}) = 9 - 4\sqrt{2}, \qquad f(-\sqrt{2}, \sqrt{2}) = 9 + 4\sqrt{2}.$$

Compare. The absolute minimum is 3 at $(1,-1)$ and the absolute maximum is $9 + 4\sqrt{2}$ at $(-\sqrt{2}, \sqrt{2})$.

2.9 Lagrange multipliers

2.9.1 The method

Problem

Minimize or maximize $f(P)$ subject to the constraint $g(P) = k$ (one constraint), where f, g are differentiable. Geometrically, we look for the extreme values of f on the level set $g = k$.

Lagrange multiplier condition

At an extremum P_0 of f restricted to the level set $g = k$, with $\nabla g(P_0) \neq \vec{0}$, there is a

number λ such that

$$\boxed{\nabla f(P_0) = \lambda\, \nabla g(P_0)}$$

i.e. $f_x = \lambda g_x$, $f_y = \lambda g_y$ (and $f_z = \lambda g_z$ in $\mathbb{R}^3$), together with $g(P_0) = k$.

Proof (two variables; the three-variable case is identical). Let $\vec{r}(t)$ be any differentiable curve lying in the level curve $g = k$ with $\vec{r}(0) = P_0$. The function $t \mapsto f(\vec{r}(t))$ then has a local extremum at $t = 0$, so

$$0 = \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t))\Big|_{t=0} = \nabla f(P_0) \cdot \vec{r}\,'(0).$$

Also $g(\vec{r}(t)) = k$ gives $\nabla g(P_0) \cdot \vec{r}\,'(0) = 0$. Hence both $\nabla f(P_0)$ and $\nabla g(P_0)$ are perpendicular to every tangent vector of the level curve at P_0. The tangent space is one-dimensional, so its orthogonal complement is one-dimensional; if $\nabla g(P_0) \neq \vec{0}$ then $\nabla f(P_0)$ must be a scalar multiple of $\nabla g(P_0)$. ∎

Strategy

Solve the system

$$\nabla f = \lambda \nabla g, \qquad g = k$$

for the candidate points and λ, then evaluate f at all candidates and compare. If the constraint set is closed and bounded, the extrema exist by the extreme value theorem.

Worked example 1: a railway station

Three cities are at $A(5, 2)$, $B(-4, 4)$, $C(-1, -3)$, and a railway track follows $y = x^3 + 1$. A station is to be built on the track minimizing the sum of the squares of the distances to the three cities.
Minimize

$$f = (x-5)^2 + (y-2)^2 + (x+4)^2 + (y-4)^2 + (x+1)^2 + (y+3)^2$$

subject to $g = y - x^3 - 1 = 0$. The Lagrange system is

$$\begin{aligned} 2(x-5) + 2(x+4) + 2(x+1) &= -\lambda \cdot 3x^2, \\ 2(y-2) + 2(y-4) + 2(y+3) &= \lambda, \qquad y = 1 + x^3. \end{aligned}$$

The first two equations simplify to

$$2x = -\lambda x^2, \qquad 6y = \lambda + 6.$$

From $2x = -\lambda x^2$: either $x = 0$ (then $y = 1$) or $\lambda = -2/x$ (then $6x^3 = 2$, impossible together with $\lambda = -2/x$ since it forces $-6x^4 = 2$). Hence the only candidate is $(0, 1)$.

Worked example 2: extrema on a disk

Find the extreme values of $f(x, y) = x^2 + 2y^2$ on $x^2 + y^2 \leq 1$.
Interior. $f_x = 2x = 0$, $f_y = 4y = 0$ give $(0, 0)$ with $f = 0$.
Boundary. With $g = x^2 + y^2$,

$$2x = 2\lambda x, \qquad 4y = 2\lambda y, \qquad x^2 + y^2 = 1.$$

From the first equation $x = 0$ or $\lambda = 1$.
If $x = 0$ then $y = \pm 1$ and $f = 2$. If $\lambda = 1$ then $2y = 0$, so $y = 0$ and $x = \pm 1$, giving $f = 1$. Therefore

$$\min_D f = 0 \text{ at } (0, 0), \qquad \max_D f = 2 \text{ at } (0, \pm 1).$$

Alternatively, on the circle $f = x^2 + 2y^2 = 1 + y^2$ with $0 \leq y^2 \leq 1$, which gives the same answer without any multiplier.

2.9.2 Two constraints

Two constraints

To extremize $f(x,y,z)$ subject to $g(x,y,z)=k$ and $h(x,y,z)=l$, solve

$$\boxed{\nabla f = \lambda \nabla g + \mu \nabla h, \qquad g = k, \qquad h = l}$$

Geometrically, the constraint set is a curve (the intersection of two surfaces); ∇f must lie in the plane spanned by ∇g and ∇h, which is the plane perpendicular to that curve.

Worked example

Find the maximum of $f(x,y,z) = x+2y+3z$ on the ellipse $\{x-y+z=1\} \cap \{x^2+y^2=1\}$.

Take $g = x-y+z$, $h = x^2+y^2$. The system is

$$1 = \lambda + 2\mu x, \qquad 2 = -\lambda + 2\mu y, \qquad 3 = \lambda, \qquad x-y+z = 1, \qquad x^2+y^2 = 1.$$

Thus $\lambda = 3$, hence $2\mu x = -2$ and $2\mu y = 5$, i.e. $x = -1/\mu$ and $y = 5/(2\mu)$. Substituting in $x^2+y^2=1$:

$$\frac{1}{\mu^2} + \frac{25}{4\mu^2} = 1 \Longrightarrow \mu^2 = \frac{29}{4} \Longrightarrow \mu = \pm\frac{\sqrt{29}}{2}.$$

Then $x = \mp 2/\sqrt{29}$, $y = \pm 5/\sqrt{29}$ and $z = 1 - x + y = 1 \pm 7/\sqrt{29}$. The values of f are

$$f = x + 2y + 3z = 3 \pm \sqrt{29},$$

so the maximum is $3+\sqrt{29}$ and the minimum is $3-\sqrt{29}$.
Check by substitution. Substituting $z = 1-x+y$ reduces the problem to maximizing $x+2y+3(1-x+y) = -2x+5y+3$ on $x^2+y^2=1$, a one-constraint problem.

Common mistakes

- Lagrange multipliers find *candidate* points; one must still compare values (and check the boundary when the constraint set has a boundary).
- The condition $\nabla g \neq \vec{0}$ is essential: if $\nabla g = \vec{0}$ at a candidate, the level set may be singular there and the condition may fail.
- A critical point of f restricted to a level set need not be a critical point of f in $\mathbb{R}^2$; do not mix the two tests.

2.10 Extra: the Hessian and the multivariable mean value theorem

2.10.1 The Hessian matrix

Local extrema

For a scalar field $f : \mathbb{R}^n \to \mathbb{R}$, the point $\vec{a}$ is a **local minimum** if $f(\vec{x}) > f(\vec{a})$ for all $\vec{x} \in B(\vec{a},\epsilon) \setminus \{\vec{a}\}$ for some $\epsilon > 0$; a **local maximum** is defined with $<$. If neither holds, $\vec{a}$ is a **saddle point**.

Hessian matrix

For a scalar field $f : \mathbb{R}^n \to \mathbb{R}$ of class C^2, the **Hessian matrix** is the $n \times n$ matrix of

second-order partial derivatives

$$H_f(\vec{x}) = \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_1 \partial x_n} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} & \cdots & \dfrac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial^2 f}{\partial x_n \partial x_1} & \dfrac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \dfrac{\partial^2 f}{\partial x_n^2} \end{pmatrix} = \nabla(\nabla f)^T.$$

If f is C^2, then by Clairaut's theorem H_f is symmetric, so all its eigenvalues are real.

Motivation (Taylor). In one variable, $f(a+h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \cdots$. The first-order term is signed, so it can never produce a local extremum while $f'(a) \neq 0$; the second-order term is unsigned, so $f'(a) = 0$ together with $f''(a) \neq 0$ gives a local extremum. The Hessian plays the role of f'' in several variables.

For a unit direction $\vec{u} = \langle u, v \rangle$ put $g(t) = f(a+tu, b+tv)$. Then

$$g'(t) = uf_x + vf_y, \qquad g''(t) = u^2 f_{xx} + 2uvf_{xy} + v^2 f_{yy} = \begin{pmatrix} u & v \end{pmatrix} H_f \begin{pmatrix} u \\ v \end{pmatrix}.$$

Since $\vec{a}$ is a critical point, the first-order terms vanish and the sign of the quadratic form g'' decides the nature of the point.

Determinant and eigenvalues

For an $n \times n$ matrix A, $\det A = \prod_{i=1}^{n} \lambda_i$. In particular for a 2×2 Hessian, $|H_f| = \lambda_1 \lambda_2$; if it is positive the two eigenvalues have the same sign, and if negative they have opposite signs.

Proof. Diagonalise $A = PDP^{-1}$; then $\det A = \det P \det D \det P^{-1} = \det D = \prod_i \lambda_i$. ■

Classification of critical points

At a critical point $\vec{x}_0$ (where $\nabla f(\vec{x}_0) = \vec{0}$) the nature of the point is read off from the eigenvalues $\lambda_1, \ldots, \lambda_n$ of $H_f(\vec{x}_0)$:

local minimum	all $\lambda_i > 0$	(positive definite)
local maximum	all $\lambda_i < 0$	(negative definite)
saddle point	mixed signs	(indefinite)
inconclusive	some $\lambda_i = 0$	(singular Hessian)

Criterion via leading principal minors

Let D_k be the determinant of the k-th leading principal minor of H_f. Then

$$H_f \text{ positive definite} \iff D_1 > 0,\ D_2 > 0, \ldots, D_n > 0,$$

$$H_f \text{ negative definite} \iff (-1)^k D_k > 0 \ \ \forall k.$$

If neither holds, the quadratic form is indefinite.

Worked example: a ternary function

Consider $f(x, y, z) = x^2 + y^2 + z^2 + 4xy$.

1. *Gradient.* $\nabla f = \langle 2x + 4y,\ 2y + 4x,\ 2z \rangle = \vec{0}$ gives the only critical point $\vec{x}_0 = (0, 0, 0)$.

2. *Hessian.*
$$H_f = \begin{pmatrix} 2 & 4 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$
3. *Eigenvalues.* $\det(H_f - \lambda I) = (2-\lambda)\big[(2-\lambda)^2 - 16\big] = 0$ gives $\lambda = 2, 6, -2$: mixed signs, so $(0,0,0)$ is a saddle point.
4. *Principal minors.* $D_1 = 2 > 0$, $D_2 = \left|\begin{smallmatrix} 2 & 4 \\ 4 & 2 \end{smallmatrix}\right| = -12 < 0$, $D_3 = \det H_f = 2(-12) = -24 < 0$. Neither definiteness condition holds, confirming a saddle point.

2.10.2 The multivariable mean value theorem

Multivariable mean value theorem

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable on an open *convex* set $D \subseteq \mathbb{R}^n$. For any two distinct points $\vec{a}, \vec{b} \in D$ there is a point $\vec{c}$ on the segment joining them such that

$$\boxed{f(\vec{b}) - f(\vec{a}) = \nabla f(\vec{c}) \cdot (\vec{b} - \vec{a})}$$

Writing $\vec{c}(t) = (1-t)\vec{a} + t\vec{b}$, there exists $t_0 \in (0,1)$ with $f(\vec{b}) - f(\vec{a}) = \nabla f(\vec{c}(t_0)) \cdot (\vec{b} - \vec{a})$.

Proof. Let $g(t) = f((1-t)\vec{a} + t\vec{b}) = f(\vec{c}(t))$ on $[0,1]$. By the chain rule g is differentiable on $(0,1)$, continuous on $[0,1]$, and

$$g'(t) = \nabla f(\vec{c}(t)) \cdot (\vec{b} - \vec{a}).$$

The one-variable mean value theorem gives $t_0 \in (0,1)$ with $g'(t_0) = \dfrac{g(1) - g(0)}{1 - 0} = f(\vec{b}) - f(\vec{a})$; substituting $g'(t_0)$ proves the claim. ■

Key observations

Convexity of D keeps the segment $\vec{c}(t)$ inside D; for $n = 1$ the theorem reduces to the classical mean value theorem; the intermediate point $\vec{c}$ need not be unique; and the theorem *fails* for vector-valued maps $\vec{f} : \mathbb{R}^n \to \mathbb{R}^m$ with $m > 1$.

Application

Estimate the change of $f(x,y) = x^2 y$ from $\vec{a} = (1,2)$ to $\vec{b} = (1.1, 1.9)$. Here $\nabla f = (2xy, x^2)$ and $\vec{b} - \vec{a} = (0.1, -0.1)$. With $\vec{c} \approx (1.05, 1.95)$,

$$\Delta f \approx \nabla f(1.05, 1.95) \cdot (0.1, -0.1) = (4.095,\ 1.1025) \cdot (0.1, -0.1) = -0.29925,$$

while the exact change is $f(1.1, 1.9) - f(1,2) = 2.299 - 2 = 0.299$ (up to sign). The theorem guarantees such a $\vec{c}$ exists.

2.10.3 More on the gradient

Gradient

For a scalar field $f : \mathbb{R}^n \to \mathbb{R}$ the **gradient** is the vector field

$$\nabla f = \Big(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\Big), \qquad \nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k} \ \ (n = 3).$$

Its basic properties: $D_{\vec{u}} f = \nabla f \cdot \vec{u}$ for unit $\vec{u}$; it points in the direction of steepest ascent; it is orthogonal to the level sets $f = \text{const}$; and $\vec{F} = \nabla f$ makes $\vec{F}$ conservative.

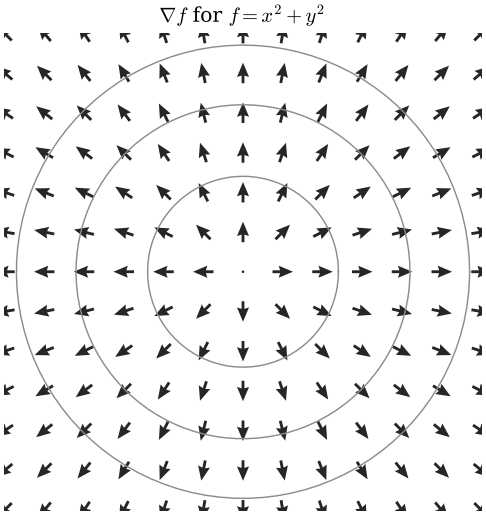


Figure 7: The gradient field of $f(x, y) = x^2 + y^2$: the arrows point radially outwards, perpendicular to the circular level curves.

2.10.4 Extra: divergence, curl and the Laplacian

Divergence

For a vector field $\vec{F} : \mathbb{R}^n \to \mathbb{R}^n$ the **divergence** is the scalar field

$$\nabla \cdot \vec{F} = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i}, \qquad \nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \quad (n = 3).$$

Its physical meaning:

- $\nabla \cdot \vec{F} > 0$: a **source** (net outward flux);
- $\nabla \cdot \vec{F} < 0$: a **sink** (net inward flux);
- $\nabla \cdot \vec{F} = 0$: a **solenoidal** field (incompressible flow);
- Gauss' theorem: $\iiint_V (\nabla \cdot \vec{F})\, \mathrm{d}V = \iint_S \vec{F} \cdot \mathrm{d}\vec{S}$, the divergence theorem.

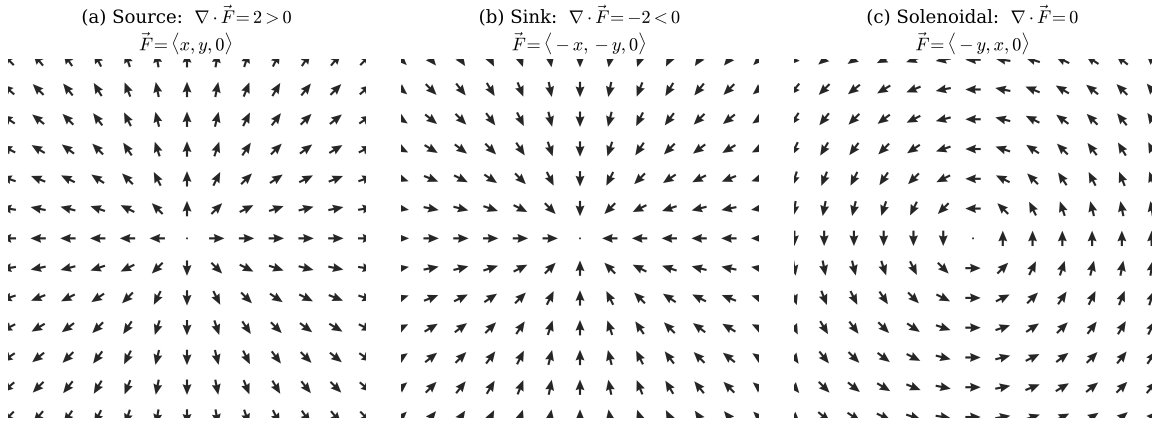


Figure 8: The three fundamental cases of divergence: (a) a source $\vec{F} = \langle x, y, 0 \rangle$ with $\nabla \cdot \vec{F} = 2 > 0$; (b) a sink $\vec{F} = \langle -x, -y, 0 \rangle$ with $\nabla \cdot \vec{F} = -2 < 0$; (c) a solenoidal field $\vec{F} = \langle -y, x, 0 \rangle$ with $\nabla \cdot \vec{F} = 0$.

Curl

For a 3D vector field $\vec{F} = \langle F_x, F_y, F_z \rangle$ the **curl** is the vector field

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \Big(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\Big)\vec{i} + \Big(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\Big)\vec{j} + \Big(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\Big)\vec{k}.$$

Its key properties:

- it measures the **circulation density** (rotation per unit area);
- an **irrotational** field has $\nabla \times \vec{F} = \vec{0}$, and on a simply connected domain this makes it conservative;
- Stokes' theorem: $\displaystyle\iint_S (\nabla \times \vec{F}) \cdot \mathrm{d}\vec{S} = \oint_C \vec{F} \cdot \mathrm{d}\vec{r}.$

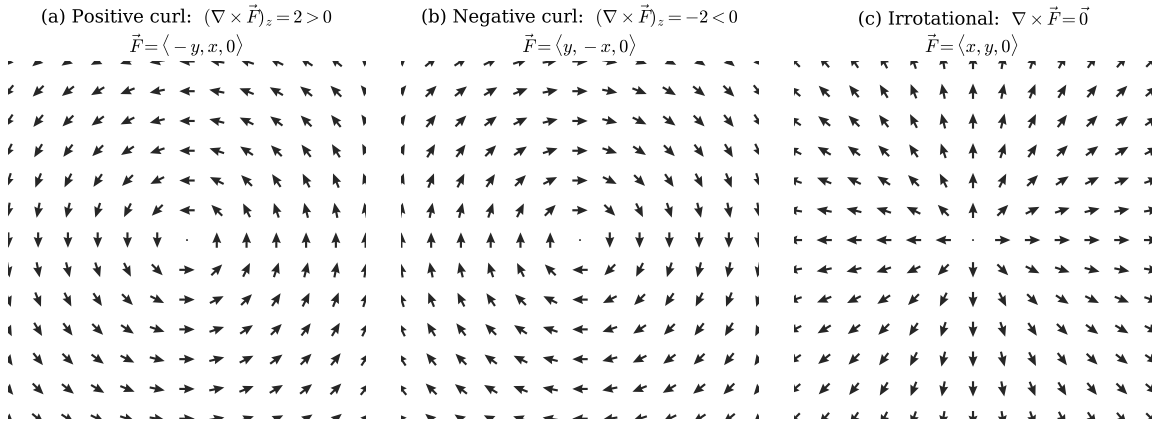


Figure 9: The three fundamental cases of curl: (a) positive curl $\vec{F} = \langle -y, x, 0 \rangle$ with $(\nabla \times \vec{F})_z = 2 > 0$; (b) negative curl $\vec{F} = \langle y, -x, 0 \rangle$ with $(\nabla \times \vec{F})_z = -2 < 0$; (c) an irrotational field $\vec{F} = \langle x, y, 0 \rangle$ with $\nabla \times \vec{F} = \vec{0}$.

Vector calculus identities

- $\nabla \times (\nabla f) = \vec{0}$: the curl of every gradient vanishes;
- $\nabla \cdot (\nabla \times \vec{F}) = 0$: the divergence of every curl vanishes;
- **scalar Laplacian:** $\nabla^2 f = \nabla \cdot (\nabla f) = \displaystyle\sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}$;
- **vector Laplacian:** $\nabla^2 \vec{F} = \nabla(\nabla \cdot \vec{F}) - \nabla \times (\nabla \times \vec{F})$.

Physical applications

Fluid dynamics (density ρ, velocity field $\vec{v}$, pressure p):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \qquad \text{(continuity equation)},$$

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = -\frac{\nabla p}{\rho} + \vec{g} \qquad \text{(Euler's equation)}.$$

Electromagnetism (Maxwell's equations):

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

3 Multiple Integrals

Definite integrals of one variable solve the area problem and the distance problem. Double integrals solve the volume problem and the mass problem for a lamina; triple integrals do the same in space. The theory is built from Riemann sums exactly as in one variable, and the practical computations are reduced to iterated one-variable integrals by Fubini's theorem.

3.1 Double integrals over rectangles

3.1.1 The definition

Partitions and Riemann sums

Let $R = [a, b] \times [c, d]$ be a closed rectangle and let $f : R \to \mathbb{R}$ be bounded. Divide $[a, b]$ into m subintervals $[x_{i-1}, x_i]$ of length $\Delta x = (b - a)/m$ and $[c, d]$ into n subintervals $[y_{j-1}, y_j]$ of length $\Delta y = (d - c)/n$. This divides R into mn subrectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j], \qquad \Delta A = \Delta x \, \Delta y.$$

Choose a **sample point** $(x^*_{ij}, y^*_{ij}) \in R_{ij}$ and form the **double Riemann sum**

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij}) \, \Delta A.$$

The double integral

If the Riemann sums tend to a limit L as $m, n \to \infty$ *independently of the choices of the sample points*, we say that f is **integrable** on R and write

$$\boxed{\iint_R f(x, y) \, \mathrm{d}A = \lim_{m,n \to \infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x^*_{ij}, y^*_{ij}) \, \Delta A}$$

The quantity $\mathrm{d}A$ is the **area element**. If $f \geq 0$ and f is continuous, the integral is the volume of the solid $\{(x, y, z) : 0 \leq z \leq f(x, y), \ (x, y) \in R\}$: the sum adds up the volumes of the thin boxes of height $f(x^*_{ij}, y^*_{ij})$ and base ΔA.

Integrability

Every continuous function on a closed rectangle is integrable. More generally, a bounded function with discontinuities only on a finite union of smooth curves is integrable.

Worked example: estimating a volume

Estimate $\displaystyle\iint_R (16 - x^2 - 2y^2) \, \mathrm{d}A$ for $R = [0, 3] \times [0, 2]$ using $m = n = 2$ and the upper right corners.

Here $\Delta A = 1.5 \cdot 1 = 1.5$ and the sample points are

$$(0.75, 0.5), \ (2.25, 0.5), \ (0.75, 1.5), \ (2.25, 1.5).$$

The sum is
$$1.5\Big[\big(16-\frac{9}{16}-\frac{1}{2}\big)+\big(16-\frac{81}{16}-\frac{1}{2}\big)+\big(16-\frac{9}{16}-\frac{9}{2}\big)+\big(16-\frac{81}{16}-\frac{9}{2}\big)\Big] = 1.5\cdot 34.25 = 51.375.$$
Refining the partition gives the exact value $\displaystyle\int_0^3\int_0^2 (16-x^2-2y^2)\,\mathrm{d}y\,\mathrm{d}x = 44$.

Midpoint rule

With $m=n$ and (x_{ij}^*, y_{ij}^*) the centre $(\bar{x}_i, \bar{y}_j)$ of each subrectangle,
$$\iint_R f\,\mathrm{d}A \approx \sum_{i=1}^{m}\sum_{j=1}^{n} f(\bar{x}_i,\bar{y}_j)\,\Delta A.$$
For smooth f the error is of order $1/m^2$, so the midpoint rule is much more accurate than a random sample of the same size.

Average value

The **average value** of an integrable f on R is
$$f_{\text{ave}} = \frac{1}{\text{area}(R)}\iint_R f(x,y)\,\mathrm{d}A.$$

3.1.2 Properties

Properties of the double integral

Let f, g be integrable on R and $c\in\mathbb{R}$. Then

(i) $\displaystyle\iint_R c\,\mathrm{d}A = c\ \text{area}(R)$;

(ii) $\displaystyle\iint_R [f+g]\,\mathrm{d}A = \iint_R f\,\mathrm{d}A + \iint_R g\,\mathrm{d}A$;

(iii) $\displaystyle\iint_R cf\,\mathrm{d}A = c\iint_R f\,\mathrm{d}A$;

(iv) if $f\ge g$ on R, then $\displaystyle\iint_R f\,\mathrm{d}A \ge \iint_R g\,\mathrm{d}A$;

(v) if $R = R_1\cup R_2$ where R_1, R_2 overlap only on their boundaries, then
$$\iint_R f\,\mathrm{d}A = \iint_{R_1} f\,\mathrm{d}A + \iint_{R_2} f\,\mathrm{d}A.$$

Proof of (ii)–(iv). For any partition and any choice of sample points,
$$\sum_{i,j}(f+g)(x_{ij}^*,y_{ij}^*)\Delta A = \sum_{i,j} f(x_{ij}^*,y_{ij}^*)\Delta A + \sum_{i,j} g(x_{ij}^*,y_{ij}^*)\Delta A.$$
Passing to the limit (the limits of both sums exist by hypothesis) gives (ii). Part (iii) follows from $\sum cf\Delta A = c\sum f\Delta A$. For (iv), every Riemann sum for f dominates the corresponding Riemann sum for g, and limits preserve non-strict inequalities. Part (v) follows by choosing partitions compatible with the common boundary. ∎

3.2 Iterated integrals and Fubini's theorem

Fubini's theorem

If f is continuous on $R = [a,b] \times [c,d]$, then

$$\iint_R f(x,y)\,\mathrm{d}A = \int_a^b \left[\int_c^d f(x,y)\,\mathrm{d}y\right]\mathrm{d}x = \int_c^d \left[\int_a^b f(x,y)\,\mathrm{d}x\right]\mathrm{d}y$$

The inner integral is computed with the other variable held constant. The two orders of integration always give the same answer for continuous f.

Proof. Let f be continuous on $R = [a,b] \times [c,d]$. Since R is compact, f is *uniformly continuous* on R: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x,y) - f(x',y')| < \varepsilon$ whenever both (x,y) and (x',y') lie in R and $\sqrt{(x-x')^2 + (y-y')^2} < \delta$.

Define, for each $x \in [a,b]$, the cross-sectional area function

$$A(x) = \int_c^d f(x,y)\,\mathrm{d}y.$$

We claim that A is continuous on $[a,b]$. Indeed, for $|x - x_0| < \delta$,

$$|A(x) - A(x_0)| = \left|\int_c^d \big(f(x,y) - f(x_0,y)\big)\mathrm{d}y\right| \le \int_c^d \big|f(x,y) - f(x_0,y)\big|\mathrm{d}y < \varepsilon(d-c),$$

so A is uniformly continuous, hence integrable, on $[a,b]$.

Now take a uniform partition of R with m subintervals in x and n in y: $a = x_0 < x_1 < \cdots < x_m = b$, $c = y_0 < y_1 < \cdots < y_n = d$, with $\Delta x = (b-a)/m$, $\Delta y = (d-c)/n$, $\Delta A = \Delta x\,\Delta y$. Choose sample points $x_i^* \in [x_{i-1}, x_i]$ and $y_j^* \in [y_{j-1}, y_j]$. The double Riemann sum can be grouped as

$$S_{mn} = \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta y\,\Delta x = \sum_{i=1}^{m}\left(\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta y\right)\Delta x.$$

For each fixed x_i^*, the inner sum $\displaystyle\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta y$ is a Riemann sum for $\displaystyle\int_c^d f(x_i^*, y)\,\mathrm{d}y = A(x_i^*)$. Since $f(x_i^*, \cdot)$ is continuous on $[c,d]$, this integral exists. Let $\varepsilon > 0$ be arbitrary. By uniform continuity of f, there exists $\delta > 0$ such that $|f(x,y) - f(x',y)| < \varepsilon$ whenever $|x - x'| < \delta$, and there exists $\eta > 0$ such that $\Delta y < \eta$ implies

$$\left|\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta y - A(x_i^*)\right| < \varepsilon \quad \text{for all } i.$$

(The latter is just the definition of the one-variable Riemann integral applied to $f(x_i^*, \cdot)$.) Therefore for sufficiently fine partitions,

$$\left|S_{mn} - \sum_{i=1}^{m} A(x_i^*)\Delta x\right| \le \sum_{i=1}^{m}\left|\sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta y - A(x_i^*)\right|\Delta x < \varepsilon\sum_{i=1}^{m}\Delta x = \varepsilon(b-a).$$

As $m, n \to \infty$, the left side vanishes because ε was arbitrary. Now $\displaystyle\sum_{i=1}^{m} A(x_i^*)\Delta x$ is a

Riemann sum for $\displaystyle\int_a^b A(x)\,\mathrm{d}x$, which exists since A is continuous. Hence

$$\lim_{m,n\to\infty} S_{mn} = \int_a^b A(x)\,\mathrm{d}x = \int_a^b \left[\int_c^d f(x,y)\,\mathrm{d}y\right]\mathrm{d}x.$$

But the left limit is also the double integral by definition. The other order of integration is proved identically by exchanging the roles of x and y. ■

Worked example

Compute $\displaystyle\iint_R (x-3y^2)\,\mathrm{d}A$ for $R=[0,2]\times[1,2]$.

Integrating in y first,

$$\int_1^2 (x-3y^2)\,\mathrm{d}y = \Big[xy-y^3\Big]_1^2 = (2x-8)-(x-1) = x-7,$$

hence

$$\iint_R (x-3y^2)\,\mathrm{d}A = \int_0^2 (x-7)\,\mathrm{d}x = \Big[\frac{x^2}{2}-7x\Big]_0^2 = 2-14 = -12.$$

Integrating in x first gives the same:

$$\int_0^2 (x-3y^2)\,\mathrm{d}x = 2-9y^2, \qquad \int_1^2 (2-9y^2)\,\mathrm{d}y = \Big[2y-3y^3\Big]_1^2 = (4-24)-(2-3) = -19.$$

A warning about the notation

The symbols $\mathrm{d}y\,\mathrm{d}x$ mean "integrate in y first, then in x"; the order of the differentials is the reverse of the order of the integral signs when read from the outside in. Reversing the order without changing the limits generally changes the value.

3.3 Double integrals over general regions

3.3.1 Type I and type II regions

Type I regions

$D\subseteq\mathbb{R}^2$ is of **type I** if

$$D=\{(x,y): a\le x\le b,\ g_1(x)\le y\le g_2(x)\},$$

where g_1, g_2 are continuous and $g_1\le g_2$ on $[a,b]$. Then

$$\boxed{\iint_D f(x,y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,\mathrm{d}y\,\mathrm{d}x}$$

Type II regions

D is of **type II** if

$$D=\{(x,y): c\le y\le d,\ h_1(y)\le x\le h_2(y)\},$$

and then

$$\boxed{\iint_D f(x,y)\,\mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,\mathrm{d}x\,\mathrm{d}y}$$

A region may be of both types; if it is of neither, cut it into pieces of the two types and use additivity.

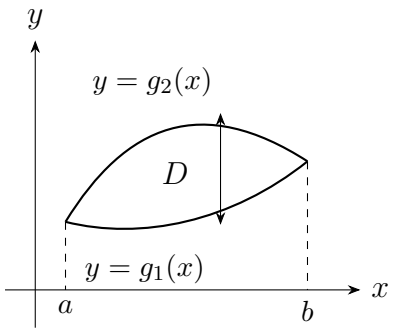


Figure 10: A type I region: the arrow indicates a vertical slice at fixed x.

Worked example: a triangular region

Evaluate $\iint_D (x+2y)\,\mathrm{d}A$, where D is the triangle with vertices $(0,0)$, $(1,1)$, $(0,1)$.
The hypotenuse is $y=x$ and the top is $y=1$, so $D=\{(x,y): 0\le x\le 1,\ x\le y\le 1\}$ is of type I:

$$\int_0^1\int_x^1 (x+2y)\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \Big[xy+y^2\Big]_{y=x}^{y=1}\mathrm{d}x = \int_0^1 \big(x+1-x^2-x^2\big)\mathrm{d}x$$
$$= \int_0^1 \big(1+x-2x^2\big)\mathrm{d}x = \Big[x+\frac{x^2}{2}-\frac{2x^3}{3}\Big]_0^1 = \frac{5}{6}.$$

Worked example: reversing the order

Express $\displaystyle\int_0^1\int_x^1 e^{y^2}\,\mathrm{d}y\,\mathrm{d}x$ as an integral that can be evaluated.
The region is $0\le x\le 1$, $x\le y\le 1$, i.e. the triangle $0\le y\le 1$, $0\le x\le y$. Hence

$$\int_0^1\int_x^1 e^{y^2}\,\mathrm{d}y\,\mathrm{d}x = \int_0^1\int_0^y e^{y^2}\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 y\,e^{y^2}\,\mathrm{d}y = \Big[\frac{1}{2}e^{y^2}\Big]_0^1 = \frac{e-1}{2}.$$

Area and volume

$$\text{area}(D) = \iint_D 1\,\mathrm{d}A, \qquad \text{volume}(E) = \iint_D \big[f_2(x,y)-f_1(x,y)\big]\mathrm{d}A$$

for the solid $E=\{(x,y,z): f_1(x,y)\le z\le f_2(x,y),\ (x,y)\in D\}$.

Worked example: area between two parabolas

Find the area of the region bounded by $y=x^2$ and $y=2x-x^2$.
The curves meet where $x^2=2x-x^2$, i.e. $x=0$ or $x=1$. Hence

$$A=\int_0^1 \big[(2x-x^2)-x^2\big]\mathrm{d}x = \int_0^1 (2x-2x^2)\,\mathrm{d}x = \frac{1}{3}.$$

3.4 Double integrals in polar coordinates

3.4.1 Polar rectangles

Polar coordinates

A point has polar coordinates (r, θ) with $x = r\cos\theta$, $y = r\sin\theta$, $r \geq 0$. A **polar rectangle**

$$R = \{(r, \theta) : a \leq r \leq b,\ \alpha \leq \theta \leq \beta\}$$

is a sector-like region in the plane.

The area element

$$\boxed{\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta}$$

and for f continuous on the polar region D,

$$\boxed{\iint_D f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\,\mathrm{d}r\,\mathrm{d}\theta}$$

where the r-limits may depend on θ for a general region.

Derivation. Divide $[\alpha, \beta]$ into n subintervals $[\theta_{j-1}, \theta_j]$ of width $\Delta\theta = (\beta - \alpha)/n$, and $[a, b]$ into m subintervals $[r_{i-1}, r_i]$ of width $\Delta r = (b - a)/m$. In polar coordinates the subrectangles are not Cartesian rectangles but thin annular sectors: the ij-th subrectangle has radial distance from r_{i-1} to r_i and angular width $\Delta\theta$. Its area is the difference of two circular sectors of angle $\Delta\theta$ and radii r_i, r_{i-1}:

$$\Delta A_{ij} = \frac{1}{2}r_i^2\Delta\theta - \frac{1}{2}r_{i-1}^2\Delta\theta = \frac{r_i + r_{i-1}}{2}\cdot(r_i - r_{i-1})\cdot\Delta\theta = r_{i-\frac{1}{2}}\,\Delta r\,\Delta\theta,$$

where $r_{i-\frac{1}{2}} = \dfrac{1}{2}(r_i + r_{i-1})$ is the mid-radius. As $m, n \to \infty$ this midpoint tends to the same value as any r_i or r_{i-1}, so $\Delta A \to r\,\mathrm{d}r\,\mathrm{d}\theta$.

Now take sample points (r_i^*, θ_j^*) in each subrectangle and form the polar Riemann sum

$$\sum_{i,j} f(r_i^*\cos\theta_j^*,\ r_i^*\sin\theta_j^*)\,\Delta A_{ij} = \sum_{i,j} f(\cdots)\, r_{i-\frac{1}{2}}\,\Delta r\,\Delta\theta.$$

Replacing $r_{i-\frac{1}{2}}$ by r_i^* introduces an error of at most $f_{\max}\max_i |r_{i-\frac{1}{2}} - r_i^*|\Delta r\Delta\theta \to 0$ as the mesh refines. Passing to the limit, the double integral becomes the iterated polar integral with integrand $f(r\cos\theta, r\sin\theta)\,r$ and measure $\mathrm{d}r\,\mathrm{d}\theta$. ∎

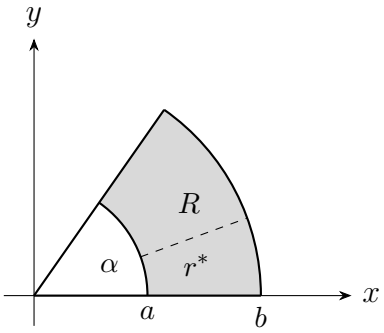


Figure 11: A polar rectangle and the area element $\mathrm{d}A \approx r\,\mathrm{d}r\,\mathrm{d}\theta$.

Worked example: a disk

Compute $\iint_D (3x + 4y^2)\,\mathrm{d}A$ for the upper half disk $D = \{x^2 + y^2 \le 1,\ y \ge 0\}$.

In polar coordinates $0 \le r \le 1$, $0 \le \theta \le \pi$, and

$$\iint_D (3x + 4y^2)\,\mathrm{d}A = \int_0^{\pi} \int_0^1 \big(3r\cos\theta + 4r^2 \sin^2\theta\big) r\,\mathrm{d}r\,\mathrm{d}\theta.$$

Separate:

$$\int_0^1 3r^2\,\mathrm{d}r \int_0^{\pi} \cos\theta\,\mathrm{d}\theta + \int_0^1 4r^3\,\mathrm{d}r \int_0^{\pi} \sin^2\theta\,\mathrm{d}\theta = 1 \cdot 0 + 1 \cdot \frac{\pi}{2} = \frac{\pi}{2}.$$

Worked example: the Gaussian integral

Show that $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x = \sqrt{\pi}$.

Let $I = \displaystyle\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x$. Then

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\mathrm{d}y\right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{d}A.$$

In polar coordinates, using $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$ and $0 \le r < \infty$, $0 \le \theta \le 2\pi$,

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi \cdot \Big[-\frac{1}{2}e^{-r^2}\Big]_0^{\infty} = 2\pi \cdot \frac{1}{2} = \pi.$$

Since $I > 0$, $I = \sqrt{\pi}$.

Area of a region bounded by a polar curve

If $D = \{(r,\theta) : \alpha \le \theta \le \beta,\ 0 \le r \le r(\theta)\}$ with $r(\theta) \ge 0$ and r continuous, then

$$\boxed{\text{area}(D) = \frac{1}{2}\int_{\alpha}^{\beta} r(\theta)^2\,\mathrm{d}\theta}$$

Proof. Divide $[\alpha, \beta]$ into n subintervals $[\theta_{j-1}, \theta_j]$ of width $\Delta\theta$. The region D_j between $r = 0$ and $r = r(\theta)$ for $\theta_{j-1} \le \theta \le \theta_j$ is a thin sector whose area is approximately the area of a circular sector of angle $\Delta\theta$ and radius $r(\theta_j^*)$ for some sample point $\theta_j^* \in [\theta_{j-1}, \theta_j]$:

$$\text{area}(D_j) \approx \frac{1}{2} r(\theta_j^*)^2\,\Delta\theta.$$

(This is the exact area of a circular sector of radius $r(\theta_j^*)$ and angle $\Delta\theta$; it approximates the area of D_j because r is continuous and so does not change much over $[\theta_{j-1}, \theta_j]$.) Summing and passing to the limit,

$$\text{area}(D) = \lim_{n\to\infty} \sum_{j=1}^{n} \frac{1}{2} r(\theta_j^*)^2 \Delta\theta = \frac{1}{2}\int_{\alpha}^{\beta} r(\theta)^2\,\mathrm{d}\theta.$$

Alternatively, this is exactly the double integral $\iint_D 1\,\mathrm{d}A$ in polar coordinates: integrating $f \equiv 1$ over $0 \le r \le r(\theta)$, $\alpha \le \theta \le \beta$ gives

$$\text{area}(D) = \int_{\alpha}^{\beta} \int_0^{r(\theta)} r\,\mathrm{d}r\,\mathrm{d}\theta = \int_{\alpha}^{\beta} \frac{r(\theta)^2}{2}\,\mathrm{d}\theta.$$

■

Worked example: the cardioid

The cardioid $r = 1 + \cos\theta$ $(0 \le \theta \le 2\pi)$ bounds the area

$$A = \frac{1}{2}\int_0^{2\pi} (1+\cos\theta)^2 \,\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi} \Big(1 + 2\cos\theta + \cos^2\theta\Big)\mathrm{d}\theta = \frac{1}{2}\Big(2\pi + 0 + \pi\Big) = \frac{3\pi}{2},$$

where $\displaystyle\int_0^{2\pi} \cos\theta \,\mathrm{d}\theta = 0$ and $\displaystyle\int_0^{2\pi} \cos^2\theta \,\mathrm{d}\theta = \pi$.

3.5 Applications of double integrals

Mass, moments and centre of mass

Let a lamina occupy D with **density** $\rho(x,y) \ge 0$. Then

$$m = \iint_D \rho \,\mathrm{d}A, \qquad M_x = \iint_D y\rho \,\mathrm{d}A, \qquad M_y = \iint_D x\rho \,\mathrm{d}A,$$

and the **centre of mass** is $(\bar{x}, \bar{y}) = (M_y/m,\ M_x/m)$. For a **uniform** lamina (ρ constant) the centre of mass is the **centroid**, and the density cancels:

$$\bar{x} = \frac{1}{A}\iint_D x \,\mathrm{d}A, \qquad \bar{y} = \frac{1}{A}\iint_D y \,\mathrm{d}A, \qquad A = \iint_D 1 \,\mathrm{d}A.$$

For a solid E with density $\rho(x,y,z)$ the same formulas hold with $\mathrm{d}V$ and a triple integral.

Justification. Split D into small pieces of area ΔA and mass $\Delta m \approx \rho\,\Delta A$. The first moment about the x-axis is $\sum y\,\Delta m \to \displaystyle\iint_D y\rho \,\mathrm{d}A$, and the centre of mass of a finite system of masses is $(\sum x_i m_i / \sum m_i,\ \sum y_i m_i / \sum m_i)$. Passing to the limit gives the formulas. ■

Moments of inertia

The **moments of inertia** of the lamina about the axes and about the origin are

$$I_x = \iint_D y^2\rho \,\mathrm{d}A, \qquad I_y = \iint_D x^2\rho \,\mathrm{d}A, \qquad I_0 = I_x + I_y = \iint_D (x^2+y^2)\rho \,\mathrm{d}A.$$

The **radius of gyration** about the origin is $\bar{\rho} = \sqrt{I_0/m}$.

Worked example: centroid of a semicircular lamina

Find the centroid of the half disk $x^2 + y^2 \le a^2$, $y \ge 0$.

By symmetry $\bar{x} = 0$. Also $A = \dfrac{1}{2}\pi a^2$ and

$$\iint_D y \,\mathrm{d}A = \int_0^{\pi}\int_0^{a} r\sin\theta \cdot r \,\mathrm{d}r\,\mathrm{d}\theta = \frac{a^3}{3}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta = \frac{2a^3}{3}.$$

Hence

$$\bar{y} = \frac{2a^3/3}{\pi a^2/2} = \frac{4a}{3\pi}.$$

Probability

A **joint density function** of two random variables is a function $f:\mathbb{R}^2\to[0,\infty)$ with

$$\iint_{\mathbb{R}^2} f(x,y)\,\mathrm{d}A = 1.$$

The probability that (X,Y) lies in D is $P((X,Y)\in D)=\displaystyle\iint_D f\,\mathrm{d}A$. The **marginal densities** are

$$f_1(x)=\int_{-\infty}^{\infty} f(x,y)\,\mathrm{d}y, \qquad f_2(y)=\int_{-\infty}^{\infty} f(x,y)\,\mathrm{d}x.$$

Worked example

$f(x,y)=c(1-x^2-y^2)$ for $x^2+y^2\le 1$, and 0 otherwise. Then

$$1=\int_0^{2\pi}\int_0^1 c(1-r^2)r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi c\Big[\frac{r^2}{2}-\frac{r^4}{4}\Big]_0^1 = \frac{\pi c}{2},$$

so $c=2/\pi$.

3.6 Surface area

3.6.1 The formula

Area of a graph

If f has continuous partial derivatives on D, the area of the surface $z=f(x,y)$, $(x,y)\in D$, is

$$\boxed{A(S)=\iint_D \sqrt{1+\big[f_x(x,y)\big]^2+\big[f_y(x,y)\big]^2}\,\mathrm{d}A}$$

For a parametric surface $\vec{r}(u,v)$ over a parameter domain D, the area is

$$A(S)=\iint_D |\vec{r}_u\times\vec{r}_v|\,\mathrm{d}u\,\mathrm{d}v.$$

Derivation. Partition D into small rectangles R_{ij} of side lengths $\Delta x,\Delta y$. The part of S above R_{ij} is nearly the parallelogram spanned by the two tangent vectors at the lower-left corner,

$$\vec{r}_x\,\Delta x=\langle 1,0,f_x\rangle\Delta x, \qquad \vec{r}_y\,\Delta y=\langle 0,1,f_y\rangle\Delta y.$$

Its area is

$$|\vec{r}_x\times\vec{r}_y|\,\Delta x\,\Delta y=\big|\langle -f_x,-f_y,1\rangle\big|\,\Delta A=\sqrt{1+f_x^2+f_y^2}\,\Delta A.$$

Summing over the partition and passing to the limit gives the integral. ∎

Worked example: part of a plane

Find the area of the part of the plane $z=2x+3y$ lying above the rectangle $[0,1]\times[0,2]$.

$f_x=2$, $f_y=3$, so $\sqrt{1+4+9}=\sqrt{14}$ and

$$A=\iint_D \sqrt{14}\,\mathrm{d}A=\sqrt{14}\cdot 2=2\sqrt{14}.$$

Worked example: a paraboloid

Find the area of the part of $z=x^2+y^2$ below $z=9$.

Here D is the disk $x^2+y^2\le 9$, $f_x=2x$, $f_y=2y$, so

$$A=\iint_D \sqrt{1+4x^2+4y^2}\,\mathrm{d}A=\int_0^{2\pi}\int_0^3 \sqrt{1+4r^2}\,r\,\mathrm{d}r\,\mathrm{d}\theta.$$

With $u=1+4r^2$, $\mathrm{d}u=8r\,\mathrm{d}r$:

$$A=2\pi\cdot\frac{1}{8}\int_1^{37}u^{1/2}\,\mathrm{d}u=\frac{\pi}{4}\cdot\frac{2}{3}\Big[u^{3/2}\Big]_1^{37}=\frac{\pi}{6}\big(37\sqrt{37}-1\big).$$

3.7 Triple integrals

3.7.1 Definition and Fubini's theorem

Triple integral

For a bounded function f on a box $B=[a,b]\times[c,d]\times[e,g]$, partition B into lmn small boxes of volume $\Delta V=\Delta x\,\Delta y\,\Delta z$, choose sample points and form

$$\sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x^*_{ijk},y^*_{ijk},z^*_{ijk})\,\Delta V.$$

If the limit as $l,m,n\to\infty$ exists independently of the choices, it is the **triple integral**

$$\boxed{\iiint_B f(x,y,z)\,\mathrm{d}V=\lim \sum\sum\sum f(x^*_{ijk},y^*_{ijk},z^*_{ijk})\,\Delta V}$$

Continuous functions are integrable, and for such f Fubini's theorem allows the three integrations to be performed in any order:

$$\iiint_B f\,\mathrm{d}V=\int_a^b\int_c^d\int_e^g f(x,y,z)\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x.$$

Simple regions

A **type 1** region is

$$E=\{(x,y,z):(x,y)\in D,\ u_1(x,y)\le z\le u_2(x,y)\},$$

and then

$$\iiint_E f\,\mathrm{d}V=\iint_D\left[\int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,\mathrm{d}z\right]\mathrm{d}A.$$

Type 2 and type 3 regions are defined analogously by projecting onto the xz- and yz-planes.

Worked example: volume of a tetrahedron

Find the volume of the tetrahedron bounded by the planes $x=0$, $y=0$, $z=0$ and $x+y+z=1$.

The projection onto the xy-plane is the triangle D with vertices $(0,0),(1,0),(0,1)$, and $0\le z\le 1-x-y$. Hence

$$V=\iint_D(1-x-y)\,\mathrm{d}A=\int_0^1\int_0^{1-x}(1-x-y)\,\mathrm{d}y\,\mathrm{d}x=\int_0^1\frac{(1-x)^2}{2}\,\mathrm{d}x=\frac{1}{6}.$$

Worked example: a triple integral

Evaluate $\iiint_E z\,dV$ where E is bounded by $z=0$, $z=y$ and $y=1-x^2$.

Project onto the xy-plane: $D=\{(x,y):-1\le x\le 1,\ 0\le y\le 1-x^2\}$, and $0\le z\le y$. Then

$$\iiint_E z\,dV=\int_{-1}^{1}\int_{0}^{1-x^2}\int_{0}^{y} z\,dz\,dy\,dx=\int_{-1}^{1}\int_{0}^{1-x^2}\frac{y^2}{2}\,dy\,dx=\int_{-1}^{1}\frac{(1-x^2)^3}{6}\,dx.$$

Expanding $(1-x^2)^3=1-3x^2+3x^4-x^6$ and integrating,

$$\frac{1}{6}\Big[2-2+\frac{6}{5}-\frac{2}{7}\Big]=\frac{1}{6}\cdot\frac{32}{35}=\frac{16}{105}.$$

3.8 Triple integrals in cylindrical and spherical coordinates

3.8.1 Cylindrical coordinates

Cylindrical coordinates

(r,θ,z) with

$$x=r\cos\theta,\qquad y=r\sin\theta,\qquad z=z,$$

and

$$\boxed{dV=r\,dz\,dr\,d\theta}$$

Derivation. A small cylindrical box has base a polar rectangle of area $r\,dr\,d\theta$ (Section 15.4) and height dz. Since the base and the height are perpendicular, $dV=(\text{base area})\cdot\text{height}=r\,dr\,d\theta\,dz$. ■

Worked example

Find the volume of the solid bounded by the paraboloid $z=x^2+y^2$ and the plane $z=4$.

In cylindrical coordinates the solid is $0\le\theta\le 2\pi$, $0\le r\le 2$, $r^2\le z\le 4$:

$$V=\int_0^{2\pi}\int_0^{2}\int_{r^2}^{4} r\,dz\,dr\,d\theta=2\pi\int_0^2 r(4-r^2)\,dr=2\pi\Big[2r^2-\frac{r^4}{4}\Big]_0^2=2\pi(8-4)=8\pi.$$

3.8.2 Spherical coordinates

Spherical coordinates

(ρ,θ,ϕ) with

$$x=\rho\sin\phi\cos\theta,\qquad y=\rho\sin\phi\sin\theta,\qquad z=\rho\cos\phi,$$

where $\rho\ge 0$ is the distance to the origin, $\phi\in[0,\pi]$ is the angle from the positive z-axis (the *polar angle*) and θ is the usual azimuthal angle. Then

$$\boxed{dV=\rho^2\sin\phi\,d\rho\,d\phi\,d\theta}$$

Derivation. Consider the *spherical wedge* $\rho_1\le\rho\le\rho_2$, $\theta_1\le\theta\le\theta_2$, $\phi_1\le\phi\le\phi_2$. Its volume is the difference of two cone-sectors:

$$\Delta V=\frac{\Delta\theta}{2\pi}\cdot\frac{2\pi}{3}\big(\rho_2^3-\rho_1^3\big)\big(\cos\phi_1-\cos\phi_2\big)=\frac{\Delta\theta}{3}\big(\rho_2^3-\rho_1^3\big)\big(\cos\phi_1-\cos\phi_2\big).$$

By the mean value theorem applied to $\rho \mapsto \rho^3$ and to $\phi \mapsto -\cos\phi$,

$$\rho_2^3 - \rho_1^3 = 3(\rho^*)^2 \Delta\rho, \qquad \cos\phi_1 - \cos\phi_2 = \sin(\phi^*)\,\Delta\phi,$$

so $\Delta V = (\rho^*)^2 \sin\phi^* \,\Delta\rho\,\Delta\phi\,\Delta\theta$. For small increments the starred values coincide with ρ, ϕ, giving $\mathrm{d}V = \rho^2 \sin\phi \,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta$. ∎

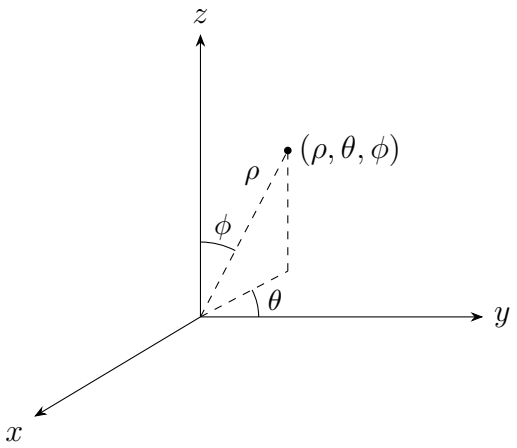


Figure 12: Spherical coordinates: ρ distance, ϕ from the z-axis, θ in the xy-plane.

Worked example: volume of a sphere

The ball of radius a is $0 \le \rho \le a$, $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$, so

$$V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho^2 \sin\phi \,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta = \frac{a^3}{3}\cdot 2 \cdot 2\pi = \frac{4}{3}\pi a^3.$$

Worked example: a cone

Find the volume of the solid above the cone $z = \sqrt{x^2+y^2}$ and inside the sphere $x^2+y^2+z^2 = 1$.

The cone is $\phi = \pi/4$ and the sphere is $\rho = 1$; the solid is $0 \le \rho \le 1$, $0 \le \phi \le \pi/4$. Hence

$$V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{1} \rho^2 \sin\phi \,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta = \frac{1}{3}\cdot 2\pi \cdot \left(1 - \frac{\sqrt{2}}{2}\right) = \frac{\pi}{3}\big(2 - \sqrt{2}\big).$$

3.9 Change of variables

3.9.1 Transformations and the Jacobian

Transformation

A **transformation** is a map $T : D \subseteq \mathbb{R}^2 \to \mathbb{R}^2$, $T(u,v) = (x(u,v), y(u,v))$. Its image of a region S is denoted $R = T(S)$. T is **one-to-one** if distinct points of S have distinct images; then the inverse T^{-1} exists on R.

Jacobian

The **Jacobian** of T is

$$\boxed{\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix} = x_u y_v - x_v y_u}$$

The notation is deliberately suggestive of a ratio of differentials, and indeed

$$\mathrm{d}x\,\mathrm{d}y = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| \mathrm{d}u\,\mathrm{d}v.$$

3.9.2 The change of variables formula

Change of variables in a double integral

Let T be a C^1 transformation which is one-to-one on the interior of S and whose Jacobian is nonzero there, and let $R = T(S)$. Then for continuous f,

$$\boxed{\iint_R f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint_S f\big(x(u,v),y(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right| \mathrm{d}u\,\mathrm{d}v}$$

In three variables, with $T(u,v,w) = (x,y,z)$,

$$\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w,$$

where the 3×3 Jacobian is the determinant of the matrix of partial derivatives.

Justification for linear maps. First take a linear transformation

$$x = au + bv, \qquad y = cu + dv, \qquad J = \det\begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The image of the unit square is the parallelogram spanned by $\langle a, c\rangle$ and $\langle b, d\rangle$, whose area is the length of their cross product in $\mathbb{R}^3$:

$$|\langle a, c, 0\rangle \times \langle b, d, 0\rangle| = |\langle 0, 0, ad - bc\rangle| = |J|.$$

Hence a region of area $\Delta u\,\Delta v$ maps to a region of area $|J|\,\Delta u\,\Delta v$, and summing gives the formula for linear maps (this also shows why the absolute value appears: orientation may be reversed).

For a general C^1 map, on a small rectangle around (u_0, v_0) the linear approximation is

$$T(u,v) \approx T(u_0,v_0) + DT(u_0,v_0)\begin{pmatrix} u-u_0 \\ v-v_0 \end{pmatrix},$$

and the local area magnification is $|\det DT(u_0,v_0)|$ by the linear case. Summing the contributions, i.e. forming Riemann sums, and passing to the limit gives the formula. ■

Worked example: an elliptical region

Find the area of the ellipse $x^2/a^2 + y^2/b^2 \le 1$ by a change of variables.

Use $x = au$, $y = bv$; then $x^2/a^2 + y^2/b^2 = u^2 + v^2 \le 1$. The Jacobian is

$$\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} = ab,$$

so

$$\text{area} = \iint_{u^2+v^2\le 1} ab\,\mathrm{d}u\,\mathrm{d}v = ab\pi.$$

Worked example: a nonlinear change of variables

Evaluate $\displaystyle\iint_R xy\,\mathrm{d}A$ where R is the region bounded by $y = x$, $y = 3x$, $xy = 1$, $xy = 4$.

Let $u = xy$, $v = y/x$; then $1 \le u \le 4$, $1 \le v \le 3$. Solving, $x = \sqrt{u/v}$ and $y = \sqrt{uv}$, so

$$x_u = \frac{1}{2\sqrt{uv}}, \quad x_v = -\frac{\sqrt{u}}{2v^{3/2}}, \quad y_u = \frac{\sqrt{v}}{2\sqrt{u}}, \quad y_v = \frac{\sqrt{u}}{2\sqrt{v}}.$$

Then

$$\frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{2\sqrt{uv}} \cdot \frac{\sqrt{u}}{2\sqrt{v}} - \Big(-\frac{\sqrt{u}}{2v^{3/2}}\Big)\frac{\sqrt{v}}{2\sqrt{u}} = \frac{1}{4v} + \frac{1}{4v} = \frac{1}{2v}.$$

Hence

$$\iint_R xy \,\mathrm{d}A = \int_1^3 \int_1^4 u \cdot \frac{1}{2v} \,\mathrm{d}u \,\mathrm{d}v = \frac{1}{2} \cdot \frac{15}{2} \cdot \ln 3 = \frac{15}{4} \ln 3.$$

Worked example: polar coordinates via the Jacobian

For $x = r\cos\theta$, $y = r\sin\theta$,

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \det \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r\cos^2\theta + r\sin^2\theta = r,$$

so $\mathrm{d}A = r \,\mathrm{d}r \,\mathrm{d}\theta$, in agreement with Section 15.4.

Worked example: spherical coordinates via the Jacobian

For $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, $z = \rho\cos\phi$, a direct computation gives

$$\frac{\partial(x,y,z)}{\partial(\rho,\theta,\phi)} = \rho^2 \sin\phi.$$

(Compute the 3×3 determinant by expanding along the third column $\langle \cos\phi\cos\theta, \cos\phi\sin\theta, -\sin\phi \rangle$.) Hence $\mathrm{d}V = \rho^2 \sin\phi \,\mathrm{d}\rho \,\mathrm{d}\theta \,\mathrm{d}\phi$ as before.

Checklist for change of variables

- The Jacobian must be *nonzero* on the interior of S (otherwise the transformation collapses area and the formula fails).
- Use the *absolute value* $|\partial(x,y)/\partial(u,v)|$; the orientation may be reversed.
- Sketch both regions: the new region S is often a rectangle, which is the whole point of the substitution.
- In polar coordinates do not forget the factor r; in spherical coordinates the factor is $\rho^2 \sin\phi$.

4 Vector Calculus

This final chapter unifies the differential and integral calculus of vector fields. Its four central results — the fundamental theorem for line integrals, Green's theorem, Stokes' theorem and the divergence theorem — are all generalizations of the fundamental theorem of calculus, and each says that something integrated over a region equals something integrated over its boundary.

4.1 Vector fields

Definition

A **vector field** on a subset $D \subseteq \mathbb{R}^n$ is a map which assigns to each point $P \in D$ a vector $\vec{F}(P)$. In the plane,

$$\vec{F}(x,y) = P(x,y)\vec{i} + Q(x,y)\vec{j} = \langle P(x,y),\, Q(x,y) \rangle,$$

in space,

$$\vec{F}(x,y,z) = P\vec{i} + Q\vec{j} + R\vec{k} = \langle P, Q, R\rangle,$$

where P, Q, R are the **component functions**. The vector field is **continuous** (or C^1, etc.) if each component is continuous (or C^1, etc.).

Examples

(a) $\vec{F}(x,y) = -y\vec{i} + x\vec{j}$: at each point the arrow is perpendicular to the position vector and points counterclockwise; this is the velocity field of a rotation.
(b) $\vec{F}(x,y) = x\vec{i} + y\vec{j}$: the arrows point radially outwards.
(c) **Gradient field.** If f is a scalar function, then

$$\nabla f = \langle f_x, f_y, f_z\rangle$$

is a vector field, called the **gradient field** of f. A vector field $\vec{F}$ is **conservative** if $\vec{F} = \nabla f$ for some **potential function** f.

4.2 Line integrals

4.2.1 Line integrals of scalar functions

Definition with respect to arc length

Let C be a smooth curve in $\mathbb{R}^3$ parametrized by $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$, traversed once, and let f be defined on C. Divide C into n subarcs of lengths Δs_i, choose a sample point (x_i^*, y_i^*, z_i^*) on the i-th subarc and form

$$\sum_{i=1}^{n} f(x_i^*, y_i^*, z_i^*)\,\Delta s_i.$$

If the limit exists independently of the choices, it is the **line integral of f along C with respect to arc length**:

$$\boxed{\int_C f(x,y,z)\,\mathrm{d}s = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*, y_i^*, z_i^*)\,\Delta s_i}$$

Evaluation

If f is continuous and C is C^1, then

$$\boxed{\int_C f\,\mathrm{d}s = \int_a^b f(\vec{r}(t))\,|\vec{r}\,'(t)|\,\mathrm{d}t}$$

In particular, the **arc length** of C is $\displaystyle\int_C \mathrm{d}s = \int_a^b |\vec{r}\,'(t)|\,\mathrm{d}t$. The integral $\displaystyle\int_C f\,\mathrm{d}s$ is unchanged if C is traversed in the opposite direction.

Justification. Recall the definition of arc length from Chapter 12: divide $[a,b]$ into n subintervals $[t_{i-1}, t_i]$ of width Δt_i, with $\max_i \Delta t_i \to 0$ as $n \to \infty$, and let $s = \sum_i |\vec{r}(t_i) - \vec{r}(t_{i-1})|$. We will show that this limit equals $\displaystyle\int_a^b |\vec{r}\,'(t)|\,\mathrm{d}t$.

Since $\vec{r}$ is C^1, for each i the mean value theorem applied componentwise gives sample

points $t_{i,j}^* \in (t_{i-1}, t_i)$, $j = 1, 2, 3$, such that

$$\vec{r}(t_i) - \vec{r}(t_{i-1}) = \langle x'(t_{i,1}^*)\Delta t_i,\; y'(t_{i,2}^*)\Delta t_i,\; z'(t_{i,3}^*)\Delta t_i \rangle.$$

By continuity of $\vec{r}\,'$, all three components x', y', z' change by at most ε over $[t_{i-1}, t_i]$ once $\max_i \Delta t_i$ is small enough, so

$$\big|\,|\vec{r}(t_i) - \vec{r}(t_{i-1})| - |\vec{r}\,'(t_i^*)\Delta t_i|\,\big| < C\varepsilon \Delta t_i$$

for some constant C and any sample point $t_i^* \in [t_{i-1}, t_i]$. Summing over i,

$$\left| \sum_i |\vec{r}(t_i) - \vec{r}(t_{i-1})| - \sum_i |\vec{r}\,'(t_i^*)|\Delta t_i \right| < C\varepsilon \sum_i \Delta t_i = C\varepsilon(b-a).$$

Since ε is arbitrary, the Riemann sum $\displaystyle\sum_i |\vec{r}\,'(t_i^*)|\Delta t_i$ converges to the same limit as the chord-sum, which is exactly $\displaystyle\int_a^b |\vec{r}\,'(t)|\,\mathrm{d}t$.

Now the line integral $\displaystyle\int_C f\,\mathrm{d}s$ is defined as the limit of $\displaystyle\sum_i f(\vec{r}(t_i^*))\Delta s_i$, where $\Delta s_i = |\vec{r}(t_i) - \vec{r}(t_{i-1})|$ and t_i^* is any sample point in $[t_{i-1}, t_i]$. Replacing Δs_i by $|\vec{r}\,'(t_i^*)|\Delta t_i$ introduces an error of at most $f_{\max} \cdot C\varepsilon\Delta t_i$ per term, which vanishes in the limit. Hence

$$\int_C f\,\mathrm{d}s = \int_a^b f(\vec{r}(t))\,|\vec{r}\,'(t)|\,\mathrm{d}t.$$

■

Worked example

Evaluate $\displaystyle\int_C (2 + x^2 y)\,\mathrm{d}s$ where C is the upper half of the unit circle.

Parametrize $\vec{r}(t) = \langle \cos t, \sin t \rangle$, $0 \le t \le \pi$; then $|\vec{r}\,'(t)| = 1$ and

$$\int_C (2 + x^2 y)\,\mathrm{d}s = \int_0^{\pi} \big(2 + \cos^2 t \sin t\big)\mathrm{d}t = 2\pi + \Big[-\frac{\cos^3 t}{3}\Big]_0^{\pi} = 2\pi + \frac{2}{3}.$$

4.2.2 Line integrals of vector fields

Definition and work

Let $\vec{F}$ be a continuous vector field on a smooth curve C parametrized by $\vec{r}(t)$, $a \le t \le b$. The **line integral of $\vec{F}$ along C** is

$$\boxed{\int_C \vec{F} \cdot \mathrm{d}\vec{r} = \int_C \vec{F} \cdot \vec{T}\,\mathrm{d}s = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\,\mathrm{d}t}$$

where $\vec{T} = \vec{r}\,'/|\vec{r}\,'|$ is the unit tangent. In components,

$$\int_C \vec{F} \cdot \mathrm{d}\vec{r} = \int_C P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z.$$

If $\vec{F}$ is a force field, this integral is the **work** done by $\vec{F}$ along C. For a closed curve C we write $\displaystyle\oint_C$ and call the value the **circulation**.

Worked example: work

Find the work done by $\vec{F} = \langle x^2, -xy\rangle$ along the unit circle from $(1,0)$ to $(0,1)$ in the counterclockwise direction.
$\vec{r}(t) = \langle \cos t, \sin t\rangle$, $0 \le t \le \pi/2$, $\vec{r}\,'(t) = \langle -\sin t, \cos t\rangle$. Then

$$W = \int_0^{\pi/2} \big\langle \cos^2 t,\, -\cos t \sin t\big\rangle \cdot \langle -\sin t, \cos t\rangle \,\mathrm{d}t = \int_0^{\pi/2} \big(-\cos^2 t \sin t - \cos^2 t \sin t\big)\mathrm{d}t$$

$$= -2\int_0^{\pi/2} \cos^2 t \sin t \,\mathrm{d}t = -2\Big[-\frac{\cos^3 t}{3}\Big]_0^{\pi/2} = -\frac{2}{3}.$$

Orientation and additivity

Reversing the orientation of C changes the sign:

$$\int_{-C} \vec{F}\cdot \mathrm{d}\vec{r} = -\int_C \vec{F}\cdot \mathrm{d}\vec{r}.$$

If $C = C_1 \cup C_2$ is a union of curves joined end to end, then $\displaystyle\int_C = \int_{C_1} + \int_{C_2}$.

Proof. If C is parametrized by $\vec{r}(t)$, $a \le t \le b$, then $-C$ is parametrized by $\vec{r}(a+b-t)$, whose derivative is $-\vec{r}\,'(a+b-t)$. Hence

$$\int_{-C} \vec{F}\cdot \mathrm{d}\vec{r} = \int_a^b \vec{F}(\vec{r}(a+b-t)) \cdot \big(-\vec{r}\,'(a+b-t)\big)\mathrm{d}t = -\int_b^a \vec{F}(\vec{r}(u))\cdot \vec{r}\,'(u)\,\mathrm{d}u,$$

and reversing the limits of integration gives the claim. ∎

Worked example: three paths

Compute $\displaystyle\int_C \vec{F}\cdot\mathrm{d}\vec{r}$ for $\vec{F} = \langle -xy,\, x^2\rangle$ from $(0,0)$ to $(2,1)$ along (a) the segment $y = x/2$, (b) the parabola $y = x^2/4$, (c) the broken path along the axes.
(a) $\vec{r}(t) = \langle 2t, t\rangle$, $0 \le t \le 1$: $\vec{r}\,' = \langle 2, 1\rangle$ and

$$\vec{F}\cdot\vec{r}\,' = \big\langle -2t^2,\, 4t^2\big\rangle \cdot \langle 2,1\rangle = -4t^2 + 4t^2 = 0,$$

so the integral is 0.
(b) $\vec{r}(t) = \langle 2t, t^2\rangle$: $\vec{r}\,' = \langle 2, 2t\rangle$ and

$$\vec{F}\cdot\vec{r}\,' = \big\langle -2t^3,\, 4t^2\big\rangle \cdot \langle 2, 2t\rangle = -4t^3 + 8t^3 = 4t^3,$$

so the integral is $\displaystyle\int_0^1 4t^3\,\mathrm{d}t = 1$.
(c) Along $y = 0$, $0 \le x \le 2$: $\vec{F} = \langle 0, x^2\rangle$, $\mathrm{d}\vec{r} = \langle \mathrm{d}x, 0\rangle$, contribution 0. Along $x = 2$, $0 \le y \le 1$: $\vec{F} = \langle -2y, 4\rangle$, $\mathrm{d}\vec{r} = \langle 0, \mathrm{d}y\rangle$, contribution $\displaystyle\int_0^1 4\,\mathrm{d}y = 4$. Total 4.
The three answers differ, so this field is not conservative.

4.3 The fundamental theorem for line integrals

Fundamental theorem for line integrals

If C is a smooth curve parametrized by $\vec{r}(t)$, $a \le t \le b$, and f is a differentiable

function whose gradient ∇f is continuous on C, then

$$\boxed{\int_C \nabla f \cdot \mathrm{d}\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))}$$

Proof. Let $g(t) = f(\vec{r}(t))$. By the chain rule,

$$g'(t) = \nabla f(\vec{r}(t)) \cdot \vec{r}\,'(t),$$

so by the fundamental theorem of calculus,

$$\int_C \nabla f \cdot \mathrm{d}\vec{r} = \int_a^b \nabla f(\vec{r}(t)) \cdot \vec{r}\,'(t)\,\mathrm{d}t = \int_a^b g'(t)\,\mathrm{d}t = g(b) - g(a) = f(\vec{r}(b)) - f(\vec{r}(a)).$$

■

Worked example

For $f(x, y) = x^2 y - y^3$, $\nabla f = \langle 2xy,\, x^2 - 3y^2 \rangle$. Along any curve from $(1, 0)$ to $(2, 1)$,

$$\int_C \nabla f \cdot \mathrm{d}\vec{r} = f(2, 1) - f(1, 0) = 3 - 0 = 3.$$

4.3.1 Path independence and conservative fields

Path independence

$\int_C \vec{F} \cdot \mathrm{d}\vec{r}$ is **independent of path** in a domain D if its value depends only on the endpoints, for every pair of points and every piecewise-smooth curve in D joining them. Equivalently,

$$\oint_C \vec{F} \cdot \mathrm{d}\vec{r} = 0$$

for every closed curve C in D.

Equivalence of the two conditions

If the integral is path independent and C is closed with base point A, write $C = C_1 \cup (-C_2)$ for two paths from A back to A. Then

$$\oint_C \vec{F} \cdot \mathrm{d}\vec{r} = \int_{C_1} \vec{F} \cdot \mathrm{d}\vec{r} - \int_{C_2} \vec{F} \cdot \mathrm{d}\vec{r} = 0.$$

Conversely, if all closed integrals vanish and C_1, C_2 join A to B, then $C_1 \cup (-C_2)$ is closed, so $\int_{C_1} - \int_{C_2} = 0$.

Conservative fields are path independent

If $\vec{F} = \nabla f$ on a domain D, then $\int_C \vec{F} \cdot \mathrm{d}\vec{r}$ is independent of path in D.

Proof. This is immediate from the fundamental theorem: the value is $f(B) - f(A)$, which depends only on the endpoints. ■

Path independent fields are conservative

If $\vec{F} = \langle P, Q \rangle$ is continuous and $\int_C \vec{F} \cdot \mathrm{d}\vec{r}$ is independent of path in an *open* domain D,

then $\vec{F}$ is conservative on D: there is f with $\nabla f = \vec{F}$.

Proof. Fix $A = (a, b) \in D$ and define, for $(x, y) \in D$,

$$f(x, y) = \int_{C_{(x,y)}} \vec{F} \cdot \mathrm{d}\vec{r},$$

where $C_{(x,y)}$ is any path in D from A to (x, y); the definition is unambiguous by hypothesis. We show $f_x = P$.

Take (x, y) and a small increment h so that the segment from (x, y) to $(x + h, y)$ lies in D; denote it by L. Then

$$f(x + h, y) - f(x, y) = \int_L \vec{F} \cdot \mathrm{d}\vec{r} = \int_L P\,\mathrm{d}x + Q\,\mathrm{d}y.$$

On L we have $\mathrm{d}y = 0$, so

$$f(x + h, y) - f(x, y) = \int_0^h P(x + t, y)\,\mathrm{d}t.$$

Dividing by h and letting $h \to 0$, the mean value theorem for integrals (or continuity of P) gives

$$f_x(x, y) = \lim_{h \to 0} \frac{1}{h} \int_0^h P(x + t, y)\,\mathrm{d}t = P(x, y).$$

The same argument with a vertical segment gives $f_y = Q$. Hence $\nabla f = \langle P, Q \rangle = \vec{F}$. ■

The curl test

Let $\vec{F} = \langle P, Q \rangle$ be C^1 on a domain D. If $\vec{F}$ is conservative, then

$$\boxed{\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}}$$

throughout D. Conversely, if D is **simply connected** (every closed curve in D encloses only points of D; informally, D has no holes) and $P_y = Q_x$ on D, then $\vec{F}$ is conservative on D.

Proof of the necessity. If $\vec{F} = \nabla f$, then $P = f_x$ and $Q = f_y$. Since P, Q are C^1, Clairaut's theorem gives

$$P_y = f_{xy} = f_{yx} = Q_x.$$

■

Sufficiency (complete proof). Let C be any piecewise-smooth simple closed curve in D. Since D is *simply connected*, the bounded region R enclosed by C is contained entirely in D (there are no holes that could force R to contain a point not in D). Apply Green's theorem to R:

$$\oint_C \vec{F} \cdot \mathrm{d}\vec{r} = \oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathrm{d}A = \iint_R 0\,\mathrm{d}A = 0,$$

where we used the hypothesis $\partial P/\partial y = \partial Q/\partial x$ on all of D. Therefore *every simple closed curve* in D has zero circulation.

Now let C_1 and C_2 be any two piecewise-smooth paths in D with the same endpoints A and B. Then $C_1 \cup (-C_2)$ is a closed curve starting and ending at A (possibly with self-intersections, but it can be homotoped to a simple closed curve in D using the simply-connectedness, or decomposed into a collection of simple closed curves all lying in D). By the same Green-theorem argument,

$$0 = \oint_{C_1 \cup (-C_2)} \vec{F} \cdot \mathrm{d}\vec{r} = \int_{C_1} \vec{F} \cdot \mathrm{d}\vec{r} + \int_{-C_2} \vec{F} \cdot \mathrm{d}\vec{r} = \int_{C_1} \vec{F} \cdot \mathrm{d}\vec{r} - \int_{C_2} \vec{F} \cdot \mathrm{d}\vec{r}.$$

Hence $\int_{C_1} = \int_{C_2}$: the integral is path independent. By the preceding theorem (path independent $\Rightarrow$ conservative), $\vec{F}$ is conservative on D. ■

Why simply connected matters.

Consider $\vec{F}(x,y) = \langle -y/(x^2+y^2),\, x/(x^2+y^2)\rangle$ on $D = \mathbb{R}^2 \setminus \{(0,0)\}$ (the punctured plane). A direct computation gives $\partial P/\partial y = \partial Q/\partial x$ everywhere on D, yet $\oint_{x^2+y^2=1} \vec{F}\cdot \mathrm{d}\vec{r} = 2\pi \neq 0$. The field is *not* conservative on the punctured plane. The hypothesis of simple connectedness fails precisely because the unit circle encloses the origin, which is not in D.

The hypothesis "simply connected" cannot be dropped

On $D = \mathbb{R}^2 \setminus \{(0,0)\}$ take

$$\vec{F} = \frac{-y\,\vec{i} + x\,\vec{j}}{x^2+y^2} = \Big\langle \frac{-y}{x^2+y^2},\, \frac{x}{x^2+y^2} \Big\rangle.$$

A direct computation gives $P_y = Q_x$ on D. But along the unit circle,

$$\oint_C \vec{F}\cdot \mathrm{d}\vec{r} = \int_0^{2\pi} \big(-\sin t, -\cos t\big)\cdot\langle -\sin t, \cos t\rangle \,\mathrm{d}t = \int_0^{2\pi} 1\,\mathrm{d}t = 2\pi \neq 0.$$

The domain has a hole, so the field is not conservative there.

Finding a potential function

Show that $\vec{F} = \langle 2xy,\, x^2 - 3y^2\rangle$ is conservative and find f.

$P_y = 2x = Q_x$ and the domain is $\mathbb{R}^2$, which is simply connected, so a potential exists. Integrate $f_x = 2xy$ in x:

$$f(x,y) = x^2y + g(y).$$

Then $f_y = x^2 + g'(y)$ must equal $x^2 - 3y^2$, so $g'(y) = -3y^2$ and $g(y) = -y^3 + C$. Hence

$$f(x,y) = x^2y - y^3 + C.$$

A three-variable example

Show that $\vec{F} = \langle y^2,\, 2xy + e^{3z},\, 3ye^{3z}\rangle$ is conservative and find f.

The domain is $\mathbb{R}^3$, which is simply connected. Integrate $f_x = y^2$:

$$f = xy^2 + g(y,z).$$

Then $f_y = 2xy + g_y = 2xy + e^{3z}$, so $g_y = e^{3z}$ and $g = ye^{3z} + h(z)$. Finally

$$f_z = 3ye^{3z} + h'(z) = 3ye^{3z} \Longrightarrow h'(z) = 0.$$

Hence $f(x,y,z) = xy^2 + ye^{3z} + C$.

4.4 Green's theorem

4.4.1 Statement

Green's theorem

Let C be a positively oriented (counterclockwise), piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P, Q have continuous

partial derivatives on an open region containing D, then

$$\boxed{\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A}$$

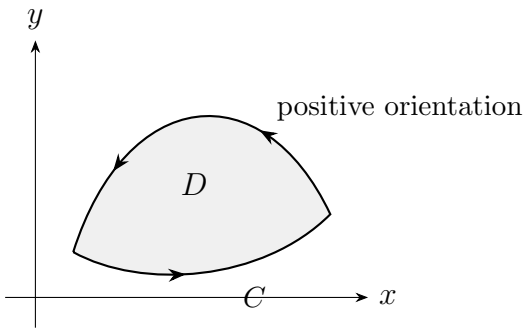


Figure 13: Green's theorem: the boundary is traversed counterclockwise, keeping D on the left.

Consequences

$$\text{area}(D) = \oint_C x\,\mathrm{d}y = -\oint_C y\,\mathrm{d}x = \frac{1}{2}\oint_C \big(x\,\mathrm{d}y - y\,\mathrm{d}x\big).$$

Proof. Take $(P, Q) = (0, x)$: then $Q_x - P_y = 1$ and Green's theorem gives $\oint_C x\,\mathrm{d}y = \iint_D 1\,\mathrm{d}A = \text{area}(D)$. Take $(P, Q) = (-y, 0)$ for the second formula. Adding the two formulas and dividing by 2 gives the third. ■

4.4.2 Proof for regions of both types

Proof of the first half

Suppose D is simultaneously of type I and type II. Then

$$\oint_C P\,\mathrm{d}x = -\iint_D \frac{\partial P}{\partial y}\,\mathrm{d}A.$$

Proof. Write $D = \{(x, y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$ and split C into the lower curve C_1 (from left to right) and the upper curve C_2 (from right to left); the vertical sides contribute nothing to $\oint P\,\mathrm{d}x$ because $\mathrm{d}x = 0$ there. Then

$$\oint_C P\,\mathrm{d}x = \int_{C_1} P\,\mathrm{d}x + \int_{C_2} P\,\mathrm{d}x = \int_a^b P(x, g_1(x))\,\mathrm{d}x - \int_a^b P(x, g_2(x))\,\mathrm{d}x.$$

On the other hand, by Fubini's theorem,

$$\iint_D \frac{\partial P}{\partial y}\,\mathrm{d}A = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}\,\mathrm{d}y\right]\mathrm{d}x = \int_a^b \Big[P(x, g_2(x)) - P(x, g_1(x))\Big]\mathrm{d}x.$$

Comparing the two expressions gives the identity. ■

Proof of the second half. Write $D = \{(x, y) : c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$ and split C into the left curve D_1 (from bottom to top) and the right curve D_2 (from top to

bottom); the horizontal sides contribute nothing to $\oint Q\,\mathrm{d}y$. Then

$$\oint_C Q\,\mathrm{d}y = \int_c^d Q(h_2(y),y)\,\mathrm{d}y - \int_c^d Q(h_1(y),y)\,\mathrm{d}y = \int_c^d \Big[Q(h_2(y),y) - Q(h_1(y),y)\Big]\mathrm{d}y,$$

and by Fubini,

$$\iint_D \frac{\partial Q}{\partial x}\,\mathrm{d}A = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x}\,\mathrm{d}x\right]\mathrm{d}y = \int_c^d \Big[Q(h_2(y),y) - Q(h_1(y),y)\Big]\mathrm{d}y.$$

Adding the two halves gives Green's theorem. For a general region, cut it into finitely many pieces of this type; the integrals along the cuts cancel because each cut is traversed twice in opposite directions. ■

4.4.3 Extended versions: regions with holes

Green's theorem for a region with holes

Suppose ∂D consists of finitely many simple closed curves $C_1,\dots,C_n$, each oriented so that D lies on the left as it is traversed (the outer boundary counterclockwise, the inner boundaries clockwise). Then

$$\boxed{\sum_{k=1}^{n} \oint_{C_k} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\mathrm{d}A}$$

Proof for an annulus. Let D be the region between an outer curve C_1 and an inner curve C_2, and cut D along a smooth curve Γ joining the two boundaries. The cut turns D into a simply connected region D' whose boundary consists of C_1, C_2 and two copies of Γ. Applying Green's theorem to D',

$$\oint_{\partial D'} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_{D'} \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\mathrm{d}A = \iint_D (\cdots)\,\mathrm{d}A,$$

because D' differs from D only by a set of area zero. The two copies of Γ are traversed in opposite directions, so their contributions cancel, and the remaining boundary is C_1 (counterclockwise) together with C_2 (clockwise). Repeating the cut for each hole gives the general statement. ■

Worked example: an annulus

For $\vec{F}$ with $Q_x - P_y = 1$ and D the annulus $1 \le x^2+y^2 \le 4$, the boundary consists of the circle C_1 of radius 2 (counterclockwise) and C_2 of radius 1 (clockwise). Hence

$$\oint_{C_1} \vec{F}\cdot\mathrm{d}\vec{r} + \oint_{C_2} \vec{F}\cdot\mathrm{d}\vec{r} = \iint_D 1\,\mathrm{d}A = \pi(4-1) = 3\pi.$$

For $\vec{F} = \tfrac{1}{2}\langle -y, x\rangle$ the left-hand side is $\pi\cdot 4 - \pi\cdot 1 = 3\pi$, as predicted.

Worked example

Evaluate $\oint_C (3y - e^{\sin x})\,\mathrm{d}x + \big(7x + \sqrt{y^4+1}\big)\mathrm{d}y$ where C is the circle $x^2+y^2=9$ traversed counterclockwise.

The direct integral is intractable, but Green's theorem applies with $P = 3y - e^{\sin x}$, $Q = 7x + \sqrt{y^4+1}$:

$$Q_x - P_y = 7 - 3 = 4,$$

so

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D 4\,\mathrm{d}A = 4 \cdot \pi \cdot 3^2 = 36\pi.$$

Worked example: area of an ellipse

The ellipse $x = a\cos t$, $y = b\sin t$ $(0 \le t \le 2\pi)$ bounds the area

$$A = \frac{1}{2}\oint_C (x\,\mathrm{d}y - y\,\mathrm{d}x) = \frac{1}{2}\int_0^{2\pi} \big(a\cos t \cdot b\cos t + b\sin t \cdot a\sin t\big)\mathrm{d}t = \frac{ab}{2}\int_0^{2\pi} \mathrm{d}t = \pi ab.$$

4.4.4 Limitations of Green's theorem and winding numbers

The field $\vec{F}(x,y) = \dfrac{-y}{x^2+y^2}\vec{i} + \dfrac{x}{x^2+y^2}\vec{j}$ on $\mathbb{R}^2 \setminus \{(0,0)\}$ has $\operatorname{curl}\vec{F} = Q_x - P_y = 0$ everywhere in its domain, yet its circulation around the unit circle is $2\pi \neq 0$, so Green's theorem does not give the value one would naively expect. The reason is that the domain is not simply connected: it has a hole at the origin. To repair the computation, surround the origin by a small circle Γ_ϵ and cut the annulus by two segments L_1, L_2, obtaining two simply connected regions R_1, R_2 to which Green's theorem applies:

$$\oint_{C_1} \vec{F}\cdot\mathrm{d}\vec{r} + \oint_{L_2} \vec{F}\cdot\mathrm{d}\vec{r} - \oint_{\Gamma_{\epsilon_1}} \vec{F}\cdot\mathrm{d}\vec{r} - \oint_{L_1} \vec{F}\cdot\mathrm{d}\vec{r} = 0,$$

$$\oint_{C_2} \vec{F}\cdot\mathrm{d}\vec{r} - \oint_{L_1} \vec{F}\cdot\mathrm{d}\vec{r} - \oint_{\Gamma_{\epsilon_2}} \vec{F}\cdot\mathrm{d}\vec{r} - \oint_{L_2} \vec{F}\cdot\mathrm{d}\vec{r} = 0.$$

Adding the two equations, the integrals along L_1, L_2 cancel and $\Gamma_{\epsilon_1} \cup \Gamma_{\epsilon_2} = \Gamma_\epsilon$, so

$$\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = \oint_{C_1} \vec{F}\cdot\mathrm{d}\vec{r} + \oint_{C_2} \vec{F}\cdot\mathrm{d}\vec{r} = \oint_{\Gamma_\epsilon} \vec{F}\cdot\mathrm{d}\vec{r} = 2\pi.$$

On a circle of radius ϵ, $\vec{r}(t) = \langle \epsilon\cos t, \epsilon\sin t\rangle$, and a direct computation gives $\displaystyle\oint_{\Gamma_\epsilon} \vec{F}\cdot\mathrm{d}\vec{r} = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\,\mathrm{d}t = 2\pi$, independent of ϵ.

Winding number

The **winding number** of a closed curve C around a point P is the integer giving the total number of times C winds around P counterclockwise (negative if clockwise):

$$n = \frac{1}{2\pi}\oint_C \mathrm{d}\theta \qquad \Longleftrightarrow \qquad \oint_C \vec{F}\cdot\mathrm{d}\vec{r} = 2\pi n.$$

Thus $\dfrac{1}{2\pi}\displaystyle\oint_C \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2+y^2}$ is the winding number of C around the origin.

4.4.5 Self-intersecting curves

A closed curve C that self-intersects can be decomposed into n simple closed curves $C_1, \ldots, C_n$; applying Green's theorem to each piece and adding gives

$$\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = \sum_{i=1}^{n} \oint_{C_i} \vec{F}\cdot\mathrm{d}\vec{r}.$$

For the field above, if C makes k loops around the origin then $\displaystyle\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = 2\pi k$; that is, the winding number of C about the origin is k.

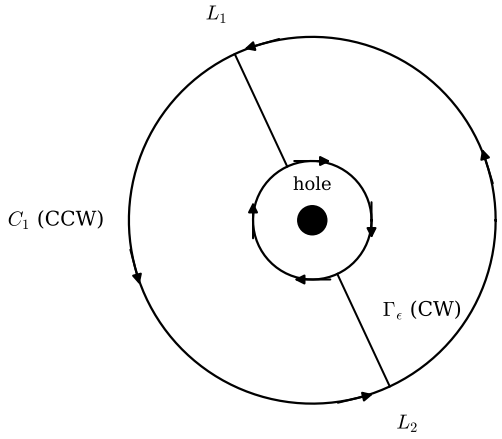


Figure 14: Green's theorem fails on a domain with a hole. Cutting the region and inserting a small clockwise circle Γ_ϵ isolates the singularity at the origin; the integrals along the connecting segments L_1, L_2 cancel.

4.5 Curl and divergence

Definitions

For a vector field $\vec{F} = \langle P, Q, R\rangle$ with C^1 components,

$$\boxed{\operatorname{curl}\vec{F} = \nabla \times \vec{F} = \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z},\ \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x},\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle}$$

$$\boxed{\operatorname{div}\vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}}$$

Curl is a vector field; divergence is a scalar function. For a plane field $\vec{F} = \langle P, Q, 0\rangle$,

$$\operatorname{curl}\vec{F} = \left\langle 0,\ 0,\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle,$$

so Green's theorem may be written $\displaystyle\oint_C \vec{F}\cdot \mathrm{d}\vec{r} = \iint_D (\operatorname{curl}\vec{F})\cdot\vec{k}\,\mathrm{d}A.$

Identities

For a C^2 scalar function f and a C^2 vector field $\vec{F}$,

$$\boxed{\operatorname{curl}(\nabla f) = \vec{0}, \qquad \operatorname{div}(\operatorname{curl}\vec{F}) = 0}$$

Proof. The first component of $\operatorname{curl}(\nabla f)$ is

$$\frac{\partial}{\partial y}(f_z) - \frac{\partial}{\partial z}(f_y) = f_{zy} - f_{yz} = 0$$

by Clairaut's theorem; the other components are similar. For the second identity,

$\operatorname{div}(\operatorname{curl}\vec{F}) = (R_y - Q_z)_x + (P_z - R_x)_y + (Q_x - P_y)_z = R_{yx} - Q_{zx} + P_{zy} - R_{xy} + Q_{xz} - P_{yz} = 0,$

again by equality of mixed partials. ∎

Curl and conservative fields

If $\vec{F}$ is a C^1 vector field on a simply connected domain, then

$$\vec{F} \text{ is conservative } \iff \operatorname{curl}\vec{F} = \vec{0}.$$

Proof. If $\vec{F} = \nabla f$ then $\operatorname{curl}\vec{F} = \operatorname{curl}(\nabla f) = \vec{0}$. Conversely, if $\operatorname{curl}\vec{F} = \vec{0}$, then in particular $P_y = Q_x$, $Q_z = R_y$ and $R_x = P_z$; using the argument of the previous section component by component (or Stokes' theorem in the next section), all closed line integrals vanish, so a potential exists. ∎

Worked example

For $\vec{F} = \langle xz,\, xyz,\, -y\rangle$:

$$\operatorname{curl}\vec{F} = \Big\langle \frac{\partial(-y)}{\partial y} - \frac{\partial(xyz)}{\partial z},\ \frac{\partial(xz)}{\partial z} - \frac{\partial(-y)}{\partial x},\ \frac{\partial(xyz)}{\partial x} - \frac{\partial(xz)}{\partial y}\Big\rangle = \langle -1 - x,\ x,\ yz\rangle,$$

and $\operatorname{div}\vec{F} = z + xz$.

4.6 Parametric surfaces and surface integrals

4.6.1 Parametric surfaces

Parametrization

A **parametric surface** is a map $\vec{r}\colon D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$,

$$\vec{r}(u,v) = x(u,v)\vec{i} + y(u,v)\vec{j} + z(u,v)\vec{k}.$$

If $\vec{r}_u \times \vec{r}_v \neq \vec{0}$, the surface is **smooth**; the vector $\vec{r}_u \times \vec{r}_v$ is normal to the surface and the tangent plane at $\vec{r}(u_0, v_0)$ is spanned by $\vec{r}_u, \vec{r}_v$.

Graphs and standard parametrizations

- A graph $z = g(x,y)$ has the parametrization $\vec{r}(x,y) = \langle x, y, g(x,y)\rangle$, with
$$\vec{r}_x \times \vec{r}_y = \langle -g_x, -g_y, 1\rangle.$$
- A sphere of radius a: $\vec{r}(\phi,\theta) = \langle a\sin\phi\cos\theta,\ a\sin\phi\sin\theta,\ a\cos\phi\rangle$.
- A cylinder $x^2 + y^2 = a^2$: $\vec{r}(\theta, z) = \langle a\cos\theta, a\sin\theta, z\rangle$.

Surface area

$$\boxed{A(S) = \iint_D |\vec{r}_u \times \vec{r}_v|\,\mathrm{d}u\,\mathrm{d}v}$$

For a graph $z = g(x,y)$ over D this becomes

$$A(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2}\,\mathrm{d}A.$$

4.6.2 Surface integrals of scalar functions

Definition

Let S be a smooth parametric surface and f continuous on S. Then

$$\boxed{\iint_S f(x,y,z)\,\mathrm{d}S = \iint_D f(\vec{r}(u,v))\,|\vec{r}_u \times \vec{r}_v|\,\mathrm{d}u\,\mathrm{d}v}$$

For a graph $z = g(x, y)$,

$$\iint_S f \,\mathrm{d}S = \iint_D f(x, y, g(x, y))\sqrt{1 + g_x^2 + g_y^2}\,\mathrm{d}A.$$

Justification. Partition D into small rectangles; the corresponding patch of S is nearly the parallelogram spanned by $\vec{r}_u\Delta u$ and $\vec{r}_v\Delta v$, of area $|\vec{r}_u \times \vec{r}_v|\Delta u\Delta v$. Summing f times these areas and passing to the limit gives the formula. ■

Worked example

Evaluate $\iint_S z \,\mathrm{d}S$ where S is the part of the paraboloid $z = x^2 + y^2$ below $z = 4$.

$g_x = 2x$, $g_y = 2y$, so $\mathrm{d}S = \sqrt{1 + 4x^2 + 4y^2}\,\mathrm{d}A$ over the disk $x^2 + y^2 \le 4$. In polar coordinates,

$$\iint_S z \,\mathrm{d}S = \int_0^{2\pi}\int_0^2 r^2\sqrt{1 + 4r^2}\, r \,\mathrm{d}r\,\mathrm{d}\theta.$$

With $u = 1 + 4r^2$, $r^2 = (u - 1)/4$, $r\,\mathrm{d}r = \mathrm{d}u/8$:

$$= 2\pi \cdot \frac{1}{32}\int_1^{17}(u - 1)u^{1/2}\,\mathrm{d}u = \frac{\pi}{16}\Big[\frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2}\Big]_1^{17} = \frac{\pi}{8}\Big[\frac{17^{5/2}}{5} - \frac{17^{3/2}}{3} + \frac{2}{15}\Big].$$

4.6.3 Oriented surfaces and flux

Orientation

A smooth surface is **orientable** if a continuous unit normal $\vec{n}$ can be chosen. The two possible orientations are $\vec{n}$ and $-\vec{n}$; for a graph $z = g(x, y)$ the **upward** orientation is

$$\vec{n} = \frac{\langle -g_x, -g_y, 1\rangle}{\sqrt{1 + g_x^2 + g_y^2}}.$$

For a parametric surface, the orientation induced by the parametrization is

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}.$$

Flux integral

The **flux** of $\vec{F}$ across an oriented surface S is

$$\boxed{\iint_S \vec{F}\cdot\mathrm{d}\vec{S} = \iint_S \vec{F}\cdot\vec{n}\,\mathrm{d}S = \iint_D \vec{F}(\vec{r}(u, v))\cdot(\vec{r}_u \times \vec{r}_v)\,\mathrm{d}u\,\mathrm{d}v}$$

For an upward-oriented graph $z = g(x, y)$,

$$\iint_S \vec{F}\cdot\mathrm{d}\vec{S} = \iint_D \big(-Pg_x - Qg_y + R\big)\mathrm{d}A.$$

Worked example

Find the flux of $\vec{F} = \langle z, y, x\rangle$ upward through the unit sphere.
With $\vec{r}(\phi, \theta) = \langle \sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi\rangle$, a computation gives

$$\vec{r}_\phi \times \vec{r}_\theta = \sin\phi\,\langle \sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi\rangle,$$

which points outwards. Then

$$\vec{F}\cdot(\vec{r}_\phi \times \vec{r}_\theta) = \sin\phi\big(\cos\phi\sin\phi\cos\theta + \sin^2\phi\sin^2\theta + \sin\phi\cos\phi\cos\theta\big),$$

The terms containing $\cos\theta$ integrate to zero over $0 \le \theta \le 2\pi$, and

$$\int_0^{2\pi} \sin^2\theta \,\mathrm{d}\theta = \pi, \qquad \int_0^{\pi} \sin^3\phi \,\mathrm{d}\phi = \frac{4}{3},$$

so the flux is $\pi \cdot \dfrac{4}{3} = \dfrac{4\pi}{3}$. The divergence theorem predicts $\iiint_E \operatorname{div}\vec{F}\,\mathrm{d}V = \iiint_E 1\,\mathrm{d}V = \dfrac{4\pi}{3}$.

4.7 Stokes' theorem

Stokes' theorem

Let S be an oriented piecewise-smooth surface bounded by a simple closed piecewise-smooth curve C whose orientation is *consistent* with that of S (walking along C with your head pointing in the direction of $\vec{n}$, the surface lies on your left). If $\vec{F}$ is C^1 on an open region containing S, then

$$\boxed{\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = \iint_S \operatorname{curl}\vec{F}\cdot\mathrm{d}\vec{S}}$$

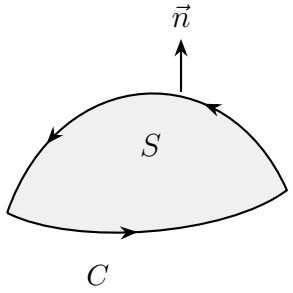


Figure 15: Stokes' theorem: the orientation of C is consistent with the normal $\vec{n}$ of S.

Two corollaries

(i) If $\operatorname{curl}\vec{F} = \vec{0}$ on a simply connected region containing S, then $\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = 0$.

(ii) The value of $\iint_S \operatorname{curl}\vec{F}\cdot\mathrm{d}\vec{S}$ depends only on the boundary curve C, not on the particular surface S spanning C.

Proof of Stokes' theorem for a graph

Let S be the graph $z = g(x,y)$ over D, with upward orientation, and let $\vec{F} = \langle P, Q, R\rangle$. The boundary C of S projects to the boundary ∂D of D, traversed counterclockwise. Parametrize C by $\vec{r}(t) = \langle x(t), y(t), g(x(t), y(t))\rangle$. Then

$$\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = \oint_{\partial D} \Big[P + Rg_x\Big]\mathrm{d}x + \Big[Q + Rg_y\Big]\mathrm{d}y,$$

because $z = g(x,y)$ makes $\mathrm{d}z = g_x\,\mathrm{d}x + g_y\,\mathrm{d}y$. Apply Green's theorem on D with

$P_1 = P + Rg_x$ and $Q_1 = Q + Rg_y$:

$$\oint_C \vec{F}\cdot \mathrm{d}\vec{r} = \iint_D \left[\frac{\partial}{\partial x}\big(Q + Rg_y\big) - \frac{\partial}{\partial y}\big(P + Rg_x\big)\right] \mathrm{d}A.$$

Expanding with the chain rule, where P, Q, R are evaluated at $(x, y, g(x,y))$ and ∂_x denotes the total derivative in x,

$$\partial_x\big(Q + Rg_y\big) - \partial_y\big(P + Rg_x\big) = Q_x - P_y + Q_z g_x - P_z g_y + R_x g_y - R_y g_x,$$

the two terms involving Rg_{xy} cancelling. On the other hand, with $\vec{n}\,\mathrm{d}S = \langle -g_x, -g_y, 1\rangle\,\mathrm{d}A$,

$$\begin{aligned}\operatorname{curl}\vec{F}\cdot\langle -g_x, -g_y, 1\rangle &= -(R_y - Q_z)g_x - (P_z - R_x)g_y + (Q_x - P_y)\\ &= Q_x - P_y + R_x g_y - R_y g_x + Q_z g_x - P_z g_y.\end{aligned}$$

The two integrands coincide, so Green's theorem gives Stokes' theorem for graphs. A general surface is a union of graph patches, and the integrals over the internal edges cancel because each edge is traversed in opposite directions. ∎

Worked example

Evaluate $\displaystyle\oint_C \vec{F}\cdot\mathrm{d}\vec{r}$ for $\vec{F} = \langle -y^2,\, x,\, z^2\rangle$ on the curve of intersection of the plane $y+z=2$ and the cylinder $x^2+y^2=1$, oriented counterclockwise when viewed from above.

Choose S as the part of the plane $y+z=2$ inside the cylinder, with upward normal. Since $z = 2-y$, $g_y = -1$, $g_x = 0$ and

$$\vec{n}\,\mathrm{d}S = \langle 0, 1, 1\rangle\,\mathrm{d}A.$$

Also

$$\operatorname{curl}\vec{F} = \left\langle \frac{\partial(z^2)}{\partial y} - \frac{\partial x}{\partial z},\ \frac{\partial(-y^2)}{\partial z} - \frac{\partial(z^2)}{\partial x},\ \frac{\partial x}{\partial x} - \frac{\partial(-y^2)}{\partial y}\right\rangle = \langle 0,\, 0,\, 1 + 2y\rangle.$$

On S, $\vec{n}\,\mathrm{d}S = \langle 0,1,1\rangle\,\mathrm{d}A$, so

$$\operatorname{curl}\vec{F}\cdot\langle 0,1,1\rangle = 1 + 2y.$$

Therefore

$$\oint_C \vec{F}\cdot\mathrm{d}\vec{r} = \iint_D (1+2y)\,\mathrm{d}A = \pi + 0 = \pi,$$

since $\displaystyle\iint_D 1\,\mathrm{d}A = \pi$ and $\displaystyle\iint_D y\,\mathrm{d}A = 0$ by symmetry over the unit disk.

4.8 The divergence theorem

Divergence theorem (Gauss)

Let E be a simple solid region (a region of type 1, 2 and 3 at once) with positively oriented boundary surface S (outward normal), and let $\vec{F}$ be a C^1 vector field on an open region containing E. Then

$$\boxed{\iint_S \vec{F}\cdot\mathrm{d}\vec{S} = \iiint_E \operatorname{div}\vec{F}\,\mathrm{d}V}$$

Proof for $F = \langle 0, 0, R\rangle$. Write $E = \{(x,y,z) : (x,y)\in D,\ u_1(x,y) \le z \le u_2(x,y)\}$, decompose $S = S_1 \cup S_2 \cup S_3$ into the bottom, top and lateral surface, and note that on the lateral surface the normal $\vec{n}$ is horizontal, i.e. $\vec{n}\cdot\vec{k} = 0$, so there $\vec{F}\cdot\vec{n} = R\,\vec{k}\cdot\vec{n} = 0$.

Hence

$$\iint_S \vec{F}\cdot\vec{n}\,\mathrm{d}S = \iint_{S_1} R\,\vec{k}\cdot\vec{n}\,\mathrm{d}S + \iint_{S_2} R\,\vec{k}\cdot\vec{n}\,\mathrm{d}S.$$

On the top S_2, $\vec{n}$ has positive z-component and $\vec{k}\cdot\vec{n}\,\mathrm{d}S = \mathrm{d}A$; on the bottom S_1, the outward normal points downwards and $\vec{k}\cdot\vec{n}\,\mathrm{d}S = -\mathrm{d}A$. Therefore

$$\iint_S \vec{F}\cdot\vec{n}\,\mathrm{d}S = \iint_D R(x,y,u_2(x,y))\,\mathrm{d}A - \iint_D R(x,y,u_1(x,y))\,\mathrm{d}A.$$

On the other hand, by Fubini's theorem,

$$\iiint_E \frac{\partial R}{\partial z}\,\mathrm{d}V = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} \frac{\partial R}{\partial z}\,\mathrm{d}z\right]\mathrm{d}A = \iint_D \Big[R(x,y,u_2) - R(x,y,u_1)\Big]\mathrm{d}A.$$

The two expressions are equal. Applying the same argument to the components $\langle P,0,0\rangle$ and $\langle 0,Q,0\rangle$ (using the other two type descriptions of E) and adding gives the theorem. For a general region, cut it into finitely many simple regions; the fluxes through the internal cuts cancel. ■

Worked example

Find the flux of $\vec{F} = \langle x^3,\, y^3,\, z^3\rangle$ across the unit sphere.

$\operatorname{div}\vec{F} = 3x^2 + 3y^2 + 3z^2 = 3\rho^2$, so by the divergence theorem and spherical coordinates,

$$\iint_S \vec{F}\cdot\mathrm{d}\vec{S} = \iiint_E 3\rho^2\,\mathrm{d}V = 3\int_0^{2\pi}\int_0^{\pi}\int_0^1 \rho^4\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta = 3\cdot\frac{1}{5}\cdot 4\pi = \frac{12\pi}{5}.$$

Worked example: computing a hard flux

Evaluate $\displaystyle\iint_S \vec{F}\cdot\mathrm{d}\vec{S}$, where $\vec{F} = \langle xy,\, yz,\, zx\rangle$ and S is the boundary of the cylinder $x^2 + y^2 \le 1$, $0 \le z \le 1$.

$\operatorname{div}\vec{F} = y + z + x$. By symmetry the integrals of x and y over the cylinder vanish, and

$$\iiint_E z\,\mathrm{d}V = \int_0^{2\pi}\int_0^1\int_0^1 z\,r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{\pi}{2}.$$

Hence the flux is $\pi/2$.

4.8.1 Limitations of the divergence theorem

The divergence theorem requires $\vec{F}$ to be defined and C^1 on the *whole* region of integration. If $\vec{F}$ has a singularity inside that region, the theorem fails.

Worked example: a point source

Consider the radial field

$$\vec{F}(x,y,z) = \frac{1}{r^2}\,\hat{r} = \frac{1}{(x^2+y^2+z^2)^{3/2}}\,\langle x,y,z\rangle, \qquad r = \sqrt{x^2+y^2+z^2},$$

whose arrows point outwards everywhere with magnitude $1/r^2$. For $r > 0$ a direct computation gives

$$\nabla\cdot\vec{F} = 0,$$

but $\vec{F}$ is undefined at the origin: the domain $\mathbb{R}^3 \setminus \{\vec{0}\}$ carries a singularity.

Surface integral. Let S be the sphere of radius R centred at the origin. There $\hat{n} = \dfrac{1}{R}\langle x, y, z\rangle$, so

$$\vec{F} \cdot \hat{n} = \frac{1}{R^3}\langle x, y, z\rangle \cdot \frac{1}{R}\langle x, y, z\rangle = \frac{x^2 + y^2 + z^2}{R^4} = \frac{1}{R^2},$$

and therefore

$$\iint_S \vec{F} \cdot \hat{n} \, \mathrm{d}S = \frac{1}{R^2} \cdot 4\pi R^2 = 4\pi.$$

Volume integral. On the other hand the divergence theorem would predict

$$\iiint_V \nabla \cdot \vec{F} \, \mathrm{d}V = \iiint_V 0 \, \mathrm{d}V = 0.$$

The two sides disagree ($4\pi \neq 0$) precisely because $\vec{F}$ is not defined at the origin, so the theorem does not apply. The non-zero flux is independent of R and measures the strength of a **point source** at the origin; rigorously the divergence is a Dirac delta concentrated there.

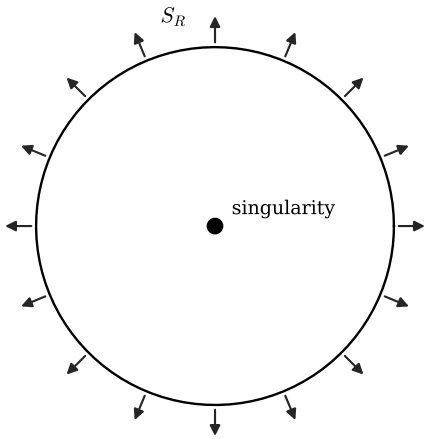


Figure 16: A point source at the origin: the field $\vec{F} = \hat{r}/r^2$ is undefined at $\vec{0}$, so the divergence theorem cannot be applied over any region containing the origin.

Summary of the four theorems

- **FTC for line integrals:** $\displaystyle\int_C \nabla f \cdot \mathrm{d}\vec{r} = f(B) - f(A)$: a 1-dimensional integral equals the value at the 0-dimensional boundary.
- **Green:** $\displaystyle\oint_C \vec{F} \cdot \mathrm{d}\vec{r} = \iint_D (\operatorname{curl} \vec{F}) \cdot \vec{k} \, \mathrm{d}A$: a line integral over a plane curve equals a double integral over the enclosed region.
- **Stokes:** $\displaystyle\oint_C \vec{F} \cdot \mathrm{d}\vec{r} = \iint_S \operatorname{curl} \vec{F} \cdot \mathrm{d}\vec{S}$: a line integral equals the flux of the curl through any surface spanning the curve.
- **Divergence:** $\displaystyle\iint_S \vec{F} \cdot \mathrm{d}\vec{S} = \iiint_E \operatorname{div} \vec{F} \, \mathrm{d}V$: the flux through a closed surface equals the triple integral of the divergence.

In each case the "boundary operator" ∂ satisfies $\partial\partial = 0$, which is why $\operatorname{curl} \nabla f = \vec{0}$ and $\operatorname{div} \operatorname{curl} \vec{F} = 0$.

Common mistakes

- Green's theorem requires the *positive* orientation; reversing C changes the sign of the integral.
- Stokes' theorem requires the orientation of C to be consistent with $\vec{n}$; otherwise a minus sign is lost.
- The divergence theorem applies only to *closed* surfaces with the *outward* normal; for an open surface one must use Stokes or a direct computation.
- The test $P_y = Q_x$ guarantees conservativity only on a simply connected domain.

4.9 Useful table of the integral theorems

Theorem / formula	Equation
Fundamental theorem for line integrals	$\displaystyle\int_C \nabla f \cdot \mathrm{d}\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$
Green's theorem (tangent / curl form)	$\displaystyle\oint_C \vec{F} \cdot \mathrm{d}\vec{s} = \iint_D (\nabla \times \vec{F}) \cdot \vec{k}\, \mathrm{d}A$
Stokes' theorem	$\displaystyle\oint_{\partial S} \vec{F} \cdot \mathrm{d}\vec{s} = \iint_S (\nabla \times \vec{F}) \cdot \mathrm{d}\vec{S}$
Green's theorem (normal / divergence form)	$\displaystyle\oint_C \vec{F} \cdot \vec{n}\, \mathrm{d}s = \iint_D (\nabla \cdot \vec{F})\, \mathrm{d}A$
Divergence theorem	$\displaystyle\iint_{\partial D} \vec{F} \cdot \mathrm{d}\vec{S} = \iiint_D (\nabla \cdot \vec{F})\, \mathrm{d}V$
Line integral of a scalar function	$\displaystyle\int_C f(x,y,z)\, \mathrm{d}s = \int_a^b f(\vec{r}(t))\, \lVert \vec{r}\,'(t) \rVert\, \mathrm{d}t$
Surface integral of a scalar function	$\displaystyle\iint_S f(x,y,z)\, \mathrm{d}S = \iint_D f(u,v)\, \lVert \vec{r}_u \times \vec{r}_v \rVert\, \mathrm{d}A$